

TECHNICAL MANUAL

**AIR FORCE METROLOGY AND  
CALIBRATION PROGRAM**

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# SECTION 1

## INTRODUCTION

### 1.1 **PURPOSE.**

This Technical Order (TO) contains the methods and procedures for the management of the Air Force Metrology and Calibration (AFMETCAL) Program. This TO implements AFMAN 21-113, *Air Force Metrology and Calibration (AFMETCAL) Management*.

### 1.2 **GENERAL.**

The AFMETCAL Program is a Department of the Air Force (DAF) program that provides measurement standards and equipment, professional and technical metrologists, Performing Work Centers (PWCs), a system of worldwide Precision Measurement Equipment Laboratory (PMEL) facilities, measurement equipment users, calibration data and integrated planning. This program ensures the reliability and accuracy of systems, subsystems, and equipment. The program provides for the calibration and repair of Test, Measurement and Diagnostic Equipment (TMDE). It also ensures measurement traceability of the TMDE through the Air Force Primary Standards Laboratory (AFPSL) to National Institute of Standards and Technology (NIST) or other AFLCMC/WNM (AFMETCAL) approved sources. To accomplish this traceability, the AFMETCAL program requires that DAF users and Precision Measurement Equipment Laboratories (PMELs) obtain calibration service from DAF PMELs or the AFPSL. AFMETCAL must approve calibration service from other sources. The responsibilities of organizations in the AFMETCAL Program are identified in AFMAN 21-113.

**1.2.1 AFLCMC/WNM (AFMETCAL).** AFLCMC/WNM, referred to as AFMETCAL throughout this document, serves as the DAF technical authority on metrology issues and is the DAF single point of contact for calibration services and traceability of measurements to the International System of Units through NIST or other AFMETCAL approved sources. AFMETCAL makes TMDE calibration and repair responsibility determinations. AFMETCAL publishes responsibility determinations in TO 33K-1-100-1 and TO 33K-1-100-2 and Calibration and Measurement Summary (CMS) TOs. TO 33K-1-100-2 and the workload summary tables in the CMS TOs are electronic data tables displayable with the Air Force Calibration Authority Viewer (AFCAV) application included with these TOs. The PMEL Automated Management System (PAMS), Facility and Equipment Maintenance System (FEMS) and LabMet systems maintain authorized facsimiles of TO 33K-1-100-2 and certain CMS workload summary tables. All references to TO 33K-1-100-2 and CMS TOs apply equally to the corresponding authorized electronic facsimiles maintained in PAMS, FEMS and LabMet. AFMETCAL Points of Contact (POC) are available on the AFMETCAL SharePoint page. This page should be used to determine the appropriate AFMETCAL personnel/office to be contacted for specific issues addressed by this TO.

#### 1.2.1.1 Key documents referenced above:

- a. TO 33K-1-100-1 (Calibration Procedure for Maintenance Data Collection Codes and Calibration Measurement Summaries)
- b. TO 33K-1-100-2 (TMDE Calibration Notes, Calibration Interval, Technical Order, And Work Unit Code Reference Guide)

### NOTE

These TOs are distributed via CD (33K-1-100-2-CD-1), which contains a desktop version of AFCAV. They are also available to all AF users via a web-based version (WebAFCAV), which is available on the Metrology Web Site (MetWeb) at url <https://metweb.afmetcal.af.mil>. In order to access WebAFCAV, submit a request for a MetWeb account by submitting the customized DD Form 2875 downloaded from this site. On this form, identify a supporting TODO account that has established a requirement for 33K-1-100-2-CD-1.

**1.2.1.2 Calibration Repair Network.** The AFMETCAL Program has been recognized as a Calibration Repair Network under the Repair Network Integration (RNI) initiative. Under RNI, the AFMETCAL Director is also known as the Product Repair Manager, AFMETCAL's Plans and Analysis Section is the Repair Network Manager, the PMEL MAJCOM Functional Managers (MFM) serve as the MAJCOM leads, and PMEL Managers are referred to as Node Managers.

**1.2.2 Air Force Primary Standards Laboratory (AFPSL).** The Air Force Primary Standards Laboratory (AFPSL) maintains the Air Force Measurement Standards. The standards are traceable to the International System of Units through

NIST or other sources as approved by AFMETCAL. Air Force Measurement Standards are used to ensure the accuracy and traceability of Base Measurement Standards. Base Measurement Standards are provided to PMELs. Unless specified elsewhere throughout this TO, the following clarifies what requirements apply to the AFPSL:

**1.2.2.1** Section 3 OPERATION. All paragraphs with the exception of para 3.10.8. The AFPSL will provide customer support IAW contract requirements.

**1.2.2.2** Section 4 CALIBRATION ASSISTANCE AND SUPPORT AGREEMENTS. All paragraphs (as applicable).

**1.2.2.3** Section 5 PREPARATION AND USE OF CALIBRATION FORMS AND LABELS. All paragraphs (as applicable).

**1.2.2.4** Section 7 PMEL ASSESSMENT PROGRAM. All paragraphs. To determine the AFPSL certification status, the AFPSL will be assessed by AFMETCAL IAW Section 7 of this TO and AFI90-201, *AF Inspection System*. However, the PT/MAP assessment will be limited to interservice or interdepartmental proficiency testing processes/results (if applicable).

**1.2.2.5** Section 8 PMEL FACILITY OPERATIONAL REQUIREMENTS. Not applicable. AFPSL facility requirements will be documented in an AFPSL Operating Instruction maintained by AFMETCAL. The AFPSL will submit ECS documentation IAW contract requirements.

**1.2.2.6** Section 9 QUALITY PROGRAM (QP). All paragraphs. The AFPSL will submit QP documents to AFMETCAL IAW contract requirements (this includes any QP upload requirements associated with the AFMAN 21-113 TMDE Self-Assessment Communicator (SAC) in the Manager's Internal Control Toolset (MICT)).

**1.2.2.7** Section 10 MANAGEMENT SYSTEM (MS). All paragraphs. The AFPSL will submit MS documents to AFMETCAL IAW contract requirements (this includes any MS upload requirements associated with the AFMAN 21-113 SAC in MICT).

**1.2.2.8** Section 11 PROFICIENCY TESTING/MEASUREMENT ASSURANCE PROGRAM (PT/MAP) AND ACCEPTANCE TESTING. AFMETCAL manages the PT/MAP program with the assistance of the AFPSL. The AFPSL may also perform interservice or interdepartmental proficiency testing as directed by AFMETCAL. The AFPSL will participate in Acceptance Testing IAW contract requirements.

**1.2.3** Precision Measurement Equipment Laboratories (PMELs).

Headquarters AF/A4LM approves the establishment of PMELs at selected installations. The PMEL is the base-level AFMETCAL Program focal point. It is the activity authorized to possess and use base measurement standards. Operational details of the program are set forth in Section 3 of this TO and related command directives.

**1.2.4** Metrology & Calibration Flight (MCF).

Metrology & Calibration Flights (MCF) are currently established at the three Air Logistics Complexes (ALC) located at Ogden, UT (OO-ALC), Oklahoma City, OK (OC-ALC), and Warner Robins, GA (WR-ALC). The MCF role in the Calibration Repair Network is limited to specific regional workload responsibilities as defined in each applicable Note Code. They have a unique function and mission separate from the co-located Type IIA PMEL. Operational details of the MCFs are set forth in Section 13 of this TO. Unless specifically stated in this TO or excluded in Section 13, the use of the term "PMEL" throughout this TO and AFMAN 21-113 is synonymous with MCF.

**1.2.5** Other DoD Services Calibration Laboratories. The use of other DOD services calibration laboratories is authorized when approved by the applicable MFM and AFMETCAL. The Army, Navy and Marine Corps operate calibration laboratories using similar measurement techniques and management concepts. These laboratories provide measurement traceability to the International System of Units through NIST. Units requesting support from another service shall follow the guidance in Section 4, which includes the interservice support agreement guidance.

**1.2.6** Air Force Enterprise Contracted (AFEC) PMEL Services Contract. The AFEC PMEL Services contract is administered by Air Combat Command's (ACC) Acquisition Management and Integration Center (AMIC). The AFEC PMEL contract is a multi-command, single contract providing services to base-level PMELs designated as "Contract" in Table 12-3. These PMELs are part of the Calibration Repair Network and follow their PWS, this TO and AFMAN 21-113 unless otherwise identified in Section 14 of this TO.



### 1.3 SCOPE.

This TO applies to all activities that possess, use, calibrate, certify, or maintain DAF TMDE, which includes TMDE belonging to the United States Space Force (USSF).

### 1.4 DEFINITIONS.

The following definitions apply for the purpose of this TO.

**1.4.1 Absolute Measurement Standard.** Designated measurement standards based on the absolute value of natural physical constants whose values can be accurately repeated under controlled conditions. Also known as an Intrinsic Standard. Examples are Monochromatic (Single Wavelength) Light Source, Josephson Junction Voltage Standards and Hall Effect Magnetic Field Standards.

**1.4.2 Air Force Primary Standards Laboratory (AFPSL).** The highest-level laboratory in the AFMETCAL Program. It maintains DAF measurement standards certified by the NIST, the US Naval Observatory (USNO), or other AFMETCAL approved sources.

**1.4.3 Calibration.** Calibration is a comparison between equipment items, one of which is a measurement standard of known accuracy, to detect, correlate, adjust and report any variation in the accuracy of the other item(s).

**1.4.4 Calibration, Limited.** Calibration of TMDE to less than the accuracy or functional capabilities specified in the authorized calibration procedure or data.

**1.4.5 Calibration, Special.** Calibration of TMDE to all specifications called for in the authorized calibration procedure plus additional requirements.

**1.4.6 Calibration and Measurement Summary (CMS).** A TO which identifies calibration support necessary to ensure the operational readiness of a specific weapon system, weapon subsystem or mission. The summary describes the calibration concept and is calibration authority for the applicable weapon system, weapon subsystem or mission. The CMS is printed as a weapon system or 33K series technical order. AFMETCAL maintains Air Force CMSs (see paragraph 2.5.3).

**1.4.7 Calibration and Measurement Requirement Summary (CMRS).** A CMRS is a three category, in-line summary of measurement parameters. Normally, a Program Office or Department of Defense (DOD) system contractor prepares and maintains the CMRS (DI-QCIC-80278B). A CMRS identifies all measurement requirements within a specific system or item of equipment. The CMRS further displays the proposed solutions for maintaining the system measurement requirements within stated limits. It is also used to identify the need for new calibration standards. This data is provided to AFMETCAL, 813 Irving-Wick Dr W, Horton Building, Heath OH, 43056-1199. AFMETCAL uses it for ensuring calibration supportability and planning for PMEL support for a particular weapon system/subsystem (see paragraph 2.5.2).

**1.4.8 Calibration Certificate or Calibration Report.** Also referred to as, "Report of Measurement". It is a document containing data relevant to the calibration of a specific test instrument or standard. The AFMETCAL Program does not differentiate between a calibration certificate and a calibration report provided the document contains the required information to interpret the calibration results for the specific calibration.

**1.4.9 Calibration Chart.** A chart prepared for a specific item. It shows the difference between the nominal value and the actual value of a measurement or group of measurements.

**1.4.10 Central Collection Point.** Organizational established collection point for TMDE transportation to the supporting PMEL. May be established at installations without an DAF PMEL to serve as the installation's focal point for the coordination and transportation of TMDE to/from the supporting PMEL with applicable provisions typically outlined in a support agreement.

**1.4.11 Certification.** The documented designation that standards and TMDE have been calibrated and meet established technical requirements.

**1.4.12 Commercial Calibration.** Calibration of TMDE obtained from a commercial source where the TMDE already has a published calibration determination, but the calibrating activity (PMEL or USER) asks for an exception. This should be considered only after organic (DAF) support options have been exhausted. AFMETCAL approval is required prior to obtaining the calibration from a commercial source. Commercial calibrations will be funded by the user.

**1.4.13 Contingency Calibration Operation.** Deployed calibration capability may include but is not limited to Rapid Assistance Support for Calibration (RASCAL), Transportable Field Calibration Unit (TFCU), mobile calibration teams, combined operations with a foreign military, etc. To limit the need for routine calibrations, units should deploy with sufficient quantities of TMDE with calibration certifications covering the projected duration of the deployment. Deployed units may coordinate support with the nearest geographical PMEL for emergency or mission essential calibration and repair.

**1.4.14 Contract Calibration.** Calibration obtained from a contracted source, funded by the user or an Item/Program Manager. Neither the PMEL nor AFMETCAL centrally manages or funds these calibrations. These items are identified with a calibration responsibility of "CONTRACT". All CONTRACT TMDE shall be coordinated with the PMEL for reporting purposes.

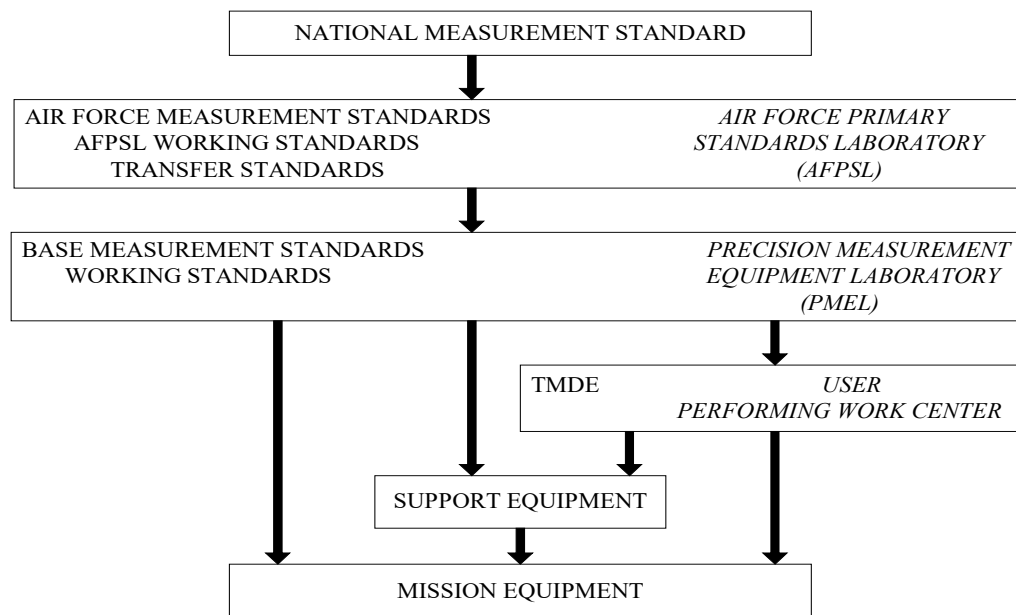
**1.4.15 Correction Chart.** A chart prepared for a specific item. The chart shows corrections to apply to indicated values to obtain true value. In instances where an item is charted and the item's accuracy or uncertainty is applied, the charted values can be used through the item's calibration interval. However, in instances where a better accuracy or uncertainty of another standard is "transferred" and being used, the charted values may only be used to complete the immediate calibration requiring the transfer/charting. The AFPSL is authorized to assign an interval only after a documented engineering analysis is completed that supports the transfer of accuracy or uncertainty to the assigned interval.

**1.4.16 Equipment Types:**

- a. Mission Equipment. Part of operational ground or airborne systems.
- b. Non-TMDE. SE not meeting the TMDE definition. This equipment does not require calibration and is the responsibility of the user.
- c. Precision Measurement Equipment (PME). For purposes of this TO, PME is the same as TMDE. This is the former title for TMDE.
- d. Support Equipment (SE). A broad category of equipment and tools used to verify and maintain mission equipment. SE can include equipment used to verify the operation of other support equipment. SE is categorized as either TMDE or Non-TMDE. The AFMETCAL Program is primarily concerned with TMDE support equipment.
- e. Test, Measurement and Diagnostic Equipment (TMDE). Devices used to maintain, evaluate, measure, calibrate, test, inspect, diagnose, or otherwise examine materials, supplies, equipment, and systems to identify or isolate actual or potential malfunction, or decide if they meet operational specifications established in technical documents. ISO/IEC 17025 refers to this equipment as "measuring and test equipment."

**1.4.17 ISO/IEC Standard 17025.** The international standard for commercial metrology, "General Requirements for the Competence of Calibration and Testing Laboratories." It includes the relevant requirements of ISO 9000 series standards. The AFMETCAL program does not approve this standard as the sole requirement for operating a DAF PMEL.

**1.4.18 Manager.** The individual responsible for day-to-day management and operation of the PMEL and its associated work centers, such as production control (logistics), quality program and material control. Synonymous with PMEL Manager, PMEL Site Manager, PMEL superintendent, PMEL branch chief, TMDE branch chief, TMDE flight chief, TMDE section chief and Laboratory Manager. The PMEL manager is referred to as the PMEL node manager under RNI.



**Figure 1-1. Measurement Traceability of the Air Force Metrology and Calibration Program.**

**1.4.19 Measurement or Reference Standards.** Equipment by which the accuracy and traceability of a precision measurement is derived. The typical chain of measurement standards is shown in Figure 1-1. The following are types of standards:

- a. Air Force Measurement Standards. Equipment certified by NIST or other AFMETCAL approved sources. These standards are used by the AFPSL as a basic measurement reference for the DAF.
- b. Air Force Primary Standards Laboratory (AFPSL) Working Standards. TMDE certified by the AFPSL and used to calibrate other TMDE.
- c. Base Measurement Standards. Equipment certified by the AFPSL or AFMETCAL approved sources. These standards are used by a DAF PMEL as a measurement reference. Base Measurement Standards are designated in TO 33K-1-100-2.
- d. Exchange Standard. A calibration standard calibrated by the AFPSL through a calibrate-and-replace methodology to minimize PMEL down time. An exchange standard is calibrated by the AFPSL and sent to a PMEL to replace a like base measurement standard before its calibration has expired. The exchange standard whose calibration interval is expiring is returned to the AFPSL for calibration, and then forwarded to another PMEL. TO 33K-2-11, *Detailed Instruction and Information Pertaining to Air Force Primary Standards, Laboratory Calibration Exchange Standards, and Artifact Proficiency Testing Standards, PMELs Only*, contains detailed instructions and information for Exchange Standards.
- e. National Measurement Standards. Equipment or physical constants identified and normally used by NIST. These standards serve as a basic measurement reference for use throughout the United States and at U.S. installations/activities overseas.
- f. Working Standards. TMDE certified by PMEL and used to calibrate other TMDE.
- g. Transfer Standard. A measurement standard calibrated by the AFPSL and circulated to multiple PMELs to calibrate base measurement standards or working standards. A transfer standard is calibrated by the AFPSL, forwarded to a PMEL, used by the PMEL to calibrate a base or working standard, and returned to the AFPSL or forwarded to the next PMEL as directed. TO 33K-2-11 contains detailed instructions and information for Transfer Standards.

**1.4.20 Metrology.** The science or system of weights and measures used to determine conformance to technical requirements. This includes the development of standards and systems for absolute and relative measurements.

**1.4.21 Precision Measurement Equipment Laboratory (PMEL).** A laboratory authorized to own and use base measurement standards to maintain working standards. The working standards are used along with PMEL-owned TMDE to maintain (troubleshoot, align, repair, and calibrate) TMDE designated as PMEL responsibility. Authorizations for PMEL equipment and facility requirements are tailored to meet specific requirements for supported missions. PMELs are the base-level link for measurement transfer and maintenance self-sufficiency for all systems in the DAF. PMEL types are:

- a. Type IIA. A base-level PMEL with a 68 °F room providing support to an Air Logistics Complex (ALC) and/or a large geographical area. Type IIA PMELs supporting ALCs are operated by AFMC. PACAF theater support command operates the Type IIA PMELs at Kadena AB, JA and Elmendorf AFB, AK. USAF theater support command operates the Type IIA PMEL at RAF Feltwell, GB.
- b. Type IIB. A base-level PMEL established to support aircraft, missiles, ground systems and/or other equipment on base or in the local area.
- c. Type IIC. A base-level PMEL with a 68°F room supporting research, development, test or evaluation programs, as well as other operational and support functions.
- d. Type III. A PMEL supporting a specific mission. It normally receives calibration support from a Type II( ) PMEL. With the exception of Rapid Assistance Support for Calibration (RASCAL) sets, a Type III PMEL is not authorized at an installation where a Type II( ) PMEL exists.
- e. Type IV. A PMEL supporting a specific weapon system. It uses a transportable measurement system in fixed and deployed locations. A Type IV PMEL receives calibration support from a Type II( ) PMEL. It may be located at an installation where a Type II( ) PMEL exists. Type IV PMELs are no longer authorized in the USAF metrology program. They still exist, however, in FMS country metrology programs.

**1.4.22 PMEL Mobile and On-Site Measurement Capabilities.**

- a. Rapid Assistance Support for Calibration (RASCAL). A portable PMEL designed for rapid deployment to support emergency calibration needs. It consists of five environmentally controlled shelters containing measurement standards and other TMDE. The shelters are joined together when in use. They are taken apart and airlifted to the needed location.
- b. Jet Engine Test Stand Calibrator (JETSC). TMDE and accessories needed to perform on-site calibration of jet engine test stands, housed in a trailer, or set of ruggedized cases.
- c. Portable Automatic Test Equipment Calibrator (PATEC). A PATEC is used to calibrate Automatic Test Equipment (ATE) on-site. A PATEC consists of a set of portable standards. A Core PATEC meets the calibration needs of a variety of test stations.
- d. Transportable Field Calibration Unit (TFCU). The TFCU consists of working standards in hardened cases selected to function within given performance tolerances as long as the equipment is operated within the published environmental restrictions of the TFCU. The TFCU is used to calibrate TMDE located at remote sites or bases.

**1.4.23 PMEL Quality Assurance (PQA) Evaluator.** Shall be a highly qualified 2P071 or equivalent when possible. PQA evaluators are selected by PMEL management and appointed in writing. PQA evaluators are the only personnel designated to perform Quality Reviews and Process Reviews applicable to the Quality Program.

**1.4.24 Proficiency Testing/Measurement Assurance Program (PT/MAP).** The PT/MAP is a process for assessing laboratory capability by means of inter-laboratory data comparisons and exchange standard intercomparisons. Results from PT/MAP are an indication of a laboratory's competence and the integrity of measurement standards and can be an integral part of the assessment process.

**1.4.25 Rainbowed Equipment.** Non-prepositioned equipment that is sourced from multiple units, left in place, and used by multiple units over time to enable the best support of vulnerable AEFs while minimizing home station impact on supporting units.

**1.4.26 Reachback.** The process of obtaining products, services and applications of forces, equipment, or materiel from DAF organizations not forward deployed. This capability allows commanders to obtain or coordinate support from units not physically located with the forward force.

**1.4.27 Risk.** Any potential that negatively affects SART, capability, mission or the RNI network (based on time, cost, quality, performance, etc.). When required, risk shall be described as Low, Moderate or High.

**1.4.28 Risk Management.** The process of identifying, assessing, and prioritizing risks to develop mitigation strategies in an effort to accept, control, avoid, minimize negative impacts. Risks identified as Moderate or High must have a corrective action documented and performed.

**1.4.29 Shall.** Shall is a directive statement. Reference TO 00-5-1, *AF Technical Order System*.

**1.4.30 Test Accuracy Ratio/Test Uncertainty Ratio (TAR/TUR).** The ratio of the uncertainty (accuracy) of the TI to the uncertainty (accuracy) of the standard (for example 4:1). TAR and TUR are equivalent in the Air Force Metrology and Calibration Program, as the term Accuracy has been used for the (more correct) term Uncertainty in Air Force Calibration procedures.

**1.4.31 Traceability.** The property of a measurement result whereby the result can be related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty. Mandatory DAF traceability relates individual measurement results to the International System of Units through NIST or AFMETCAL approved sources through an unbroken chain of calibrations all having stated uncertainties as shown in Figure 1-1.

**1.4.32 User.** User refers to the using activity or owning organization responsible for calibration and maintenance of items designated "USER" responsibility in a CMS or TO 33K-1-100-2. In most cases, the 'user' and the 'owner' are the same. In general, the 'user' is the activity that physically uses the item. The 'owner' is the organization that owns the item via a supply account. The 'user' performs USER calibrations or coordinates with the PMEL for assistance when resources are not available. The PMEL will assist to identify required equipment or approved sources of support. As a last resort, the PMEL may perform the calibration. The owner is responsible for all funding associated with these efforts.

## **1.5 PROGRAM ABBREVIATIONS.**

|    |        |                                        |
|----|--------|----------------------------------------|
| a. | ABS    | Absolute Standard                      |
| b. | AEF    | Air Expeditionary Force                |
| c. | ALC    | Air Logistics Complex                  |
| d. | ASC    | Allowance Source Code                  |
| e. | CAP    | Corrective Action Plan                 |
| f. | CBU    | Calibrate Before Use                   |
| g. | CEE    | Contingency Equipment Exception        |
| h. | CPIN   | Computer Program Identification Number |
| i. | CRF    | Centralized Repair Facility            |
| j. | DAF    | Department of the Air Force            |
| k. | DPAS   | Defense Property Accountability System |
| l. | ECS    | Environmental Control System           |
| m. | ESD    | Electrostatic Discharge                |
| n. | FMS    | Foreign Military Sales                 |
| o. | GEOLOC | Geographic Location                    |
| p. | MAD    | Measurement Area Discipline            |
| q. | MCF    | Metrology & Calibration Flight         |
| r. | OEM    | Original Equipment Manufacturer        |
| s. | OWC    | Owning Work Center                     |
| t. | PAMS   | PMEL Automated Management System       |
| u. | PAS    | PMEL Activity Summary                  |
| v. | PNC    | Process Nonconformity                  |
| w. | PPP    | Public Private Partnership             |
| x. | PQA    | PMEL Quality Assurance                 |

|     |        |                                                    |
|-----|--------|----------------------------------------------------|
| y.  | PR     | Process Review                                     |
| z.  | PRP    | Process Review Plan                                |
| aa. | PT/MAP | Proficiency Testing/ Measurement Assurance Program |
| bb. | PWC    | Performing Work Center                             |
| cc. | QNC    | Quality Nonconformity                              |
| dd. | QP     | Quality Program                                    |
| ee. | QR     | Quality Review                                     |
| ff. | RADIAC | Radiation, Detection, Indication, and Computation  |
| gg. | RC     | Root Cause                                         |
| hh. | RNI    | Repair Network Integration                         |
| ii. | RNM    | Repair Network Manager                             |
| jj. | SA     | Security Assistance                                |
| kk. | SART   | Safety, Accuracy, Reliability and Traceability     |
| ll. | SATAF  | Site Activation Task Force                         |
| mm. | SBSS   | Standard Base Supply System                        |
| nn. | SICL   | See Individual Component Listing                   |
| oo. | SICW   | See Item Calibrated With                           |
| pp. | TCM    | Technical Content Manager                          |
| qq. | TI     | Test Instrument                                    |
| rr. | TPE    | Trainer Proficiency Evaluation                     |
| ss. | TRC    | Technical Repair Center                            |
| tt. | WRM    | War Reserve Material                               |
| uu. | WUC    | Work Unit Code                                     |

## SECTION 2

### AFMETCAL PROGRAM ORGANIZATION AND INTERFACE

#### 2.1 GENERAL.

The AFMETCAL Program includes AFMETCAL, PMELs and their resources, the TMDE in the DAF, which includes the USSF supported under AFMC. The directives that provide the policy and guidance are also part of the program. The purpose

of the AFMETCAL Program is to ensure that all DAF TMDE used to make measurements is accurate, reliable, provides standardized measurements, and is traceable through the AFPSL to the International System of Units through NIST or other AFMETCAL approved sources. This TMDE is used to make sure that each system and subsystem is accurate and reliable and ensures that they properly interface with other systems.

#### 2.2 AFMETCAL PROGRAM.

**2.2.1 Figure 2-1 Description.** Figure 2-1 depicts each element of the AFMETCAL Program and the interrelationship of major DAF organizations. The center circle identifies the major elements of the AFMETCAL Program: AFMETCAL Director (AFLCMC/WNM), the PMELs and TMDE users. The focus of the AFMETCAL Program is to ensure accuracy and traceability of TMDE. This TMDE is used to ensure the accuracy and reliability of all DAF systems. The resources needed to operate the AFMETCAL Program are identified in the rectangles. In cooperation with the other elements, AFMETCAL provides program guidance, PMEL standards, traceability of standards, calibration procedures, PMEL evaluations, metrology engineering support, and determines calibration requirements of systems. The PMELs are the performing work centers containing the personnel that calibrate, certify, and repair the TMDE owned and used by DAF organizations. The AFMETCAL program requires users and PMELs to obtain calibration service from DAF PMELs or the AFPSL as appropriate. AFMETCAL must approve obtaining calibration service from other sources. Users of TMDE utilize their PMEL calibrated items to verify their support equipment (test benches, simulators, etc.) as well as major systems or subsystems. Users of TMDE are assigned owning work center (OWC) codes.

**2.2.2 Resources used by AFMETCAL Program.** There are four primary groups of resources used by the AFMETCAL Program to do its mission (personnel, equipment/material, documentation, and facilities). A basic need is sufficient personnel, having the necessary education, training, technical knowledge, and experience for their assigned functions. These personnel require directives/procedures, material, and facilities. Calibration procedures provide the calibration methods. The PMEL requires parts and material to perform calibration and repair of TMDE. They also need their own measurement standards. The PMEL facility has some special requirements based on the system(s) it supports. The requirements of the program are based upon the measurement needs, and the accuracy of the measurements required of all systems supported by the DAF. The outer ring of Figure 2-1 identifies the organizations that interface with the AFMETCAL Program. They provide, or coordinate, in the provisioning of the resources for the program. The outer ring of Figure 2-1 also shows the planning required to ensure that the proper AFMETCAL Program support is available. That support is needed for all systems from the time a system is under development through its entire life cycle.

**2.2.3 Funding.** Material and personnel needs are satisfied according to the amount of funding available.

**2.2.3.1** Funds for daily operating expenses of the PMELs are provided by the owning command. They generally come from operation and maintenance (O&M) appropriations. DAF regional workload support is generally funded by O&M appropriations.

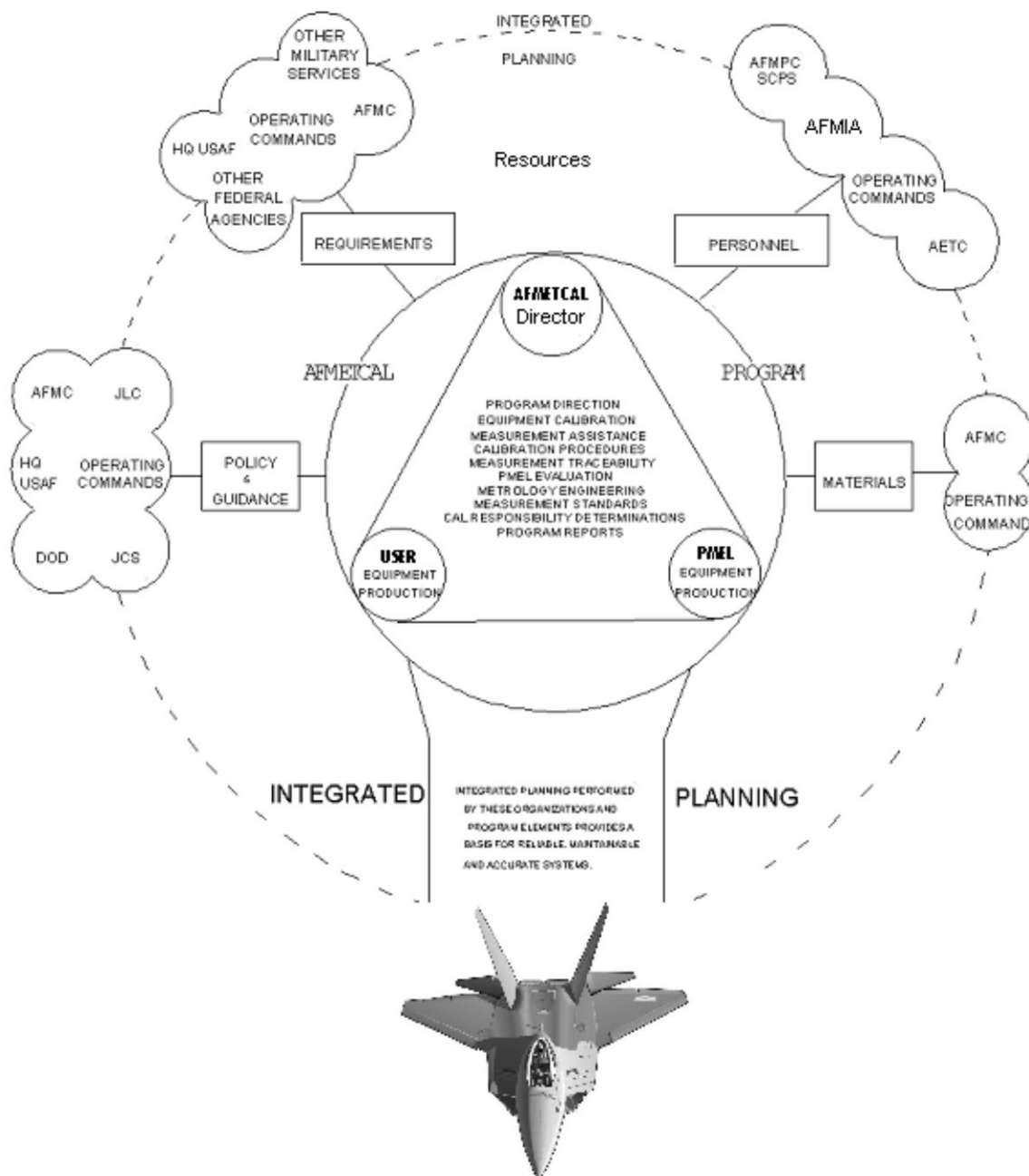
**2.2.3.2** Centrally procured investment equipment funding for equipment costing less than \$250K per item is provided to the AFMC procuring agency by MAJCOMs.

**2.2.3.3** Centrally procured expense (parts) item funding comes from the DAF Stock Fund.

**2.2.3.4** Operating command or base procured investment and expense items are generally funded from O&M and equipment funds.

**2.2.3.5** Facility funds are part of 3300 (Military Construction – DAF) appropriations. This includes major and some minor construction projects as well as facility project planning efforts.

**2.2.3.6** AFMC is responsible for funding research and development projects. This includes support required for those projects turned over to other operating commands.



**Figure 2-1. AFMETCAL Program Resources and Planning Interfaces.**

## **2.3 PROGRAM POLICY AND DIRECTION.**

HQ USAF/A4LM provides program policy in AFMAN 21-113, *Air Force Metrology and Calibration (AFMETCAL) Management*.

## **2.4 PERIODIC REPORTS UTILIZED.**

AFMETCAL and the PMEL functional managers utilize periodic reports submitted by each of the PMELs to assist in managing the program.



## **2.5 PROGRAM REQUIREMENTS.**

New systems or improvements in weapon system performance are major factors in upgrading the measurement capability of supporting PMELs. AFMETCAL has the responsibility to review the measurement requirements of all DAF systems. This normally happens through the review of Support Equipment Recommendation Data (SERD) and/or CMRS documents.

**2.5.1 Support Equipment Recommendation Data (SERD).** The SERD review process is used to justify the weapon system support equipment. SERDs are reviewed by Operating Commands, the acquisition agency, the item manager and AFMETCAL. Calibration responsibility determinations are made for approved SERDs. Calibration procedure requirements are established when necessary.

**2.5.2 Calibration and Measurement Requirement Summary (CMRS).** The CMRS documents are only required on complex systems. The completed document identifies the measurement requirements. They extend from the weapon system through the support equipment to the PMEL measurement standards.

**2.5.3 Calibration and Measurement Summary (CMS).** The CMS is published as a TO and details the measurement accuracy requirements on a complex system. AFMETCAL is responsible for the technical adequacy of the CMRS and the subsequently published CMS. AFMETCAL is also responsible for ensuring that measurement traceability exists from the national level to the system measurement. This includes the AFPSL capability. If the system has no unique or complex support requirements, the applicable support equipment is added to TO 33K-1-100-2.

**2.5.4 Measurement Upgrade Identification.** Necessary improvements in PMEL measurement capability are identified early in AFMETCAL review. This is done in the calibration responsibility determinations and the calibration procedure preparation process. If additional measurement capabilities are required, the measurement standard shall be acquired and distributed by AFMETCAL to the appropriate PMELs following coordination with MAJCOM Functional Managers (MFMs). If only national stock listed test equipment is required; the equipment shall be added to the applicable Allowance Source Code (ASC). The equipment shall be requisitioned by the PMEL. Normally, workload summaries derived from the SERD and CMRS reviews, are distributed to the operating commands. They are distributed at various times during the life cycle of the system to allow for PMEL support planning.

## **2.6 PROGRAM MATERIAL.**

Material is a broad category. It includes facilities as well as equipment, parts, and other items. It is used to support the AFMETCAL Program mission at all levels.

**2.6.1 PMEL Facility Requirements Documentation.** AFMETCAL specifies PMEL facility requirements and performs reviews of facility project documentation. Design and construction are an owning command responsibility. Facilities Criteria (FC) 4-218-01F, *Air Force Criteria for Precision Measurement Equipment Laboratory Design and Construction*, and DAFMAN 32-1084, *Standard Facility Requirements*, are some of the major publications involved. TO 00-20-14 (Section 8, PMEL Facility Operational Requirements) contains facility operating parameters for PMELs.

**2.6.2 Equipment.** Equipment is obtained through the Defense Property Accountability System (DPAS) by ASC authorizations from ALC Commodity Management, AFMETCAL or local purchase.

**2.6.3 Expense Items.** Expense items (parts, tools, furnishings, etc.) are obtained through the Standard Base Supply System (SBSS) from ALC Commodity Management, local purchase, or local manufacture.

**2.6.4 Technical Data.** Technical data is obtained from the Air Force TO System, AFMETCAL, other DOD Agencies or local purchase.

**2.6.5 Management Information System (MIS).** Software and hardware that provides inventory, processing, calibration scheduling, job data documentation and quality assurance functionality.

## **2.7 PROGRAM INTEGRATION.**

TMDE requiring support drives all resource needs and normally comes from three sources. They are (A) centrally procured items resulting from a systems acquisition, (B) centrally procured replacement buys and (C) locally procured TMDE.

**2.7.1 AFMC Centrally Procured TMDE.** Close working relationships among the System Program Directors (SPDs), AFMETCAL's measurement area and systems planners, ALC Commodity Managers, Product Group Managers, Material Group Managers, AETC/TT, and the using commands are necessary to ensure proper planning, programming, budgeting, acquisition, training, and distribution of resources needed to support this TMDE are provided.

**2.7.2 Centrally Procured Replacement TMDE.** Close working relationships among ALC Commodity Managers, AFMETCAL measurement area planners, Product Group Managers, Material Group Managers, AETC/TT, and the using commands are necessary to ensure planning, programming, budgeting, acquisition, training, and distribution of resources needed to support this TMDE are provided.

**2.7.3 Locally Procured TMDE.** The local user/buyer funds support requirements. Before contracting for TMDE, the buyer shall coordinate support requirements with PMEL. See TMDE User Responsibilities in Section 3 for more information.

## SECTION 3

### OPERATION

#### 3.1 CALIBRATION OF STANDARDS AND OTHER TMDE.

The PMELs, which are maintained, manned, and operated by the MAJCOMS, are responsible for calibrating and/or repairing standards and other TMDE specified in CMS TOs, TO 33K-1-100-1, and TO 33K-1-100-2. PMELs are also responsible for other workload that may be assigned by the MAJCOM. Policy, procedures, intervals, and responsibilities for the maintenance, calibration, and certification of standards and TMDE are identified in this TO, TO 33K-1-100-1, TO 33K-1-100-2, other applicable command directives, and weapon or support system CMS TOs. These TOs and directives shall be used for all TMDE used to make qualitative/quantitative measurements on operational systems and equipment. All TMDE having an effect on the accuracy and validity of calibrations shall be calibrated. Any standard or TMDE that has exceeded its calibration due date or is otherwise judged unreliable shall be recalled or removed from service.

**3.1.1 Valid and Accurate Calibration Requirements.** To assure valid and accurate calibrations are performed, it is necessary to:

- a. Comply with appropriate calibration authority.
- b. Use technically qualified personnel.
- c. Use equipment called for in the calibration authority or substitute equipment with required functions, ranges, and test uncertainty ratios.
- d. Perform calibrations in an environmentally controlled atmosphere.
- e. Utilize current technical data.

**3.1.2 Calibration Authority Precedence.** Calibration authority selection shall be made in accordance with the following precedence:

- |                |                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>First.</u>  | CMS* (Applicable to TMDE used solely to support a system referenced in the CMS). Where TMDE is applicable to more than one CMS and where different calibration intervals are listed, use the shortest interval listed and note in the special block which CMS was used for the interval. |
| <u>Second.</u> | TO 33K-1-100-2*.                                                                                                                                                                                                                                                                         |
| <u>Third.</u>  | TO 33K-1-100-1 Section 3-1, if applicable.                                                                                                                                                                                                                                               |
| <u>Fourth.</u> | Other official DOD calibration procedures.                                                                                                                                                                                                                                               |
| <u>Fifth.</u>  | Equipment maintenance TO.                                                                                                                                                                                                                                                                |
| <u>Sixth.</u>  | Commercial Publications or Commercial Data.                                                                                                                                                                                                                                              |

\* For PAMS, FEMS, and LabMet users: TO 33K-1-100-2, and Calibration and Measurement Summary (CMS) TOs are distributed on CD and manually updated via monthly updates posted to the AFMETCAL SharePoint. Users may also request access to the web-based version (WebAFCAV) available on MetWeb (<https://metweb.afmetcal.af.mil>). This information is also provided to the PAMS managers to automatically update the PAMS database. Thus, information contained in the PAMS system is current and calibrating organizations are authorized to use the calibration determination information contained in the PAMS system. FEMS and LabMet users must ensure that their calibration determination databases are current with those distributed via the CD and electronic updates. The corresponding precedence applies to these systems. If specifications are listed as "Table 1" in 33K-1-100-2, they must be used and can only be viewed using AFCV.

**3.1.3 Local K100.** A locally managed addendum to TO 33K-1-100-2 that contains the following inventory items listed below. NOTE: It is recommended the reason for the addition be maintained within PAMS or MIS equivalent.

- a. Items that have an AFTO Form 45 being processed. These items must be purged when they become listed in a CMS or TO 33K-1-100-2.
- b. Items that have no identifiable part number or manufacturer and have been calibrated under the provisions for "Special" calibrations.
- c. Items from FMS countries.
- d. Items that do not belong to the DAF, ANG or AFRC. (Support agreements, NASA, DOE, etc.)
- e. Locally manufactured items determined to be TMDE. (Reference TO 33K-1-100-1, Section 1)
- f. Items with Responsibility Determination Exceptions as per 3.1.5.3.

**3.1.4 AFPSL Requirements.** The AFPSL evaluates many first-article items and is responsible for acceptance testing of certain new items entering the inventory. This requires some deviation from normal directives since formal 33K procedures may not exist for such items. The process for conducting first-article and acceptance testing shall be documented.

**3.1.5 Calibration Responsibility Determinations.** TMDE calibration and maintenance responsibility determinations are managed by AFMETCAL and are published in CMS TOs and TO 33K-1-100-2.

**3.1.5.1 Request for Calibration Responsibility Determinations.** If the TMDE is not listed in the CMS or TO 33K-1-100-2 as prescribed in TO 33K-1-100-1, the PMEL shall submit an AFTO Form 45, *Request for Calibration Responsibility Determination*, to AFMETCAL in accordance with instructions in Section 5 of this TO. The PMEL shall assist the user in maintenance and calibration; however, the maintenance and calibration of the equipment shall be the responsibility of the user and the procuring agency until the calibration responsibility decision is made. While awaiting the calibration responsibility determination, as published in the CMS or 33K-1-100-2, the item may be calibrated using other DOD procedures or commercial data per the calibration authority precedence in paragraph 3.1.2. If the item is new and was calibrated by the manufacturer, the PMEL may accept the initial commercial calibration IAW paragraph 4.5.1.2.2.

**3.1.5.2 Non-DAF TMDE.** Ensure requests for calibration responsibility determinations, AFTO Form 45s, are not submitted for equipment owned and used by departments outside the DAF. Such a customer is responsible to give the PMEL the capability to support new equipment, if the PMEL does not already have it.

**3.1.5.3 Responsibility Determination Exceptions.** There may be instances where individual MAJCOMS may require exceptions from the assigned determination because the capability exists in an organization other than the designated PMEL. Requests to deviate from the assigned calibration responsibility, by having the calibration performed in an organization other than the designated PMEL, shall be processed through the PMEL MAJCOM Functional Manager (MFM). Prepare a 'Request to Deviate From the Published Calibration Determination' form, available on the AFMETCAL SharePoint, Customer Support POCs, Commercial Cals, Commercial Calibration Files. Forward the request to the PMEL MFM, who will in turn forward to the AFMETCAL email address indicated on the Request Form. AFMETCAL's concurrence or non-concurrence shall be forwarded to the MFM. If AFMETCAL concurs, the MFM may authorize the PMEL to change their local K100 via message, letter, or email. PMEL shall retain the authorization and make it available upon request. Calibration of a USER or MCF designated item by a PMEL is not considered an exception to responsibility determination.

**3.1.6 Calibration Equipment Traceability/Accuracy.** The overall program shall be designed and operated to ensure measurements are traceable through the AFPSL to the International System of Units through NIST or other AFMETCAL approved sources. The equipment required to calibrate TMDE is listed in each calibration TO. General calibration TOs may have multiple equipment requirements that are broad in order to cover the requirements for any item listed in these procedures; thus, select equipment based on the accuracy of the item being calibrated. Other equipment, which has the necessary functions, ranges, and accuracies, may be substituted to perform the calibration even though they may not have all the characteristics of the listed item. Use TO 33K-1-100-2 to assist in the identification of substitute items. When selecting or substituting equipment, an uncertainty ratio of at least 4 to 1 must be maintained unless the equipment specified in the calibration authority has an uncertainty ratio of less than 4 to 1. In that case, substitute equipment shall not lower the uncertainty ratio to less than that of the equipment specified (i.e., If the uncertainty ratio of the equipment specified in the calibration authority is greater than 4 to 1, the uncertainty ratio of the substituted standard must be at least 4 to 1. If the equipment specified has an uncertainty ratio accuracy of 2 to 1, the uncertainty ratio of the substituted equipment must be at least 2 to 1). Round uncertainty ratios to one decimal place; (i.e., 3.94 rounds to 3.9; 3.95 rounds to 4.0). CAUTION: Care must be taken in determining the accuracy of a given measurement. In many instances, the accuracy of the measurement is not the sum of accuracies of the instruments used. Modification of, or deviation from calibration procedure steps is authorized when substitute equipment is used. However, all parameters, ranges and appropriate uncertainty ratios relating to the accuracies in Table 1 of the calibration TO shall be met or exceeded. Questions regarding determination of accuracies shall be directed to the responsible engineer at AFMETCAL. DAF calibration TOs are designed to ensure calibrations are traceable through the AFPSL to the International System of Units through NIST or other AFMETCAL approved sources. DAF calibration laboratories shall ensure that any substitution of equipment and subsequent modification of procedural steps does not adversely affect the traceability of the calibration being performed.

**3.1.6.1 Calibration Software.** When using calibration software, the calibration authority for the measurement shall be established using the order of precedence noted in Paragraph 3.1.2. Normally, the calibration authority is a 33K Series calibration procedure.

**3.1.6.1.1** Calibration software provided by AFMETCAL can be released as: a) CPIN software referenced in a manual 33K procedure, b) CPIN software listed in the [Auto CTO] field of a CMS or 33K-1-100-2, or c) software package released as a 33Kx-x-xxxx-10 automated procedure and listed the [Auto CTO] field of a CMS or 33K-1-100-2. The (CPIN or 33K) automated procedure listed in the [Auto CTO] field of a CMS or 33K-1-100-2 carries the same calibration authority as the equivalent manual procedure. Thus, the calibration of the end item shall satisfy the requirements contained in the calibration authority.

**3.1.6.1.2** Use of calibration software is viewed as a controlled substitution of equipment. The control is established through the documentation, validation and verification requirements noted in Table 3-1. AFMETCAL reserves the right to disapprove use of the software, if it has not been adequately validated and verified, does not have current up to date documentation, or does not satisfy/is not equivalent to the Calibration Authority. Use of software that has not been properly documented, validated, and verified is prohibited.

#### NOTE

Spreadsheets or programmable calculator programs are not considered calibration software; however, they must be verified prior to use to ensure desired/accurate results. Verification may be accomplished by the calibrating technician prior to use, or if desired, by PQA or other technically qualified personnel who may then protect/lock the product for use by all laboratory technicians. Calibration software is divided into four classes as shown in Table 3-1.

**Table 3-1. Calibration Software Classes.**

| Software Class                                 | Developer      | Maintainer     | Distributor        | Val & Ver                | Calibration Authority | Documentation                                                                            |
|------------------------------------------------|----------------|----------------|--------------------|--------------------------|-----------------------|------------------------------------------------------------------------------------------|
| AFMETCAL<br>CPIN or<br>Technical<br>Order (TO) | AFMETCAL       | AFMETCAL       | AFMETCAL           | AFMETCAL                 | See<br>para 3.1.2     | AFMETCAL Generated &<br>Approved                                                         |
| PMEL/<br>AFPSL CPIN                            | PMEL/<br>AFPSL | AFMETCAL       | AFMETCAL           | PMEL/ AFPSL/<br>AFMETCAL | See<br>para 3.1.2     | PMEL/ AFPSL Generated -<br>AFMETCAL Approved                                             |
| PMEL/<br>AFPSL Local                           | PMEL/<br>AFPSL | PMEL/<br>AFPSL | Not<br>Authorized. | PMEL/ AFPSL              | See<br>para 3.1.2     | PMEL/ AFPSL Generated -<br>PQA Approved - Copy of Val<br>& Ver doc shall be kept on file |
| Commercial                                     | Commercial     | Acquirer       | Acquirer           | Acquirer                 | See<br>para 3.1.2     | Acquirer Responsibility - Copy<br>of Val & Ver documentation<br>shall be kept on file    |

**3.1.6.1.3** Definitions for Table:

- a. Developer: Authors software.
- b. Maintainer: Maintains software and is responsible for configuration control and software changes/updates. The Maintainer shall keep the calibration software in agreement with the Calibration Authority.
- c. Distributor: Distributes software to users. Updated copies of the calibration software shall be provided whenever the Maintainer accomplishes changes/updates.
- d. Validation & Verification: Organization responsible to validate and verify that software satisfies calibration requirements. Validation and verification shall demonstrate and document that the software operates correctly, produces the desired result, cannot be manipulated/changed during operation, and satisfies and is equivalent to the Calibration Authority.
- e. Calibration Authority: Calibration requirements as determined by AFMETCAL shall be established using the order of precedence noted in Paragraph 3.1.2.
- f. Acquirer: Organization that procures the calibration software.

**3.1.6.1.4** Calibration software documentation levels shall be determined by the maintainer but shall include as a minimum:

- a. End items to be calibrated with the software.
- b. Software identification number.
- c. Software revision status/number.

- d. Calibration authority to be cited (including version date).
- e. Software Validation and Verification (Val & Ver) Plan.
- f. Validation and Verification results. This documentation shall be maintained throughout the software's life.

#### NOTE

Software Validation and Verification shall be re-accomplished, and documentation updated if the Calibration Authority is changed and/or updated. Calibration software that has not undergone Validation and Verification against the published (official) version of the Calibration Authority or lacking current up to date documentation cannot be used.

**3.1.6.1.5 Software Verification and Validation Support.** One or several PMELs may be tasked through the Command MFM to participate in software Verification and Validation tests for AFMETCAL managed CPIN or TO calibration software. Specific guidelines detailing the Validation and Verification process will be provided to the PMEL for each test.

**3.1.7 Calibration Procedures.** AFMETCAL shall prepare, maintain, and control content, and issue 33K series technical orders and software for calibrating measurement standards, TMDE and automated test equipment defined as the responsibility of the PMEL. AFMETCAL shall publish and maintain TO 33K-1-100-1, *Calibration Procedure for Maintenance Data Collection Codes and Calibration Measurement Summaries*. AFMETCAL shall ensure:

- a. Calibration procedures contain the required range and tolerance or uncertainty of each item or unit parameter being calibrated or verified.
- b. The procedures contain a generic description of the measurement standards and equipment needed with the required parameter, range, tolerances or uncertainties, and specifications for performing the measurement, and/or representative types (manufacturer, model, option) that are capable of meeting the generic description for the measurement standards.
- c. The procedures are consistent with the accuracy required, and with any standard specifications relevant to the calibrations/verifications concerned.
- d. The calibration uncertainties are sufficiently small so that the adequacy of the measurement is not affected. Well-defined and documented measurement assurance techniques or uncertainty analyses may be used to verify the adequacy of a measurement process. If such techniques or analyses are not used, then the collective uncertainty of the measurement standards shall not exceed 25 percent (4:1) of the acceptable tolerance for each characteristic of the measuring and test equipment being calibrated or verified.

**3.1.7.1 Locally Developed Procedures.** Locally developed procedures shall not be used to perform calibration until approved by AFMETCAL and the using command (TO 00-5-1). Format of the procedures shall conform to MIL-PRF-38793 and MIL-STD-38784. The writing shall conform to the plain English standards and minimum editorial requirements listed in DAFI 33-326. The local procedures shall be routed through the PMEL's MFM to AFMETCAL. If a locally developed procedure must be changed or revised, the revised locally developed procedure shall be routed through the PMEL's MFM to AFMETCAL for approval prior to use. Do not submit Recommended Changes for locally developed procedures.

**3.1.7.2 Procedure Specifications.** 33K Series calibration procedures shall be written to meet manufacturer's specifications whenever possible. The procedure shall direct use of a Limited Certification Label if the parameters listed in Table 1 of the calibration procedure do not meet the manufacturer's specifications. This is to ensure the user of the TMDE is made aware of the deviations from manufacturer's specifications. Table 1 specifications of 33K Series TO take precedence over other specifications or tolerances listed throughout the TO. Individuals discovering contradictions between Table 1 and any other section of the TO shall comply with paragraph 3.1.8. Manufacturer's typical (or operational) specifications generally indicate the manufacturer's expected performance of a given product and are not warranted or guaranteed by the manufacturer. Specifications listed as typical in Table 1 will not normally be calibrated (nor require a limited calibration) unless directed by Section 4 of the calibration TO.

**3.1.7.3 TO Verification Process.** Laboratories shall perform TO verification on all draft calibration procedures (manual and/or automated) received from AFMETCAL for review. Guidelines detailing the verification process are outlined on MetWeb.

**3.1.7.4 Post-Publication Reviews.** AFMETCAL will perform Post-Publication Reviews on TOs under any of the following conditions:

- a. The TO has not been changed for five years.
- b. The recommendation of the Technical Content Manager (TCM).
- c. The recommendation of the TO Manager due to the number of recommended changes received and at the request of the using command PMEL MFM.

During a Post Publication Review, one or several PMELs may be tasked through the Command PMEL MFM to participate, either locally or at another site. This tasking shall be at the discretion of the TCM.

**3.1.8 Documenting and Reporting Procedure Problems.** PMEL technicians experiencing difficulty performing calibration procedures shall contact the appropriate AFMETCAL branch for assistance and submit a recommended change IAW TO 00-5-1, if necessary. When a calibration procedure (33K) contains an error that prevents successful calibration and a recommended change is submitted, the technician shall select an alternate procedure for the affected parameter(s) in compliance with paragraph 3.1.2 and document the calibration authority for the affected parameter(s) in the Special block of the certification label in accordance with paragraphs 5.4.2.3.2 and 5.4.2.3.2.1.

### **3.2 EXCEPTIONS FROM PERIODIC CALIBRATION.**

Some types of TMDE do not require periodic calibration. They include items not used to make quantitative measurements, TMDE seldom used but must be calibrated before use, items designated as not requiring calibration and TMDE that requires initial calibration only.

**3.2.1 No Calibration Required (NCR) Items.** Certain types of TMDE identified as NCR do not require calibration because they do not provide important quantitative measurement information, and/or are accessories to other TMDE or function only as an interface device. Examples are RF receivers traceable to a frequency standard; some automotive test equipment; some pressure regulators, isolators, terminations, loads, adapters, tees, circulators, and waveguide components. Exception may occur when such items present an actual workload for the PMEL. Examples: NCR items may require calibration in specific applications, or the user may request calibration of an NCR item. The user shall supply justification for the need. Bar code labels shall not be applied on NCR items that cannot be repaired unless the item requires periodic calibration by the PMEL. An AFTO Form 256, *No Calibration Required*, label or a 'white dot' shall be affixed IAW Section 5.

**3.2.2 Calibrate Before Use (CBU).** There are two criteria for CBU:

- a. TMDE identified by AFMETCAL as CBU in a CMS or TO 33K-1-100-2. These items require calibration or operational verification prior to every use rather than periodical calibration. Generally, the CMS or TO 33K-1-100-2 entry for CBU items will reference a specific 'Note Code' which may be viewed in AFCAV.
- b. TMDE that normally is not used at least once during its calibration interval need not be periodically calibrated and may be designated as CBU. Calibration of TMDE designated CBU need not be recalibrated until the TMDE is to be used even though the calibration due date has passed. CBU items shall be calibrated at least once before being designated as CBU. New TMDE shall require an initial calibration to ensure it meets requirements before being designated CBU. Previously calibrated TMDE shall not require a recalibration before being designated CBU unless requested by the user. After notifying the PMEL to update the master inventory, the user may enter "/CBU" following the DATE DUE date on the AFTO Form 99, 108, 394, or 398 as applicable and hold the item until they require it to be calibrated. After calibration of a CBU designated item, an AFTO Form 99 or 398, whichever is applicable, shall be completed and marked CBU in accordance with Section 5. The CBU designation shall be entered in the PMEL Master Inventory listing in place of the calibration interval.

#### **NOTE**

PMEL cannot make an item CBU without the user's direction or permission.

**3.2.3 No End-Item Calibration (NEC).** NEC in the calibration interval denotes TMDE designated SICL (See Individual Component Listing) in a CMS or TO 33K-1-100-2 that does not have an end-item-calibration requirement. NEC only applies to the end-item; not to components listed with separate calibration intervals.

#### **NOTE**

For scheduling systems that have not been updated to process NEC, load as NCR.

**3.2.4 Next Higher Assembly (NHA).** NHA in the calibration interval field of a CMS or TO 33K-1-100-2 denotes components that are calibrated as part of a next higher assembly. An AFTO Form 99 or 398 shall be completed and affixed to the TMDE. The calibration interval and calibration TO of the next higher assembly apply unless otherwise noted. NHA items shall not generate a separate “due calibration” in addition to the end item. The next higher assembly certification label applies to all components designated NHA.

#### **NOTE**

For scheduling systems that have not been updated to process NHA, load as NPC so only the next higher assembly is scheduled for calibration.

**3.2.5 No Periodic Calibration Required (NPC) Items.** There are two criteria for NPC:

- a. TMDE which are components of a higher assembly that are identified by AFMETCAL as NPC in a CMS or TO 33K-1-100-2 do not require periodic calibration. An AFTO Form 99 or 398 shall be completed and affixed to the TMDE. TMDE designated NPC shall be identified in the PMEL Master Inventory Listing as NPC where the frequency is normally entered.
- b. Based on user applications (see below), an individual unit of TMDE may not require periodic calibration. Even though it belongs to a part number group that normally requires calibration per a CMS or TO 33K-1-100-2, it may be locally designated as NPC by the user. The TMDE user exercises the option of designating an item NPC by contacting the PMEL with the request that a specific item be made NPC for one of the reasons listed below. An AFTO Form 99 or 398 shall be completed and affixed to the TMDE. TMDE designated as NPC shall be identified in the PMEL Master Inventory Listing as NPC where the frequency is normally entered. The TMDE user may designate an item NPC if it does not affect safety and any of the following are true:
  - (1) Its performance is verified, checked, or monitored by other certified TMDE.
  - (2) It is not used to verify equipment performance factors or make absolute measurements.
  - (3) It is used only to provide estimated or approximate values or indicate non-quantitative relative measurements.

#### **NOTE**

PMEL cannot make an item NPC without the user's direction or permission.

**3.2.6 Training Equipment.** TMDE used solely for training and not used to verify specifications or performance factors of operational weapon/support systems or does not affect safety may be identified as NPC. Calibration shall be performed if the TMDE is required for contingency purposes. Calibration after repair of TMDE used for training and designated NPC is optional based on agreement between user and the PMEL flight chief. Instructors shall remove, or annotate malfunctions implanted for training purposes prior to processing any TMDE for repair or calibration. Instructors shall ensure TMDE coded NPC with inoperative functions or ranges is processed for repair before the item becomes totally inoperative. TMDE requiring calibration shall use the interval specified in the applicable CMS, if appropriate or TO 33K-1-100-2. TMDE used for training and identified as NPC shall have an AFTO Form 99 or 398 attached. The Standard Reporting Designator (SRD) for resident training TMDE is "HTE". All other TMDE shall use normal SRD reporting codes.

**3.2.7 Initial Calibration Only (ICO).** Due to its inherent design features, TMDE designated ICO does not require periodic calibration. This TMDE shall be calibrated when initially entering service, or if necessary to meet required specifications after repair. Retain report of measurement or certificate of calibration from NIST, AFPSL, PMEL (if applicable) or original equipment manufacturer (OEM) on file in the PMEL until the TMDE is no longer needed. A properly completed certification label (and any supporting data referenced on said label) is sufficient for ICO items certified by the PMEL. The ICO category also includes certified reference materials or samples which might be obtained from NIST or other sources.

**3.2.8 Depot or Base Supply Stocks.** Except for the following situations, items in depot or base supply stocks do not require periodic calibration. Items of TMDE that are an integral part of an automatic test station and are maintained as spares in a forward supply point shall be calibrated prior to turn-in to supply and thereafter at their designated calibration interval unless another interval is prescribed in the applicable weapon CMS. The items shall be tested in the station if possible.



### **3.3 LIMITED OR SPECIAL CALIBRATION.**

A Limited or Special calibration requires the use of an AFTO Form 99 or 398 Limited TMDE Certification label. The authorization for a limited or special calibration shall be documented in the USER APPROVAL block of the AFTO Form 99 or in the INI block of the AFTO Form 398 in accordance with Section 5.

**3.3.1 Limited.** There are several situations where the PMEL may be authorized to perform a limited calibration.

**3.3.1.1 TO Directed Limited Calibration.** The procedure can contain a statement such as "The Test Instrument accuracy has been downgraded from the manufacturer's specifications due to a lack of adequate standards" and shall direct use of the limited calibration certification label.

**3.3.1.2 Limited PMEL Capability.** When a PMEL does not have the capability to calibrate the full range of specifications contained in the calibration procedure or maintenance TO, the PMEL shall advise the using activity of the extent of services available. The using activity shall decide if the available service meets the mission requirements. If the limited calibration service does not meet their needs, the using activities shall advise the PMEL, and action shall be taken in accordance with paragraph 4.1.

**3.3.1.3 Limited User Requirement.** When a user requires less than the full capability of the TMDE, the user shall specify the desired calibration points or ranges to the PMEL. The PMEL shall clearly identify the certified points or ranges (or the limitations) on the instrument's certification label (AFTO Form 99 or AFTO Form 398 or the form that accompanies them).

**3.3.1.4 Instrument Limitation.** When a function, range or specification on an instrument cannot be economically restored to original design specifications, the PMEL shall advise the using activity of the condition. The user shall then decide if the missing function is necessary to support the mission or if a replacement item is needed. If a limited calibration can permit proper mission support, the PMEL can perform the limited calibration. If not, the item shall be returned to the user, with the appropriate condition tag affixed in compliance with TO 00-20-3.

**3.3.1.5 AFPSL Limitation.** If the AFPSL cannot perform a complete calibration because the capability does not exist to calibrate the item to full manufacturer's specifications, the AFPSL may perform a limited calibration. Document all such limitations on an AFTO Form 99.

**3.3.2 Special.** When a special calibration is performed, the ability of the test instrument to maintain accuracy through the calibration interval is questionable and the user shall be notified of the inherent risk. A typical special calibration includes an extra data point(s) such as a special frequency or reading that is not in the calibration procedure. They may also be:

- a. Calibration of an item designated as NCR.
- b. Calibration of a unit to a higher accuracy than specified in the calibration procedure. In this case, the unit must be capable of providing the higher accuracy within manufacturer's specifications. Whenever possible, performance specifications shall be consistent with the manufacturer's specifications (MIL-PRF-38793, paragraph 3.2.2.2). If they are not, the calibration procedure shall identify any deviations from manufacturer's specifications.
- c. Calibration of an item that cannot be identified by manufacturer, part number or accuracy. In this case the user shall submit, in writing, a special calibration request that includes range and accuracy required and a documented reference (TO, Test Program Set, test plan, etc.). Laboratory personnel shall determine if the test instrument is capable of supporting the required measurement. If there is reasonable doubt of Test Instrument capability, the item shall not be calibrated. When calibrated, the TUR shall be at least 4:1 using the general calibration procedure as per TO 33K-1-100-1. The certified accuracy shall be annotated on the certification label.

**3.3.2.1 PMEL Special Calibration.** The PMEL shall contact AFMETCAL for direction when there is a 33K-series calibration procedure written with uncertainties downgraded from the manufacturers stated uncertainties and there is a need to certify TMDE to the manufacturer specifications.

**3.3.2.2 AFPSL Extended Calibration.** To meet DAF requirements, the AFPSL may perform special calibrations that exceed manufacturer's ranges and/or specifications. In such cases, specific details of the extended calibration shall be documented and published in the calibration procedure if the extended calibration is an ongoing occurrence. An AFTO Form 108 shall be used by the AFPSL in these instances. An AFTO Form 99 shall be used if it also has a limited parameter. An uncertainty analysis shall be performed and documented to substantiate the uncertainties published.

### 3.4 CALIBRATION INTERVALS.

The DAF calibration interval (calibration cycle) listed in a CMS or TO 33K-1-100-2 is the period of time over which the equipment shall perform its mission (or function) with a statistically derived end-of-period reliability (shall be within tolerance) of 85% or better. These intervals are established and modified as necessary from data collected on the total TMDE population through the maintenance data collection system. TMDE that has exceeded the prescribed calibration interval shall not be used. The date due, however, can be extended in certain cases. Paragraphs 3.4.10 through 3.4.19 identify those cases.

**3.4.1 Precedence for Calibration Intervals.** Calibration interval selection shall be made in accordance with the following precedence:

- a. CMS (Applicable to TMDE used solely to support a system referenced in the CMS). Where TMDE is applicable to more than one CMS and where different calibration intervals are listed, use the shortest interval listed and note in the special block which CMS for the interval was used.
- b. TO 33K-1-100-2.
- c. General procedure interval listed in TO 33K-1-100-1 Section 3-1, if applicable.
- d. Individual maintenance TOs.
- e. Commercial data.

**3.4.2 Intervals for New Items and Items Listed without a Calibration Interval.** The maximum calibration interval for TMDE is 12 months when an interval is not prescribed by a TO.

**3.4.3 Published Calibration Interval Changes.** It is not mandatory that TMDE be rescheduled or recalled for calibration or update of certification solely because a change to a calibration interval is published. Items changed from NCR, ICO, or CBU to a periodic calibration interval shall be removed from service and recalled to the PMEL and calibrated. When large quantities are involved, the recall should be staggered to maintain an even flow of workload into the PMEL.

**3.4.4 Other Reasons for Calibration Interval Changes.** TMDE users may request recalibration of their TMDE any time there is reason to question the accuracy of the instrument. Items exposed to rough handling, overload, or other severe conditions shall be removed from service and recalibrated regardless of the calibration due date. TMDE users may also request shorter calibration intervals to meet mission requirements such as Medical TMDE used by biomedical equipment repair technicians in support of medical equipment and NADCAP (*National Aerospace and Defense Contractors Accreditation Program*) items at the ALCs; these examples are not all inclusive. Users may request special calibration of NCR items if they are used to accept or reject items. Users must supply supporting laboratory with justification (such as DAF technical order, contract, or legal references) for using shortened calibration intervals. The additional calibrations may drive a cost to the user. PMEL may shorten a calibration interval based on calibration mobility team schedules, PMEL workload or excessive unscheduled maintenance.

**3.4.5 Recommending Changes to Calibration Intervals.** DAF activities that have accumulated data to substantiate a change in calibration intervals may submit this data and the recommended changes to AFMETCAL. Interval changes must be approved by AFMETCAL.

**3.4.6 War Reserve Materiel (WRM) TMDE Calibration Intervals.** A WRM kit package or containers shall be labeled with a calibration due date equal to 18 months from the earliest date calibrated item packed in accordance with Section 5. TMDE to be placed in WRM kits shall be calibrated prior to packaging. Calibration intervals for TMDE in WRM kits do not start until they are unpacked for use. Once removed from the package, all items shall revert to their normal interval. TMDE that is not removed from the package and whose accuracy is not in question need not be calibrated until the 18-month period marked on the package has expired. Any obvious damage to the equipment package would be cause to recalibrate the TMDE. Due to the many unforeseen circumstances that could arise from storing equipment in relatively uncontrolled areas, 18 months has been determined as the maximum time WRM equipment could be stored with a reasonable expectation that the equipment will operate within the specifications of the calibration procedure. In lieu of storing TMDE in WRM kits, TMDE may be individually tagged and placed in a stringently controlled, limited-access area(s). When an item of TMDE is removed from this controlled area, calibration interval reverts to the interval specified in the CMS or TO 33K-1-100-2. The following paragraph identifies items that cannot be stored in WRM kits or require special precautions.

**3.4.6.1 Items that either can't be stored in WRM kits or require special precautions:**

- a. GAS DETECTORS CONTAINING SENSORS WITH A SHORT SERVICE LIFE - Ensure the Sensor service life covers the maximum possible projected storage period.

- b. RADIAC DOSIMETERS - Those that need to be recharged periodically (usually 30 days) shall not be placed in long time storage. This includes Electronic Personal Dosimeters which require calibration at least annually.
- c. STANDARD CELLS - A period of weeks is required to obtain accurate voltage readings.
- d. CESIUM BEAM FREQUENCY STANDARD - Cesium Beam Tube would be ruined if it were turned off more than 90 days. The ion pump could not keep the tube evacuated.
- e. NI-CAD BATTERIES - Shall require recharging after 18 months in storage.
- f. TEMPERATURE BATH - Drain oil before storing.
- g. DEADWEIGHT TESTER - Drain oil before storing.
- h. VACUUM SYSTEM - Drain oil before storing.
- i. HYDRAULIC FORCE PRESS - Drain oil before storing.
- j. HOOK GAGE - Drain water before storing.
- k. PNEUMATIC PRESSURE STANDARDS AND LO PRESS GAGES - Require dry nitrogen source to use.
- l. BATTERIES - Remove from equipment prior to storage. See TO 33-1-5 for further guidance.
- m. INFRARED TARGET SIMILATOR - Drift rate must be verified on a scheduled interval.

**3.4.7 Contingency Equipment Exception (CEE) TMDE Calibration Intervals.** TMDE that is not used regularly may be placed in storage prior to use after calibration. For scheduling purposes, the calibration interval of CEE TMDE shall be 18 months. CEE TMDE shall be calibrated prior to being stored. An AFTO Form 99/108/394/398 Certification Label shall be attached in accordance with Section 5. Calibration intervals for CEE TMDE do not start until they are removed from storage. Once removed from storage, the CEE TMDE shall revert to the normal interval as detailed in the applicable CMS or TO 33K-1-100-2. CEE TMDE assets that are calibrated as part of an end-item (e.g., a system calibration) that are NOT installed back into the same end-item they were originally calibrated in, shall be re-calibrated at the time of installation to establish system integrity. See WRM paragraph 3.4.6.1 for items that cannot be stored as CEE TMDE (same list as for WRM) or require special precautions. *Note: CEE TMDE may be identified with WRM status in the PMEL MIS with a remark of CEE added to the entry if the PMEL MIS does not accommodate CEE entries.*

**3.4.8 Special Purpose Recoverable Authorized Maintenance Set (SPRAMS).** Spare Tester Replaceable Units (TRUs) stored in SPRAMS require a DD Form 1574 Serviceable Label and an AFTO Form 99/108/394/398 Certification Label attached in accordance with Section 5. For scheduling purposes, the calibration interval of TRUs in SPRAMS shall be 18 months. TRUs to be placed in SPRAMS shall be calibrated prior to being stored. Calibration intervals for TMDE in SPRAMS do not start until they are removed from storage. Once removed from storage, the TRU shall revert to the normal interval. SPRAMS assets that are calibrated as part of an end-item (e.g., a system calibration) that are NOT installed back into the same end-item they were originally calibrated in, shall be re-calibrated at the time of installation to establish system integrity.

**3.4.9 Foreign Military Sales (FMS) TMDE Interval Adjustments.** FMS TMDE calibration dates may be adjusted in the following manners:

- a. AFMETCAL may delay the start of (postdate) calibration intervals for FMS items (except for those items specified by Metrology engineers) being assembled to meet foreign country PMEL activation dates to agree with the activation date or other milestones.
- b. FMS customers may adjust calibration intervals if shipping time to country is in excess of fifteen days. The following rules apply:
  - (1) Receipt in-country must be recorded and filed.
  - (2) The number of shipping days consumed will advance calibration Due Dates when:
    - (a) Interval is equal to or less than 90 days. May be adjusted up to 45 days.
    - (b) Interval is more than 90 days. May be adjusted up to 60 days.
    - (c) No interval may be adjusted by more than 60 days.

**3.4.10 Coding of WRM.** TMDE that is subject to deployment as WRM assets shall be identified as WRM in the master inventory. This shall be accomplished by checking the WRM block of the PAMS scheduling screen. For non-PAMS users, accomplish this coding as specified in the applicable MIS documentation.

**3.4.11 Exchange Standard Intervals.** The calibration interval for Exchange Standards sent to PMEL in exchange for base measurement standards due for certification may be up to 60 days longer than the calibration interval listed in TO 33K-1-100-2 to account for administrative and transit time. The date calibrated on Exchange Standards shall reflect the actual date of calibration. The date due shall be established as the 28<sup>th</sup> day of the month in which the item is due for recertification.

These dates shall not be changed by the PMEL. AFMETCAL is responsible for authorizing calibration due date extensions for exchange standards.

**3.4.12 AFPSL Equipment Due Date Extensions.** AFMETCAL is responsible for authorizing the extension of calibration due dates of Air Force Measurement Standards used by the AFPSL.

**3.4.13 Test Project and Research & Development TMDE Due Date Extensions (excluding RADIAC equipment and safety-related measurement devices).** Test project and Research & Development TMDE that is assigned a periodic calibration interval may have its due date extended by the test or research director, if calibration of the TMDE would adversely affect a test or research & development project. The due date may be extended to the end of the test/research project. The director must notify the PMEL manager of the extension and verify that the extension is at their request, and the PMEL is no longer responsible for the accuracy of the TMDE in the test setup.

**3.4.14 Critical TMDE Due Date Extensions (excluding RADIAC equipment).** MAJCOMs are authorized to extend the calibration due date if the loss of the TMDE will delay or prevent critical mission accomplishment. Requests for extensions shall be forwarded to the OWC's command PMEL Functional Manager through the PMEL. The request shall describe the TMDE involved, the calibration due date, the specific reasons calibration cannot be accomplished as scheduled, the estimated date calibration action can be initiated, actions taken to locate a suitable alternative or substitute item, and the calibration history of the TMDE. Also include a statement from the OWC Maintenance Superintendent or equivalent indicating the understanding that calibration interval is the period of time over which TMDE maintains a statistically derived end-of-period reliability of 85% or better and that extending the interval may jeopardize TMDE accuracy and reliability.

**3.4.15 RADIAC TMDE Due Date Extensions.** Due dates are rarely extended for RADIAC TMDE. RADIAC due date extension requests will be routed through the PMEL to the command PMEL Functional Manager for approval/disapproval by AFMETCAL. Documentation will be consistent with paragraph 3.4.14 and must also document the unusual circumstance driving the request.

**3.4.16 Base Measurement Standard Due Date Extension.** The PMEL MFM is authorized to extend the calibration due date of Base Measurement Standards provided by AFMETCAL to the PMELs. Information copies of extensions shall be sent to AFMETCAL. This authorization does not apply to Exchange Standards.

**3.4.17 AFPSL TMDE Extensions.** The AFPSL may request extension of TMDE standards if the loss will delay or prevent critical support to AFPSL customers. Requests for AFPSL TMDE extensions will be coordinated with AFMETCAL.

**3.4.18 On-Site Off-Base TMDE Due Date Extensions.** The PMEL Manager is authorized to extend the calibration due date up to 1 month for off-base TMDE calibrated on-site to facilitate scheduling only after considering the calibration history of the TMDE. Contact the TCM or MFM for assistance when necessary.

**3.4.19 Return Transportation Time and TFCU Visit Due Date Adjustment.** PMEL Managers have the authority to adjust calibration due dates up to 1 month to offset return transportation time, scheduled monthly pickup and delivery, or scheduled TFCU visits. The date calibrated shall reflect the actual date of calibration. The PMEL is authorized to extend the date due up to 1 month if one of the following cases applies:

- a. TMDE is picked up by or delivered to the customer at intervals less often than weekly.
- b. TMDE is used to equip maintenance teams that are deployed to support off-base customers.
- c. To coincide with the scheduled arrival date at the deployed location for TMDE with calibration intervals of 5 months or less that are deploying in support of Air Expeditionary Force (AEF) operations.
- d. To account for return transportation time for TMDE.
- e. To coincide with the arrival of the on-site TFCU.

### **3.5 PRECISE TIME SUPPORT.**

Precise time support is now accomplished using the GPS system through the local PMEL. Repair and replacement actions for cesium standards used for frequency or time standard should be accomplished through the appropriate item manager at WR-ALC.

### 3.6 **TMDE USER RESPONSIBILITIES.**

The user shall:

- a. Appoint a TMDE coordinator. A TMDE coordinator is not required at locations where the PMEL supports only one organization. The TMDE coordinator shall attend training and maintain a TMDE coordinator file that includes:
  - (1) Name of primary TMDE coordinator and alternate.
  - (2) Current TMDE Master ID Listing.
  - (3) TMDE Quarterly Master Equipment Schedule.
  - (4) Test equipment hand receipts from PMEL (unless a Central Collection Point (CCP) has been established. In such case, PMEL-issued hand receipts will be kept on file at the CCP. The CCP will issue a local hand receipt to the TMDE coordinator).
  - (5) A copy of the base instruction, if applicable.
  - (6) Any customer handout(s) that might be provided by PMEL.

#### **NOTE**

Customers (or at least the TMDE coordinators) shall have access to and be familiar with contents of TO 33K-1-100-1; TO 33K-1-100-2; TO 00-20-14; any applicable CMS TOs; TO 00-25-234 *General Shop Practice Requirements for the Repair, Maintenance, and Test of Electrical Equipment*; TO 33-1-32, *General Instructions for Input Power Wiring of Electrical/Electronic Support Equipment*; and the Special Certification Roster (SCR) or equivalent document identifying persons designated as calibration limitation approval authority. Preferably, coordinators of large accounts shall maintain their own copies of these TOs.

#### **NOTE**

A TMDE coordinator file is not required for the PMEL's TMDE coordinator.

- b. Calibrate, certify and repair TMDE specified in any applicable CMS TOs or TO 33K-1-100-2 as USER at the specified interval. If the USER organization (same as performing work center) does not have the capability, obtain calibration and maintenance support from the lowest echelon organization having the capability. If no other organization has the capability, the PMEL may assist by providing training, workspace, technical assistance, or support, as resources allow. Organizations owning/using TMDE designated USER shall:
  - (1) Document that all personnel performing USER calibrations are properly trained and qualified on the tasks they are performing.
  - (2) Perform calibrations using authorized technical data IAW paragraph 3.1.2.
  - (3) Document maintenance actions in applicable MIS.
  - (4) Comply with local Maintenance Group (MXG) or equivalent quality program.
  - (5) Ensure calibrations are completed in an environment (temperature/humidity) that supports the specifications of standards and devices under test.
  - (6) Track due dates of TMDE designated USER and requiring re-calibration. Use appropriate MIS. USERS will ensure TMDE is not used unless calibrated. USERS will remove the TMDE from service once the calibration due date has expired. Exceptions are listed in paragraphs 3.4.10 through 3.4.18.
  - (7) Use TMDE calibrated and certified IAW Section 3 of this TO to maintain equipment designated as USER responsibility.
- c. Request approval from AFMETCAL through local PMEL prior to obtaining calibration of DAF TMDE from non-DAF sources (see Section 4).
- d. Accomplish and use applicable forms, labels, and alternate methods of certification IAW Section 5 of this TO.
- e. Return all TMDE specified as PMEL () or AFPSL responsibility in any applicable CMS TOs or TO 33K-1-100-2 to the PMEL upon receipt from supply, when scheduled for calibration, or for unscheduled maintenance. Identify TMDE not listed in any applicable CMS TOs or TO 33K-1-100-2 to the PMEL for calibration determination action and provide technical data that clearly defines weapon system requirements (i.e., F-16, KC-46, C-17 etc.). Commanders and supervisors of all activities owning and using TMDE requiring calibration are responsible to ensure this TMDE is not used unless it has been calibrated and that it is removed from service once the calibration due date has expired. Exception to this statement only as covered by paragraphs 3.4.10 through 3.4.18.
- f. Consider alternative of limited calibration, CBU, or NPC status, where possible. Calibration limitations shall be approved by the User's shop chief or designated seven level craftsman/civilian equivalents. Users shall identify

individuals authorized to approve limitations and sign or initial certification labels on the Special Certification Roster or equivalent document. TMDE coordinator and calibration limitation approving official are two distinctly separate duties.

- g. Deliver TMDE to the PMEL with all ancillary equipment (e.g., preamps, power supplies, adapters, cables, or probes) needed for the calibration. PMEL personnel shall advise the user when an item is not sufficiently complete to allow full calibration and may return the item without action if the ancillary equipment and/or technical data are not readily available.
- h. Provide proper care, handling, cleanliness, and transportation of TMDE (see paragraph 3.8). If the TI is used on a liquid system (fuel trucks, hydraulic console, liquid flow stand, etc.), ensure TI has been drained prior to scheduling into the PMEL. Also make sure the PMEL has a copy of the MSDS for the fluid used if present.
- i. Ensure that notes contained in the SPECIAL block of the certification labels are read and understood.
- j. Refer measurement standards authorized for use by activities other than the PMEL to the base PMEL for calibration. The base PMEL shall either calibrate these items or request assistance in accordance with paragraph 4.1. Measurement standards owned by research and development laboratories and used to support measurement areas peculiar to a specific PMEL may be used to calibrate TMDE that cannot be supported by the PMEL. The PMEL is responsible for ensuring these standards meet certification requirements.
- k. Maintain technical data file for equipment owned and provide it to PMEL when requested. Provide technical data with the TMDE when requested by the PMEL. The TMDE User shall ensure adequate maintenance and technical data for each item of TMDE are available and that the technical data accompany their TMDE whenever they are deployed. Notify the PMEL and provide applicable technical data when weapon system requirements have an update that may drive a change in AFCAV CMS specifications.
- l. Request assistance from the base PMEL for calibration of individual items of TMDE that require equipment and technical skills normally in the PMEL. Also request PMEL help when specified in applicable weapon system or equipment CMS TO. Ensure that TMDE used as component parts of systems or subsystems are not removed for calibration unless the system, subsystem, maintenance, or CMS TO requires it.
- m. Identify and/or coordinate any requirements for limited or special calibrations with the base PMEL. Comply with procedures for exceptions from periodic calibration as specified in paragraph 3.2. Place authorizing signature or initials on AFTO Form 99 or AFTO Form 398 in accordance with Section 5. This is to be done for TMDE that has received limited or special calibration when returned from the PMEL, or for TMDE that is to be exempted from periodic calibration.
- n. Reserved.
- o. Perform organizational maintenance on assigned TMDE. This responsibility applies to stock listed and nonstock listed TMDE in all federal supply classes.
- p. When TMDE items (except those on Redistribution Order actions) are being returned to supply stocks, notify PMEL of the turn-in. Do not remove AFTO Forms 108, 394, 99 or 398; however, do eradicate the calibration and due calibration dates from the forms. The calibration labels on items on Redistribution Order action remain valid and AFTO forms 65 and 66 stay on equipment.
- q. Leave all condition tags and other documentation including warranty tags (except the receipt) on TMDE being forwarded to the PMEL for initial calibration or acceptance testing and retain shipping containers. The initial calibration is a serviceability check on items from a vendor, technical repair center, or other base. Discrepancies found by the PMEL during initial calibration shall be reported to the users so they can submit deficiency reports back to the source of supply. A calibration procedure can serve as a guideline for an acceptance check.
- r. Provide facilities meeting temperature, humidity, and power requirements IAW this TO for TFCU or other PMEL mobile calibration operations.
- s. Ensure that torque-indicating devices, specified as PMEL responsibility in TO 33K-1-100-1/2 or any applicable CMS TO, are sent to the PMEL for scheduled calibration or unscheduled maintenance. Torque indicating devices are considered TMDE and PMELs are assigned the primary responsibility for calibration and repair.
- t. Identify special weapons unique requirements and applicable guidance (such as Nuclear Certified Equipment (NCE)/AFMAN 21-204, *Nuclear Weapons Maintenance*) which could impact calibration, calibration interval, documentation, etc. to the PMEL. NOTE: The calibration interval for NCE must not exceed the interval published in the individual TOs listed in TO 0-1-2 or TO 0-1-11N-C.
- u. Ensure that medical equipment is referred to the appropriate support activity IAW AFI 41-201, *Managing Clinical Engineering Programs*, AFMAN 41-209, *Medical Logistics Support* and TO 33K-1-100-1.
- v. Ensure that Skydrol fluid pressure gauges are flushed of all lubricating residue before being sent to PMEL for calibration. After cleaning, the gauges shall be capped and double bagged, and the AFTO Form 350 tag shall be documented stating that the gauges are clean and free of Skydrol lubricants prior to transporting these items to PMEL. The receiving PMEL shall not accept these gauges if these requirements are not met.

- w. Ensure adjustments/potentiometers that are sealed with an AFTO Form 255, "Certification Void if Broken" labels are not reset. Any unit discovered with a broken AFTO Form 255 shall be removed from service and submitted for recalibration.
- x. Ensure that components of complex TMDE (test stands, checkout consoles, etc.) are properly identified to the PMEL as part of the system and not removed for calibration merely because the components are listed individually in TO 33K-1-100-2. Removal of these components for calibration shall be accomplished only as specified in the calibration authority. Complex TMDE components that require calibration shall be calibrated in place, if feasible. Components that require calibration by the PMEL shall be calibrated in accordance with the provisions of a CMS or TO 33K-1-100-2. Typical items are pressure gages, panel meters, power supplies, frequency meters, etc. To preclude excessive down time, each component shall be calibrated at the shortest interval listed for any of the components, or at a multiple of the shortest interval, if this multiple does not exceed its own listed interval. For example, if the shortest interval is 4 months, an item with a 9-month interval may be checked every 4 months or it may be shortened to 8 months and checked every other time.
- y. Prior to the 'local purchase' of TMDE:
  - (1) Coordinate with PMEL to verify there is no existing TMDE that can be used to satisfy the requirement as soon as a part number or model number is available.
  - (2) Coordinate with PMEL to determine if the item is already listed in a CMS or TO 33K-1-100-2.
  - (3) Assist PMEL in completing the Request for Calibration Responsibility Determination upon purchase per Section 5 and provide the necessary commercial data.
  - (4) Ensure that resources and/or funding for resources needed to support locally procured TMDE will be available and/or can be acquired. This includes funding for repair and calibration when locally procured TMDE is beyond the support of the local PMEL and providing the supporting PMEL with the necessary calibration and/or adjustment software.
  - (5) Follow the guidance in DAFI 64-117, *AF Government-Wide Purchase Card Program*, when using a Government Purchase Card to local purchase TMDE.
- z. Provide PMEL with mission impact statements for AWP follow-ups.
- aa. Perform all periodic maintenance or inspections as directed by maintenance TOs.
- bb. Notify PMEL of the date items are removed from WRM packages or from CEE storage and placed into use. Complete the date due block of the certification label using the calibration interval listed in TO 33K-1-100-2 or applicable CMS at the time the item is removed from the WRM package. PMEL may be contacted for assistance in calculating the date due. Date due shall be dependent upon calibration interval stated in the TO 33K-1-100-2 or applicable CMS.
- cc. Ensure all forms, labels, and calibration correction charts received with the TMDE from the PMEL are complete. Notify the supporting PMEL of errors encountered.
- dd. Identify all TMDE that is designated as WRM or subject to deployment. This information shall be provided to servicing PMEL to be included on master inventory.
- ee. Provide PMEL information regarding the use made of RADIAC equipment to permit PMEL to determine calibration interval. To be precise, is it NDI equipment, disaster preparedness, or medical equipment?
- ff. Remove batteries from FSC 6665 RADIAC equipment being sent to a PMEL that does not affect the calibration of the unit.
- gg. Maintain warranty information on all TMDE and document start and stop dates and other warranty conditions.
- hh. Coordinate with PMEL (or organization identified as having calibration responsibility for the TMDE) prior to upgrading TMDE firmware or operational software\*. Firmware/software upgrades may impact TMDE calibration thereby voiding the usability of the previous calibration.  
\*Operational software refers to software provided by the TMDE manufacturer required for basic instrument control, or to provide basic software utility functions.
- ii. When notified by the PMEL of an "out-of-tolerance" TMDE item, the owning work center should determine the impact of the "out-of-tolerance" condition on shop production and what steps, if any, they should take (customer notification, recall of production items, etc.).
- jj. Prior to the 'lease/borrow' of TMDE:
  - (1) Coordinate with PMEL to verify there is no existing TMDE that can be used to satisfy the requirement.
  - (2) Provide the PMEL with the expected duration of the 'lease/borrow' arrangement.
  - (3) If TMDE will be 'lease/borrowed', comply with the requirements of paragraph 4.9.
- kk. Coordinate with the organization responsible for calibration prior to replacing SICL test stand NHA components with spares. A recalibration may be required.
- ll. Inform PMEL if noise dosimeters, sound level meters, or acoustic calibrators are used in the "AF Occupational Noise and Hearing Conservation Program". Those items used in support of the "AF Occupational Noise and Hearing Conservation Program" shall be identified by the User and calibration shall be performed IAW the shortest

- calibration interval listed in the appropriate Calibration Authority (see TO 00-20-14, Par 3.1.2 Calibration Authority Precedence) or AFI 48-127, *Occupational Noise and Hearing Conservation Program*.
- mm. Ensure pipettes are clean and free of biohazards or other hazardous materials before sending to PMEL. The OWC must certify the pipette is free of contaminants on an attached AFTO Form 350 tag or other official form.
- nn. Consider offsetting critical TMDE calibration due dates to allow continual coverage and minimize mission impact in the absence of those assets.

### 3.7 **INPUT PRIORITY SYSTEM.**

Items submitted to a PMEL or the AFPSL shall use the following priority system based on category and first-in, first-out within each category. The PMEL management information system allows internal use of additional priority categories to meet mission requirements. All additional priority categories used in the PMEL should be defined. The PMEL manager approves or disapproves priority calibration requests (disapprovals at contract PMELs must be coordinated through the PMEL government program office). Customers may request the following priorities:

- a. **EMERGENCY Calibration or Repair.** One-of-a-kind TMDE that is inoperable or due calibration and for which a critical job is at a work stoppage. A letter of justification signed by the OWC Maintenance Operations Officer /Maintenance Superintendent or equivalent shall accompany the TMDE. The letter may be handwritten to prevent delay, and telephone coordination between the OWC and PMEL is encouraged. The PMEL or AFPSL shall accept emergency TMDE any time and immediately place it into work, with calibration or repair action continuous until repair or calibration is completed or status of the item changes (e.g., AWP, deferred for lack of standards or technical data, etc.). The PMEL Manager may require an OWC technician familiar with the TMDE to accompany the TMDE to and remain at the PMEL to provide technical assistance until the work is completed or placed in an interim-complete status. The owning or using organization picks up the TMDE immediately upon notification of completion. For AFPSL items, the item shall be shipped to the Air Force Primary Standards Laboratory (AFPSL), Heath, Ohio, by priority means; and the supporting PMEL shall alert the AFPSL scheduler and AFMETCAL Cal & TO Management Section with a telephone call or by message of the impending arrival of the priority; and the PMEL Manager shall sign a cover letter (suggest a form letter) requesting and justifying the priority.
- b. **MISSION ESSENTIAL Calibration or Repair.** One-of-a-kind or one-deep TMDE that is part of a unit's deployment package, or is critical to daily peacetime operations, or TMDE assets falling below critical availability levels. A letter of justification signed by the OWC flight chief or equivalent shall accompany the TMDE. The OWC flight chief or equivalent may pre-identify, by letter, TMDE which meets the mission essential definition as approved by the PMEL Manager or delegated approval authority. The PMEL or AFPSL shall accept mission essential TMDE any time during duty hours and schedule it with sufficient priority to ensure the calibration/repair is complete, or the maintenance status changes (e.g., from INW to AWP, DEF, etc.), by the date and time specified by the customer. The OWC or using organization picks up the TMDE immediately upon notification of completion. For AFPSL items, the supporting base PMEL shall alert the AFPSL scheduler and AFMETCAL Cal & TO Management Section with a telephone call or by message of the impending arrival of the priority; and the PMEL Manager shall sign a cover letter (suggest a form letter) requesting and justifying the priority.
- c. **ROUTINE Calibration or Repair.** TMDE not categorized as emergency or mission essential. The PMEL or AFPSL shall accept routine TMDE during normal turn-in and pick-up hours. This is equivalent to the normal processing of TMDE in the PMEL or AFPSL.

### 3.8 **CARE OF TMDE.**

Activities owning and using TMDE are responsible for its proper handling, transportation to and from designated drop off points, care, use and cleanliness. These are important factors in ensuring that TMDE performs reliably within specified tolerances for the duration of the established calibration interval.

**3.8.1 Physical characteristics of TMDE.** Physical characteristics of TMDE range from sturdy to fragile. The sturdy can be outer cases with shock suppression mountings. The fragile can be components that can be damaged by a slight jolt, scratch, or static electricity. Special handling instructions for specific items are prescribed in maintenance TOs when handling requirements exceed normal practice. Electrostatic Discharge (ESD) control procedures are described in MIL-HDBK-263B *Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies, and Equipment*, MIL-STD-1686C, *Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies, and Equipment*, and TO 00-25-234.

**3.8.1.1** Items of TMDE shall be handled as "delicate instruments" regardless of outer physical appearance.



**3.8.1.2** TMDE that is too heavy or awkward to be carried by one person may require team lifting or mechanical assistance (DAFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards*).

**3.8.1.3** Items of TMDE shall be handled individually unless the item's size and physical characteristics allow it to be held in one hand.

**3.8.1.4** External cords, cables, accessories, and special connectors shall be secured to the equipment when delivering to PMEL for scheduled or unscheduled maintenance.

**3.8.1.5** All unmated connectors shall be kept covered with moisture-proof and vapor-proof caps. Exceptions are when stored in environmentally controlled areas or if subject to use on a daily basis. Unmated connectors on units that contain circuitry susceptible to ESD damage shall be covered with conductive caps. Do not substitute the use of tape for the caps.

**3.8.1.6** Metal-to-metal contact with TMDE items shall be avoided unless the total instrument is contained in a protective case.

**3.8.1.7** Only TMDE awaiting work, in delay status or delivery to the customer shall be stored in the PMEL receiving and shipping areas. If a portion of the area is used for other purpose(s), the area shall be well marked as to status or purpose of use.

**3.8.1.8** All TMDE shall be stored on shelves or racks unless the physical characteristics prevent this type of storage. Items of TMDE constructed so that the instrument is completely contained in a case may be stored on metal or wooden shelving without further protection. Care must be taken to protect protruding dials, knobs, or meter faces. TMDE not constructed in cases shall be protected from damage by use of suitable containers and appropriate padding. Containers, protective caps and coverings, and padding are not required for equipment in work or subject to use on a daily basis. Items of TMDE completely contained in packing cases may be stored on the floor.

**3.8.1.9** TMDE shall be removed from service if it has been subjected to overloading, mishandling, gives suspect results, or otherwise determined to be defective. Properly tag and store the TMDE to prevent further use until it can be scheduled for repair or calibration.

**3.8.2** Transportation of TMDE. Transportation of TMDE by trained personnel using a regularly assigned and specially configured vehicle will provide the maximum degree of protection to the equipment. Physical characteristics of individual TMDE shall dictate the extent of protection required during transportation. In general, protection from the elements and excessive shock and vibration is all that is required for TMDE in a protective case. Padding protection between items on the same vehicle may be required to protect knobs, dials, and meter faces from damage if the load shifts.

**3.8.3** Cleaning TMDE. The user shall clean the exterior of all TMDE immediately before transporting to the PMEL. TMDE may be returned to the user for cleaning. When an item is disassembled in the calibration area of the PMEL and found to be dirty internally, the calibrating technician shall return the item to the cleaning room. Dust removal may be done in the calibration area by vacuuming only. Use ESD approved systems when vacuuming ESD sensitive internal circuitry/equipment to remove dust. Techniques for cleaning electronic equipment are provided in TO 00-25-234. These techniques may also be applied for electromechanical and physical standards when applicable. Oxygen TMDE shall be certified as clean by the user prior to shipment to the PMEL.

#### NOTE

Oxygen gages shall contain the warning OXYGEN-USE NO OIL printed on the face of the gage by the manufacturer or by use of the OXYGEN-USE NO OIL decal part number 69A52267 (TO 15X-1-102, *General Care and Cleaning of Oxygen Gauges and Oxygen Device Related Test Equipment*) or an equivalent statement of oxygen service. If the gage does not have the OXYGEN-USE NO OIL warning or decal or equivalent statement of oxygen service printed on the face or affixed to the gage, the PMEL shall not include the "Not Oxygen Clean" statement on the certification label.

**3.8.4** PMEL Tracking of TMDE. Each item of equipment shall be labeled, marked, or otherwise identified to indicate its calibration status. Records shall be maintained for each item of equipment and all materials significant to the calibrations

or verifications performed. ABSOLUTE standards and all items having PAMS ID numbers that require calibration when used to accomplish the calibration shall be entered into the PAMS or alternate management information system traceability listing (calibration detail listing in PAMS). TMDE used for preliminary steps, but not listed as a standard in the calibration procedure, does not require listing. The records shall be stored and held secure and in confidence to the customer for a minimum period of one year or the length of one calibration interval if the calibration interval exceeds one year. The intent is to have at least the last record in the file. The records shall include as a minimum:

- a. Nomenclature
- b. Manufacturer, type identification and serial number or other unique identification
- c. Current location, where appropriate
- d. Where applicable, date calibrated, results of calibration and/or verifications, and date due calibration
- e. Details of maintenance carried out to date and planned for the future
- f. History of any damage, malfunction, modification, or repair
- g. The identity of the certifying technician

### **3.8.5 PMEL Handling of TMDE.**

- a. The PMEL shall have a system for uniquely identifying the items to be calibrated, to ensure there can be no confusion regarding the identity of such items at any time.
- b. Upon receipt of the TMDE for calibration, any abnormalities or departures from standard condition as prescribed in the relevant calibration method shall be recorded on an AFTO Form 350 or equivalent document. Where there is any doubt as to the item's suitability for calibration, where the item does not conform to the description provided, or where any limited or special calibration required is not fully specified, the PMEL shall consult the customer for further instruction before proceeding. The PMEL shall establish whether the item has received all necessary preparation, or whether the customer requires preparation to be undertaken or arranged by the PMEL.
- c. The PMEL shall separate TMDE in a workable status from TMDE in a non-workable status (i.e., awaiting customer pick-up, awaiting parts, deferred, etc.) and shall notify OWCs of the change in status.

## **3.9 PMEL MOBILE/ON-SITE MEASUREMENT CAPABILITIES.**

**3.9.1 Transportable Field Calibration Unit (TFCU).** The TFCU consists of working standards, TMDE and accessories. These items are mounted in shockproof transportable cases. The cases serve as equipment racks during operation and provide protection during transit. The TFCU is used to augment a PMEL to provide on-site calibration support.

**3.9.1.1** TFCU working standards will operate satisfactorily within temperature ranges of 18°C (64.4°F) to 28°C (82.4°F) with humidity up to 70% RH. Depending on the TFCU working standard(s) used, calibrations performed outside of these ranges may result in a degradation of accuracy.

**3.9.1.2** TMDE with accuracy requirements beyond the capability of the TFCU shall be returned to the host PMEL. TMDE that needs more than minor maintenance shall normally be returned to the host PMEL.

**3.9.1.3** A TFCU shall be considered for assignment to specific PMELs on a case-by-case basis to accomplish the following:

- a. Provide calibration support to established DAF and DOD agencies remotely located from a host base PMEL.
- b. Provide interim or emergency TMDE support at remote sites and bases where operating units have been deployed and where no support PMEL exists.
- c. Permit PMELs to maintain and operate calibration support capabilities within prescribed geographic areas.
- d. Provide calibration support at a location where an operation does not justify the establishment of a PMEL because of its limited size and workload.
- e. Provide temporary support to augment a base PMEL during facility renovations, construction, natural disasters, or loss of environmental systems.

**3.9.1.4** Requests for authorization for a TFCU or redistribution of a PATEC shall be submitted through the PMEL MFM to AFMETCAL with an information copy to HQ USAF/A4LF. The requests shall include the following:

- a. Identify the proposed owning activity, the purpose for which the TFCU (or PATEC) shall be used, and how it shall be supported.
- b. Model or part number, manufacturer, nomenclature, and quantity of TMDE that requires on-site support.
- c. The military designation, squadron or unit, and MAJCOM for each activity that shall receive TFCU support.
- d. The geographic location of each customer and distance from the host PMEL.

- e. Alternate course(s) of action to provide support if the TFCU (or PATEC) is not authorized.
- f. Statement of mission impact if a TFCU (or PATEC) is not obtained.

**3.9.1.5** PMELs that have lost the on-site customers that justified their need for a TFCU (or PATEC) shall contact AFMETCAL and the command PMEL Functional Manager for disposition instructions.

**3.9.1.6** Any calibration support requirements that cannot be met within the above guidelines shall be returned to the main operating base for support.

**3.9.2** Portable Automatic Test Equipment Calibrator (PATEC). The PATEC is normally used to calibrate ATE on-site. The PATEC consists of a set of portable standards designed to calibrate a specific ATE system. The PATEC standards are housed in portable cases. See paragraphs 3.9.1.4 and 3.9.1.5 for redistribution or turn-in process.

**3.9.2.1** When used on-site, environmental requirements are determined by the requirements of the ATE being calibrated and the operational specifications of the standards in the PATEC.

**3.9.2.2** Calibration of ATE using the PATEC shall be in accordance with 33K- or 33D- series TO, as applicable.

**3.9.2.3** The PATEC may be assigned to a designated PMEL or an organization owning the ATE because of mission security, geographic, or deployment considerations. When a PATEC is built to support specific ATE, the activity owning the ATE shall have priority use, even though the PATEC may be held by the PMEL. The PATEC shall be available at all times for scheduled or unscheduled calibration of the specified ATE.

**3.9.2.4** Supply procedures of AFI 23-101 and AFMAN 23-122 shall be followed for transfer and receipt of the equipment when the PATEC is intended to be used in the deployment mode.

**3.9.3** Rapid Assistance Support for Calibration (RASCAL). RASCAL is a deployable PMEL with a downsized measurement capability designed for rapid deployment to support emergency calibration needs. It is comprised of five NAVAIR shelters, each measuring 8 ft wide and 20 ft long. These units are put together to form four measurement area shelters and one scheduling/receiving shelter. The four measurement area shelters provide calibration capability in the DC and Low Frequency, Microwave, Dimensional, and Electro-Mechanical areas.

**3.9.3.1** RASCAL can be deployed using aircraft, air-ride trailers or ships. Once deployed, RASCAL requires a 20,000 to 25,000 pound crane or forklift to complex the units. Based on various conditions, complexing time is approximately half a day.

**3.9.3.2** RASCAL is equipped with Environmental Control Units (ECUs) designed to maintain interior temperature at 73 ±6 °F and relative humidity at <50%.

**3.9.3.3** RASCAL has the ability to be powered by either a commercial or a mobile generator type source.

**3.9.3.4** Requests for RASCAL support shall be processed through MAJCOM PMEL Functional Managers.

**3.9.4** Jet Engine Test Stand Calibrator (JETSC). TMDE and accessories needed to perform on-site calibration of jet engine test stands. They are housed in ruggedized cases, or in a trailer which has both a work and a storage area. Environmental requirements are determined by the requirements of the items being calibrated and the standards being used.

**3.9.5** Other On-Site. At times it is not possible to return TMDE to the PMEL and calibrations must be performed on-site. When performing a calibration of TMDE designated PMEL responsibility outside of the controlled environment of the PMEL, additional care must be taken.

**3.9.5.1** The temperature and humidity ranges of all standards and test items must be considered. Generally, base measurement standards should not be taken outside of the PMEL. Before taking a base measurement standard out of the PMEL, consult with the applicable AFMETCAL Engineering Section.

**3.9.5.2** Items assigned a calibration responsibility of 'PMEL (68 °F)' are calibrated at a nominal 20 °C (68 °F) and should not be calibrated outside of a controlled environment. If you have a specific need to make this type of measurement on-site, consult with the AFMETCAL Mechanical Engineering Section.

**3.9.5.3** Other measurements are performed at a nominal 73 °F (22.8 °C). On-site calibration should be held to an environment of 73 °F  $\pm$  9 °F (22.8 °C  $\pm$  5 °C) and <70% RH. If you have a need to perform an on-site calibration outside these environmental conditions, review the following:

- a. End-item to be calibrated, manufacturer's stated operating environment and applicable temperature coefficients.
- b. Any limited calibrations noted on the calibration label or certificate of the TI and/or standards used.
- c. All equipment to be used in the calibration, the manufacturer's stated operating environment, and applicable temperature coefficients for each piece of equipment.
- d. Any environmental conditions detailed in the calibration authority.
- e. When calibrating in extreme environmental conditions, care should be taken to ensure the test uncertainty ratio for each measurement performed in the calibration (including the effects of any temperature coefficients) is equal to or better than the test uncertainty ratio in the calibration authority.

**3.9.5.4** Environmental conditions must be monitored for the calibration. The calibration technician must determine how frequently the calibration environment should be monitored/checked based on the stability of the calibration environment, the length (time) of the calibration, and the sensitivity of the equipment involved in the calibration to fluctuations in the calibration environment. The calibration environment should be monitored/checked frequently enough to provide assurance that the calibration environment does not adversely impact the calibration process/results. Record the worst observed temperature/humidity in the PMEL MIS.

**3.9.5.5** In all cases, PMELs shall conform to the maintenance TO or manufacturer's data regarding any environmental limitations or temperature coefficients of any item used in a measurement setup. The calibration authority may state specific environmental conditions, which shall be met, for the calibration.

### **3.10 RESPONSIBILITIES OF PMEL MANAGERS.**

The PMEL Manager is the senior on-site manager responsible for the overall PMEL, management system, quality program and production control functions (also referred to as logistics throughout this document). The PMEL manager shall:

**3.10.1** Establish a system to control test fixtures. The system shall provide for:

- a. An identity on the test fixture that can be related to the equipment/calibration procedure for which the test fixture was fabricated.
- b. A storage area for test fixtures not permanently used in measurement setups.
- c. An index or reference file that identifies the test fixture and its storage location.

**3.10.2** Take necessary actions to minimize the late delivery of TMDE for scheduled calibration. Managers shall have local policy in the QM defining how overdues will be managed, the notification process to customers, and documentation process of those notifications. Overdue calibration notifications shall include a statement to remove TMDE from service. Notify the Commander (or equivalent) by letter when the OWC routinely fails to deliver or schedule delivery within a reasonable period following notification.

**3.10.3** Notify the OWC if a deficiency is discovered with the physical condition or documentation that affects the ability to calibrate or repair the item. Notify the customer when customer owned TMDE is found out of tolerance or improperly functioning, IAW the decision process outlined in paragraph 10.2c.

**3.10.4** Maintain a TO File on the use and operation of PMEL-owned equipment, on the handling and preparation of PMEL-owned items, and for calibrations/verifications. All instruction, standards, manuals, and reference data relevant to the work of the PMEL shall be maintained up-to-date and readily available to the staff. Manage TOs to include filing and TO improvement reports IAW TO 00-5-1 and 00-5-16. AFTO Form 57, described in Section 5.13, can be used to document data peculiar to each TO. File AFTO Forms 57 per local procedures.

**3.10.5** Identify non-PMEL TMDE, identical in part number or model to PMEL standards, to any off-base organization to which they are sent for calibration. This permits the certifying technician to determine the correct calibration procedure.

**3.10.6** When requested by the OWC prior to the local purchase or “lease/borrow” of new TMDE, verify there is no available existing TMDE that can be used to satisfy the requirement, and determine if the item is already listed in TO 33K-1-100-2.

**3.10.7** Produce quarterly TMDE schedules and annual master ID lists at least 5 workdays prior to the first duty day of the quarter/year and distribute to OWCs which do not have access to PAMS/MIS for correction and verification. TMDE schedules and master ID lists may be distributed via e-mail and/or webpage with the proper level of protection. Schedules for OWCs are not required when there are no items in maintenance or overdue status. Establish a tracking and a suspense system for return of corrected listings and ensure PAMS/MIS data are updated NLT 3 workdays of receipt of customer corrections.

**3.10.8** Take action to resolve problems with customers who fail to pick-up completed TMDE within a reasonable period.

**3.10.9** Coordinate with AFMETCAL Plans & Analysis through the PMEL MFM prior to procuring:

- a. Repair or replacements for AFMETCAL-procured equipment (i.e., base measurement standards, working standards, etc.).
- b. Standards/TMDE from other sources (i.e., item managers, local funding, etc.) which give the PMEL a capability identified by a Note Code listed in a CMS or TO 33K-1-100-2.

**3.10.10** Contact the applicable technical content manager/engineer when AFMETCAL-procured equipment is found in need of repair. Provide details on cause of failure when known. The intent is to aid AFMETCAL in determining if there is a systemic issue with the equipment or process.

**3.10.11** Maintain and perform positive maintenance cyber discipline practices defined in DAFI 21-101, TO 33K-1-1, *Cybersecurity for Test, Measurement, and Diagnostic Equipment in the AF Metrology Program* and other prescribed directives. Perform cyber maintenance tasks IAW TO 33K-1-1.

**3.10.12** Exhaust all means necessary to ensure TMDE used to support a specific weapon system and/or covered by a specific CMS are returned to service “fully certified”. Managers will maximize common metrology practices to include calibration, alignment, troubleshooting, repair, cannibalization and/or Engineering Technical Assistance Requests to return weapon system specific TMDE to service. Managers need to understand the role the PMEL fills in the life cycle sustainment of every weapon system they support. Managers are encouraged to seek MAJCOM Functional Manager, System Program Office and/or AFMETCAL engineering assistance to help in this process when necessary to meet mission requirements. When Engineering Technical Assistance Requests are used/approved, managers should consider directing submission of a recommended change to update the applicable CMS.

**3.10.13** In addition to following required DAF and base reporting policies, contact the AFMETCAL engineer if a possible system hazard has been identified with AFMETCAL procured equipment and of any corresponding mishap that occurred during normal operations. This information is used for hazard tracking and mitigation if appropriate.



## SECTION 4

### CALIBRATION ASSISTANCE AND SUPPORT AGREEMENTS

#### 4.1 REQUESTS FOR CALIBRATION ASSISTANCE.

PMELs are obligated to support their assigned workload to the greatest extent possible. If a PMEL lacks the capability to calibrate assigned DAF-owned TMDE, which includes USSF, ANG, and AFR, calibration assistance shall be requested as follows:

- a. For items with a Note Code or PMEL (68 Deg) calibration responsibility in a CMS/TO 33K-1-100-2:
  - (1) Coordinate support through the nearest geographical PMEL with the Note Code/68 Degree Room measurement capability unless specifically designated in the Note Code definition. Requests for exceptions to using the nearest geographical PMEL for primary Note Code/68 Degree Room support must be coordinated with the PMEL's MFM and the RNM.

#### NOTE

Andersen, Hickam, and Thule PMELs only: coordinate Note Code/68 Degree Room support with the applicable supporting Type IIA PMEL (see Figure 4-1). If the Type IIA PMEL is not listed in the Note Code, coordinate support through the nearest geographical PMEL with the Note Code capability.

- (2) If the nearest Note Code/68 Degree Room PMEL has lost measurement capability, the PMEL shall request support via the Calibration Support application on METWEB. The AFMETCAL RNMs will identify the PMEL with measurement capability and coordinate support.
- b. For items that have a PMEL Calibration Responsibility and the PMEL lacks capability, the PMEL shall request support via the Calibration Support application on METWEB. The AFMETCAL RNMs will identify the PMEL with measurement capability and coordinate support.
- c. For items with a PMEL 2A calibration responsibility:
  - (1) Coordinate support through the applicable Type IIA PMEL as shown in Figure 4-1.
  - (2) If the supporting Type IIA PMEL has lost measurement capability, the PMEL shall request support via the Calibration Support application on METWEB. The AFMETCAL RNMs will identify the PMEL with measurement capability and coordinate support.

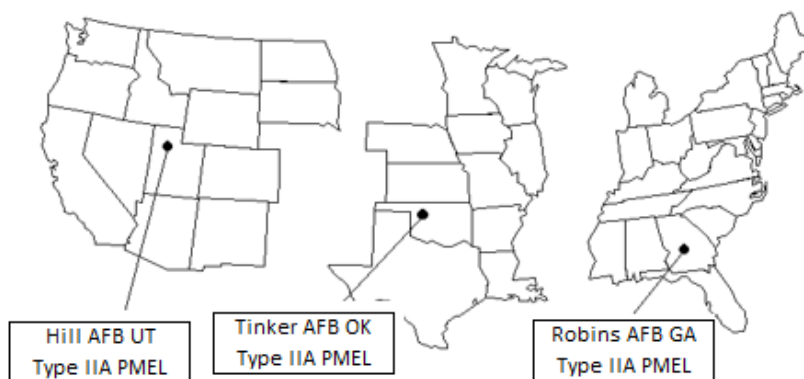


Figure 4-1. Type IIA PMEL Areas of Responsibility.

- |                                                                                                            |                                                               |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| * RAF Feltwell GB IIA PMEL supports USAFE and AFCENT (Al Udeid)                                            | * Kadena AB JA IIA PMEL supports Western Pacific areas        |
| * Elmendorf AFB IIA PMEL supports Alaska, Guam, and Hawaii                                                 | * Hill AFB IIA PMEL also supports Elmendorf AFB and Kadena AB |
| * Robins AFB IIA PMEL also supports Greenland (Thule), USAFE (including the Azores), and AFCENT (Al Udeid) |                                                               |

## NOTE

Non-DAF owned (i.e., Coast Guard, Border Patrol, FMS, contractor-owned, etc.) TMDE items are not authorized to be entered into the METWEB application. These items are typically supported via a local support agreement.

**4.1.1 AFPSL Support.** PMELs shall notify the AFPSL prior to shipment of items designated AFPSL responsibility for scheduled calibration support via the advance copy of DD Form 1149. Submit signed advanced copies via email to [afmc.afpsladvancedshippers@us.af.mil](mailto:afmc.afpsladvancedshippers@us.af.mil). Items not designated as AFPSL responsibility or unscheduled AFPSL items must be coordinated with the AFPSL scheduling section prior to shipment (DSN 366-5472/5473 or email [afpsl.scheduling@us.af.mil](mailto:afpsl.scheduling@us.af.mil)).

**4.1.2 Maintenance Support Assistance** (other than Calibration Assistance described in this section). Maintenance support assistance required by the PMEL over and above their capability shall be obtained using the instructions in TO 00-25-107.

## **4.2 TECHNICAL/ENGINEERING ASSISTANCE.**

Requests for technical and/or engineering assistance on specific metrology problems that cannot be resolved by the PMEL or TO 00-5-1 shall be submitted to AFMETCAL. PMELs requesting assistance shall furnish a comprehensive description of each problem and advise if on-site assistance is recommended.

## **4.3 CALIBRATION, REPAIR & RETURN, AND LOAN.**

TMDE may be processed between PMELs on a repair and return; calibrate and return; or repair, calibrate, and return basis. Loan of TMDE between PMELs is authorized IAW procedures outlined in AFI 23-101 and AFMAN 23-122. This maintenance concept is used when it is necessary for a PMEL to obtain maintenance assistance from an off-base PMEL. These transactions shall be coordinated between the PMELs involved. If the equipment is not to be sent to the PMEL, but to the depot or TRC, it shall be necessary to follow the procedures in TO 00-25-107 prior to movement of the equipment. After notification from the off-base PMEL that the required support can be given, the TMDE shall be delivered by courier or shipment. TMDE delivered and returned by courier shall not be processed through supply channels at either the supported or supporting PMELs location. TMDE shipped through transportation channels shall be processed in accordance with established directives. In the event the TMDE is placed in awaiting parts status, the supporting PMEL shall notify the supported PMEL immediately and provide an estimated delivery date for parts and an anticipated completion date.

## **4.4 PMEL SUPPORT AGREEMENTS.**

### **4.4.1 Definitions.**

**4.4.1.1 Interservice Support.** Action by one military service or element to provide logistic support to a different military service or element.

**4.4.1.2 Interdepartmental/Agency Support.** Provision of logistics support by the military services to departments or agencies or the United States government (other than military) with or without reimbursement.

**4.4.1.3 Host-Tenant Support Agreement.** An intraservice agreement recorded on a DD Form 1144 (IAW AFI 25-201) between active DAF, ANG or AFRC units (as required).

**4.4.1.4 Unique Support Equipment.** Equipment required solely to support the customer's equipment shall be provided by that customer if the customer is not DAF, ANG or AFRC. The DAF is not required and shall not be tasked to build up a capability at DAF expense that is used only to support another military service or federal department.

**4.4.2 Generating Agreements.** Regional support requirement external to a PMELs parent unit may require a wide variety of documented agreements. The intent of this paragraph is to assist the PMEL manager in performing their task in generating an agreement if AFI 25-201 specifies a requirement or there is a need to generate an interservice agreement per DODI 4000.19. PMEL managers require a working knowledge of support agreements and financial procedures to ensure resources are available and adequate to provide customer support. They also generate procedures to maintain communications to ensure mission support needs within a PMEL's area of responsibility are met. PMEL managers (or PMEL contract managers) shall also be aware of how and when to charge customers for services rendered. AFMC Type IIA calibration



laboratories typically are not required to generate Host Tenant support agreements when servicing other PMELs. Agreements that could be encountered are:

- a. Host-tenant (AFI 25-201, AFI 65-601, Vol 1)
- b. Interservice/Interdepartmental/Agency (DODI 4000.19).

**4.4.2.1** The following policies shall apply when generating support agreements:

- a. Support of other DAF Units. Consider:
  - (1) Facility Requirements.
  - (2) Cost, Reimbursement.
  - (3) Manpower Authorizations Required and Source of Spaces and Funding.
  - (4) Effect on PMEL parent unit's Operations Plan. Consider what shall be PMELs role.
  - (5) Training Requirements, User and PMEL.
  - (6) Equipment, Parts, and Technical Data Requirements. Customers are responsible to provide any material needed to support locally purchased items.
  - (7) Priority of Work, Customer's FAD and Precedence Rating. [USAF Programming Directive - (PD)].
  - (8) Customer's Wartime Requirements, Priorities, Exercises or Mobility support.
  - (9) Overdue Procedures.
  - (10) PMEL Assistance to Customer.
  - (11) Deficiency Reporting Procedures.
  - (12) Contract Maintenance and Calibration.
  - (13) Off-base Shipping and Transportation.
  - (14) Organizational Maintenance.
  - (15) Scheduling File Maintenance.
  - (16) Exercise or Mobility Support.
  - (17) Mission Changes.
  - (18) Unique Equipment Maintenance Cost.
  - (19) Support Requirements for Locally Procured TMDE.
  - (20) Workload.
- b. Support of non- DAF units: Agreement is required on all items listed in 4.4.2.1a and the following:
  - (1) Forms and Labels to Use.
  - (2) Calibration Intervals to Use.
  - (3) Scheduling and Management Procedures.
  - (4) Calibration Procedures to Use.
  - (5) Responsibilities.
- c. Support from non- DAF sources: Agreement is required on all items listed in 4.4.2.1a and the following:
  - (1) Forms and Labels to use.
  - (2) USAF CMS and TO 33K-1-100-2 Intervals shall be used.
  - (3) Scheduling and Management Procedures.
  - (4) USAF Calibration Authorities shall be used.
  - (5) Responsibilities.
- d. Agreements can be simplified if PMELs have a base instruction established that contains policy, assigns responsibilities, directs actions, and prescribes procedures tailored to meet local needs and is referenced in the agreement. Base regulations apply to all organizations assigned or attached to the base or tenant organizations. They also apply to off-base and remote-site organizations of the same or different parent command if the subject matter relates directly to support and services furnished them by the issuing base.

**NOTE**

Do not submit a Request for Calibration Responsibility Determination for non-DAF customer assets. Other agency customers are responsible for providing the calibration procedure to be used, or the specifications and accuracies required. Do not ask AFMETCAL to write a calibration procedure or provide special equipment. It is the customer's responsibility to provide material resources the PMEL needs to assist them unless the PMEL already has the capability.

- e. Policy, information, and procedures may be prescribed in a letter or message when there isn't time to process a publication. Letters or messages have to be replaced by a formal publication as soon as possible (DAFI 90-160).

- f. Financial Administration. Information on financial matters, budgeting reimbursement, etc., is found in AFI 65-601 Vol 1 and DFAS-DER7010-1.
- g. Support Agreement Assistance. Assistance for development of support agreements and resolution of problems can be obtained from:
  - (1) Base Level Office of Primary Responsibility (OPR) for support agreement.
  - (2) Parent Command PMEL MFM (AFMAN 21-113), or Command Support Agreement Functional Manager.
  - (3) Air Force Material Command (AFMC), AFMETCAL is the DAF point of contact for International Calibration Support.
  - (4) HQ USAF.
    - (a) Directorate of International Programs.
    - (b) International Logistics Planning and Policy.
  - (5) Support Agreement References.
    - (a) AFI 25-201, *Support Agreement Procedures*.
    - (b) AFI 38-101, *Manpower and Organization*.
    - (c) AFMAN 21-113, Air Force Metrology and Calibration Management.
    - (d) Federal Acquisition Regulation (FAR).
    - (e) DODI 4000.19, *Interservice and Intragovernmental Support*.
    - (f) DFAS-DER7070-3, *Financial Management and Accounting for Security Assistance and International Programs*.
    - (g) COMMAND 21-XX, *Maintenance Management*.
    - (h) AFI 65-601V1, *Budget Guidance and Procedures*.

**4.4.3 Negotiation, Administration and Review.** Each military service is responsible for providing or arranging for support of its own forces. Maximum practical use of support agreements shall be made when overall economies can be realized without impairing military effectiveness.

**4.4.3.1** Support agreements shall be generated only when necessary. Support agreements shall be negotiated at the lowest possible organizational level; customer to PMEL, for instance. This is with the concurrence of the base OPR for the agreement. PMEL managers shall coordinate on all agreements that include provisions of PMEL support.

#### **4.5 COMMERCIAL AND N49/59/64 CALIBRATION OF TMDE.**

**4.5.1 Commercial Calibration of TMDE.** Commercial calibration of TMDE represents a deviation from the published AF calibration determination. Therefore, the following policies pertain to commercial calibration of TMDE.

**4.5.1.1 General.** AFMETCAL is the only authority for approving commercial calibration service, unless authorized in paragraph 4.5.1.2.2 or 4.5.1.4 or the applicable CMS TO. An DAF PMEL shall calibrate TMDE in accordance with the applicable CMS, TO 33K-1-100-2. When organic (DAF) capability exists, commercial calibration shall not be purchased, and support shall be obtained from another PMEL IAW Section 4.1 of this TO. If organic capability does not exist or calibration requirements exceed the AFMETCAL program capabilities, then supporting PMELs may submit a Request for Authorization of Commercial Calibration Services. When authorized to contract any part of the calibration, the work shall be placed with a calibration source that provides a calibration certificate/report containing the information required by Data Item Description DI-QCIC-80798C as a minimum (see paragraph 4.5.1.4), except when authorized by the applicable CMS. Also, the calibration should meet the same performance specifications of the DAF calibration authority; if not, any deviation shall be documented and coordinated with the owner.

**4.5.1.2 Requests for Commercial Calibration Service.** OWCs and PWCs shall obtain PMEL MFM and AFMETCAL approval prior to procuring any calibration services from commercial sources except as noted below in paragraphs 4.5.1.2.1 and 4.5.1.2.2. The OWC or PWC shall accomplish each of the following:

- a. Verify that the part number is listed in TO 33K-1-100-2 or an applicable CMS, and:
  - (1) If not listed, assist the supporting PMEL in preparing and submitting an AFTO Form 45 in accordance with Section 5 of this TO.
  - (2) If listed, deliver the TMDE to the supporting DAF PMEL for calibration. If the supporting PMEL lacks the capability (e.g., trained personnel or equipment) to accomplish the calibration, then the supporting PMEL shall seek lateral support from another PMEL IAW paragraph 4.1 of this TO. If no lateral support source can be located, the supporting PMEL shall contact the MAJCOM PMEL MFM for approval to submit the request.

b. When approved to submit a request for commercial calibration services:

- (1) Prepare a 'Request to Deviate From the Published Calibration Determination form, which is available on the AFMETCAL SharePoint, Customer Support POCs, Commercial Cals, Commercial Calibration Files page. Forward the request to the PMEL MFM, who will in turn forward to the AFMETCAL email address indicated on the Request Form.
- (2) The requestor shall ensure and be able to provide documentation that the contractor is able to perform a calibration that meets the same performance specifications of the Air Force calibration authority, or that any deviation from the calibration authority was coordinated with the owner. AFMETCAL shall, at its discretion, determine if additional data or a site visit is required. AFMETCAL may grant approval for commercial calibration support for one or more calibration cycles.

**4.5.1.2.1 Exception for Items Designated CONTRACT.** Items listed in TO 33K-1-100-2 or an applicable CMS with a calibration responsibility of "CONTRACT" do not require AFMETCAL approval.

**4.5.1.2.2 Exception for New TMDE.** PMELs may accept the initial commercial calibration of newly procured TMDE without requesting authorization from AFMETCAL if it includes a calibration certificate/report containing the same information required by Data Item Description DI-QCIC-80798C as a minimum (see paragraph 4.5.1.4), and by verifying that the commercial calibration meets the AF calibration authority performance specifications or owner requirements. Any deviation from the AF calibration authority shall be documented and coordinated with the owner. If the PMEL does not accept the commercial calibration, calibration of the TMDE shall be IAW the applicable CMS or TO 33K-1-100-2. If the TMDE is not listed in TO 33K-1-100-2 or a CMS, refer to paragraph 3.1.5.1 prior to accepting the commercial calibration.

**4.5.1.3 Intervals for Commercially Calibrated Items.** The intervals for commercially calibrated TMDE shall be in accordance with any applicable CMS or TO 33K-1-100-2, regardless of calibration interval assigned by the calibration laboratory. The appropriate Air Force certification label shall be affixed to the TMDE IAW Paragraph 4.5.3. The calibration interval on commercially calibrated TMDE starts the date the contractor calibrated the item.

**4.5.1.4 Commercial Repair of TMDE.** Commercial repairs performed on customer TMDE shall be verified by the PMEL when the PMEL has the capability and the responsibility to calibrate the item. PMELs may accept commercial calibrations performed in conjunction with commercial repairs by verifying that the commercial calibration satisfies the User's specific calibration requirements and includes a calibration report containing the information required by Data Item Description DI-QCIC-80798C as a minimum. If the PMEL does not accept the commercial calibration, calibration of the TMDE shall be performed by the PMEL when possible. Data Item Description DI-QCIC-80798C requires the calibration certificate to include (NOTES shown below are to assist the PMEL's interpretation and verification of the requirement):

- a. Title, e.g., "Calibration Report" or "Calibration Certificate".
- b. Name and address of laboratory, and the location where the calibrations were carried out, if different from the address of the laboratory.
- c. Unique identification of the calibration report or certificate (e.g., the serial number), and on each page an identification in order to ensure that the page is recognized as a part of the calibration report or certificate, and a clear identification of the end of the calibration report or certificate.
- d. Name and address of customer.  
NOTE: A requirement for the name and address of customer may be waived with customer coordination/concurrence.
- e. Identification of the method used.  
NOTE: The calibration report or certificate may identify the method used in a manner similar to the following examples: a calibration procedure number; an ASTM, IEC, or other standard specification; or a narrative description of the method used (i.e., compared with laboratory standard X at 10 points between 10% and 100% of full scale).
- f. A description of, the condition of, and unambiguous identification of the item(s) calibrated.
- g. Date of receipt of the calibration item(s) where this is critical to the validity and application of the results, and the date(s) of performance of the calibration.
- h. Reference to the sampling plan and procedures used by the laboratory or other bodies where these are relevant to the validity or application of the results.

**NOTE**

The calibration report or certificate might not reference a sampling plan as sampling plans rarely apply to calibration; sampling plans normally are used in testing laboratories.

- i. The calibration results with, where appropriate, the units of measurement.

**NOTE**

Units of measurement would be appropriate if needed by the User (calibration factors, correction factors, etc.) to operate the TMDE.

- j. The name(s), function(s) and signature(s) or equivalent identification of person(s) authorizing the calibration report or certificate.
- k. Where relevant, a statement to the effect that the results relate only to the items calibrated.
- l. The conditions (e.g., environmental) under which the calibrations were made that have an influence on the measurement results.
- m. The uncertainty of measurement or a statement of compliance with an identified metrological specification or clauses thereof.

**NOTE**

A calibration report or certificate that contains a statement of compliance to original manufacturer's specifications (with any exceptions or limitations noted) may satisfy this requirement.

- n. Evidence that the measurements are traceable.

**4.5.1.5 Procuring TMDE Involving Commercial Calibration.** Item and program managers procuring TMDE involving commercial calibration shall submit a data call to AFMETCAL prior to contract award. AFMETCAL will review the technical specifications contained within the data call and determine if organic capability currently exists. AFMETCAL will work with the item manager/SPO office to determine the most efficient and cost effective means to ensure calibration support while complying with AFMETCAL policy. If commercial calibration is approved, AFMETCAL will provide the necessary data requirements and appropriate quality statements to the item manager/SPO to be included within the contract. Data calls shall include the following information:

- a. Nomenclature and description of the item being procured.
- b. Weapon system (if applicable).
- c. Specifications contained within a purchase description (PD), commercial item description (CID) or other statement of work.
- d. Purpose for obtaining commercial calibration.
- e. Name, office symbol, phone number and address of the requesting organization.
- f. List of possible commercial vendors that may provide commercial calibration services.
- g. Timeline in which the procurement is to be accomplished and equipment fielded.
- h. Submit data calls to the AFMETCAL Technology Applications Section electronically or by FAX.

**4.5.2 Calibration of TMDE under N49/59/64.** Calibrations of TMDE under N49/59/64 are in accordance with published AF calibration determinations and are approved by AFMETCAL. The following policies pertain to items listed within CMS TOs and TO 33K-1-100-2 with a calibration responsibility identified as N49, N59 or N64.

**4.5.2.1 General.** AFMETCAL has responsibility for management of calibration services for DAF TMDE listed as N49, N59 or N64, except for equipment requiring calibration under Consolidated Supply Activity-Maintenance (CSAG-Mx) [formerly known as Depot Maintenance Activity Group (DMAG)], or Stock Funded programs, which shall not be serviced or funded under this process unless a reimbursement agreement has been established. The PMEL shall not accept or process non-AF N49, N59 or N64 calibration workload. PMELs shall notify their customers that Air Force N49/59/64 calibration management has been centralized at AFMETCAL. To be eligible for centrally funded calibration, all items must be

operational. Any requests for short cycling (or early calibration) more than 30 days prior to date due calibration must have a valid reason and must be approved by AFMETCAL.

**4.5.2.2 N49/N59 Calibration Support.** N49/59 items will be scheduled by the local PMEL through PAMS. The PMEL scheduler will contact the appropriate service provider listed on the AFMETCAL SharePoint, Customer Support POCs, Commercial Cals, Commercial Calibration Files, Note 49 N59 POCs. The PAMS shipping module will be used to log and track the item status and location. Equipment shall be shipped from the PMEL's location directly to the service provider. Prior to shipment, maintenance data shall be entered into PAMS IAW TO 33K-1-100-1. Upon return of the calibrated N49 or N59 item, contact the responsible workload scheduler if the unit does not conform in any way or documentation is insufficient. For serviceable TMDE, complete maintenance data actions in PAMS utilizing action taken codes IAW TO 33K-1-100-1.

**4.5.2.3 N64 Calibration Support.** Calibration support for DAF TMDE may be obtained by either the owner or supporting PMEL.

- a. Owners shall contact the AFPSL Business Administration Section directly for shipping instructions. AFPSL POC information is available on the AFMETCAL SharePoint AFMETCAL Leadership & POCs page.
- b. PMELs shall submit requests and process N64 items via the AFMETCAL N64 Tracking Software located on METWEB.

#### **NOTE**

The PMEL may provide a limited calibration that meets customer requirements on an N64 designated item if they have the capability. Other deviations to N64 support require the PMEL to submit a 'Request to Deviate From the Published Calibration Determination' form (see paragraph 3.1.5.3).

**4.5.2.3.1 N64 Shipping.** Once shipping instructions are received (via the AFMETCAL N64 Tracking Software), deliver the TMDE, a completed DD Form 1149 and an appropriate reusable shipping container to the local Traffic Management Office. The DD Form 1149 shall be completed according to DAFI 24-602V2 and DAFI 21-101 to include applicable MAJCOM supplements. Ensure N64 instructions are enclosed in the shipment. Mark one copy of the DD Form 1149 "ADVANCE COPY" and send it (e-mail, FAX, or mail) to the calibration destination, requesting the shipping destination to return a signed copy of the DD Form 1149. The PMEL may accept a signed email, fax, or memo that shows how, when, and who verified receipt of the shipped asset in lieu of a returned copy of the DD Form 1149.

**4.5.2.3.2 Return from N64 Calibration.** Upon return of the calibrated equipment, the PMEL shall update the status in the AFMETCAL N64 Tracking Software to reflect receipt and upload a copy of the vendor's calibration certificate; or owning work center (if shipped directly) shall notify the AFPSL of receipt and fax a copy of the certificate per instructions provided by the AFPSL. The PMEL or owning work center shall also perform an incoming acceptance inspection (conformance assessment for physical condition, safety, documentation, and cleanliness), which includes validation of the calibration certificate meets the requirements of paragraph 4.5.1.4. Contact the AFPSL Business Administration Section if the unit does not conform in any way or documentation is insufficient; the AFPSL has the responsibility to ensure the calibration certificates/reports meet AFMETCAL requirements are corrected, as well as validating and uploading the calibration certificate for overseas PMELS.

**4.5.2.4 Intervals for N49, N59, and N64 Calibrated Items.** The intervals for N49, N59 and N64 calibrated TMDE shall be in accordance with any applicable CMS or TO 33K-1-100-2 (see Section 3 of this TO for guidance on interval exceptions such as CEE, NADCAP, NCE, WRM, medical TMDE, etc.), regardless of calibration interval assigned by the calibration laboratory. The appropriate Air Force certification label shall be affixed to the TMDE in accordance with paragraph 4.5.3. The Air Force calibration interval starts the date the contractor or DOD agency calibrated the item.

**4.5.2.5 Repair During N49/59/64 Calibration.** Incidental repair during calibration shall be accomplished and funded by AFMETCAL. Funding shall be provided for incidental repair only. When an item is received requiring major repair, AFMETCAL will contact the owning organization for fund cite and/or disposition instructions. AFMETCAL must grant approval for major repair actions.

**4.5.3 Certification Label.** The non-AF source certification label shall remain on the unit. An AF certification label shall be attached to the TMDE citing "00-20-14" in the authority block. Enter an annotation in the special block, referencing the

non-AF certification label or certificate, and any additional certification data associated with that calibration. If the non-AF calibration deviates from the Air Force calibration authority performance specifications, a limited/special AF certification label shall be attached. The PMEL or owning work center shall retain a copy of the non-AF certificate of calibration, and warranty document on file until warranty expiration. If data (other than calibration due date) contained within the certification label, calibration certificate, or certificate of traceability is incorrect, notify the appropriate workload scheduler for instructions.

#### **4.6 CALIBRATION SUPPORT FOR SITE ACTIVATION TASK FORCE (SATAF), OPERATIONAL TEST AND EVALUATION (OT&E), and INITIAL OPERATIONAL TEST AND EVALUATION (IOT&E).**

**4.6.1 General.** (Reference DODI 5000.2 AF SUP 1 and AFI 63-101/20-101.) Calibration support for activation of new sites or upgrading of operational site equipment by Air Force contractors shall be provided in accordance with host-tenant agreements established under the provisions of AFI 25-201. AFMETCAL, in coordination with the program manager and the commands involved, shall develop the necessary calibration support plans and alert the selected PMEL of the projected workload and responsibility to support the installation and checkout contractor. The responsible PMEL shall initiate action in accordance with the plan to obtain equipment and manning to support TMDE to be utilized by the installation, checkout, and acceptance teams and during the subsequent operational phase. This support is normally limited to:

- a. TMDE listed in any applicable CMS or TO 33K1-100-2 as PMEL responsibility.
- b. TMDE that is scheduled to remain on-site for use during the operational phase.
- c. The PMELs existing capability.

#### **NOTE**

Equipment, which will not be used during the operational phase, shall be identified by the PMEL as equipment requiring alternate support.

**4.6.2 Requests.** Requests for support of contractor owned TMDE shall normally be received from one of the following sources:

- a. The administrative contracting officer.
- b. The SATAF or project officer.
- c. The local base procurement officer.
- d. AFMETCAL through one or more of the above.

**4.6.3 Application.** These provisions also apply for tests on new or improved systems at system or subsystem sites.

#### **4.7 SECURITY ASSISTANCE (SA) PROGRAMS AND FOREIGN MILITARY SALES (FMS).**

FMS cases for the sale of systems to friendly nations often involve the necessity of establishing or augmenting a metrology program for that system. AFMETCAL develops specific metrology programs providing for the repair and calibration of TMDE and follow-on calibration support. All queries for FMS cases or SA Programs shall be directed to the AFMETCAL International Metrology Applications Section, 813 Irving-Wick Dr W, Heath, OH 43056-1199.

#### **4.8 TMDE WARRANTIES.**

TMDE initially issued to using activities is frequently covered by a provisional warranty from the manufacturer. These warranties vary in degree of coverage, specific service/maintenance performed, and correction of design, material, and workmanship deficiencies, which result in an item of TMDE performing below specification and contractual requirements. Warranty information for medical equipment can be found in AFI 41-201 and AFMAN 41-209.

**4.8.1 New Equipment Receipt (includes centrally procured items during their entire warranty period).** Upon receipt of new equipment, the lab (PMEL or AFPSL) shall:

- a. Determine warranty expiration date. Complete and mail any warranty cards for lab-owned equipment.
- b. Calibrate the equipment (unless commercial calibration is accepted IAW paragraph 4.5), or, if repair is required and is covered by warranty, return to customer with appropriate information required to promptly send the item through Base Procurement and contracting office IAW applicable regulations.
- c. After calibration, whenever reasonable and conditions permit, short cycle the next date due calibration so the item is due back in the lab no later than 30 days prior to warranty expiration date.
- d. Submit Product or Quality Deficiency Reports IAW TO 00-35D-54 for centrally procured items with warranties that fail calibration.

**4.8.2 Warranty Processing.** Processing of a warranty action is at the discretion of the local contracting office. If the local contracting office requires an AF Form 9, it shall be submitted.

#### **4.9 LEASED/BORROWED EQUIPMENT.**

**4.9.1 Leased/Borrowed Equipment.** The following policies pertain to calibration requirements for Leased/Borrowed TMDE that is not owned by the DAF but is used to support an DAF mission.

##### **NOTE**

TMDE owned and used by contractors performing under an DAF contract on an DAF installation to support an DAF mission shall be considered leased/borrowed equipment when determining calibration requirements.

**4.9.1.1 General.** Leased and/or Borrowed equipment is equipment used to accomplish maintenance within the DAF in accordance with prescribed DAF instructions. Such equipment may be used by organic (DAF) personnel or contractors. As such, requirements for accuracy and performance are identical to DAF owned equipment.

**4.9.1.2 Definition of Leased versus Borrowed.** Leased equipment is obtained under contract and payment is made for its use. Borrowed equipment is obtained without financial consideration for its use. There is no difference in their application in support of the mission.

**4.9.2 Calibration Requirements for Leased/Borrowed Equipment.** The User may use the equipment without an AF PMEL calibration if all of the following conditions are met:

- a. The equipment has a valid calibration label and the Certificate (or Report) of Calibration provides the information required by TO 00-20-14, Paragraph 4.5.1.4.
- b. The User will lease/borrow the equipment for a period of less than 12 months and will not lease/borrow the item in the future.

**4.9.2.1** If the equipment does not meet the requirements of Paragraph 4.9.2 and **is listed** in TO 33K-1-100-2 (or an appropriate CMS), the User shall obtain calibration services IAW 33K-1-100-2 (or the appropriate CMS). An DAF PMEL may calibrate Leased/Borrowed equipment without adjustment to avert voiding the warranty unless adjustment action is included within the support agreement. Equipment that is found out of specification will be returned to the User for shipment to the owner for adjustment.

**4.9.2.2** If the equipment does not meet the requirements of Paragraph 4.9.2 and **is not listed** in TO 33K-1-100-2 (or an appropriate CMS), then the User shall deliver it to the supporting DAF PMEL and assist the PMEL to prepare and submit an AFTO Form 45 in accordance with Section 5 of TO 00-20-14. Identify in the Remarks block of the AFTO 45 that the equipment is leased/borrowed, and the terms (length) of the lease. The User may have to reimburse the DAF for costs of calibration procedure development, additional calibration equipment required to calibrate the item, or calibration of the item itself.

**4.9.2.3** If the equipment referenced in Paragraph 4.9.2.2 will be leased/borrowed for a period of less than 12 months, the certificate (or report) of calibration does not provide the information required by TO 00-20-14, Paragraph 4.5.1.4, the item is not listed in TO 33K-1-100-2 (or an appropriate CMS), and the item will not be leased/borrowed in the future, a request for commercial calibration may be submitted IAW Paragraph 4.5.1.2.b instead of submitting an AFTO Form 45. Any request for commercial calibration should identify that the item is leased/borrowed and the term (length) of use by the DAF. The User will be responsible for the cost of the commercial calibration if the request is approved.

#### **4.10 DEPOT PUBLIC PRIVATE PARTNERSHIPS (PPP).**

**4.10.1 Public Private Partnerships (PPP) Programs.** A PPP for depot-level maintenance under section 2474 of Title 10, USC Code is a cooperative agreement between an organic depot-level maintenance activity and one or more private sector entities to perform DOD or Defense-related work and/or to utilize DOD depot facilities and equipment. Other government organizations, such as program offices, inventory control points, and material/systems/ logistics commands may be parties

to such agreements. The following paragraphs pertain to calibration support of TMDE located on a military facility in support of a PPP program.

**4.10.1.1 General.** The weapon system Program Manager and/or the base PPP office will coordinate calibration support with AFMETCAL in accordance with AFI 63-101/20-101. Support of PPP represents a deviation from normal PMEL/customer relationships. PMEL calibration support for programs defined as PPP shall be provided in accordance with base Partnering Agreements (PA) and detailed within an associated Implementation Agreement (IA). The base PPP office shall coordinate calibration support issues with AFMETCAL and the appropriate PMEL MFM. AFMETCAL, in coordination with the weapon systems program manager, the Air Logistics Complex (ALC), PMEL MFM, base PPP office, and contractor, shall develop the necessary calibration support plan and document the projected workload and responsibility to support the installation and PPP contractor. The PMEL shall review the calibration requirements and confirm that they can meet support expectations contained in both the PA and IA.

**4.10.1.2 Partnering Agreement (PA).** The PA is a detailed agreement which establishes the organizational interactions, assumptions, and processes which the stakeholders agree to follow during the partnership. The base PPP office is responsible for authoring and coordinating the PA and associated IA.

**4.10.1.3 Implementation Agreement (IA).** The IA, prepared by the PPP office, defines PMEL support to be provided the contractor as negotiated between the PMEL, PPP contractor, AFMETCAL, and the Weapon System program manager. Performance parameters including, but not limited to, turn-around-time, level of effort, and level of technical data provided may differ from current DAF requirements and should be included in the IA. Some typically governmental tasks such as CPIN assignment, configuration control, and control of technical data may be delegated to the contractor. PMEL support for PPP programs will be limited to services and equipment defined within the IA and equipment listed as PMEL or MCF calibration responsibility within an applicable Calibration Measurement Summary (CMS) and/or TO 33K1-100-2. Once approved, the responsible PMEL should initiate action to obtain the necessary equipment and manning to provide support in accordance with the IA.



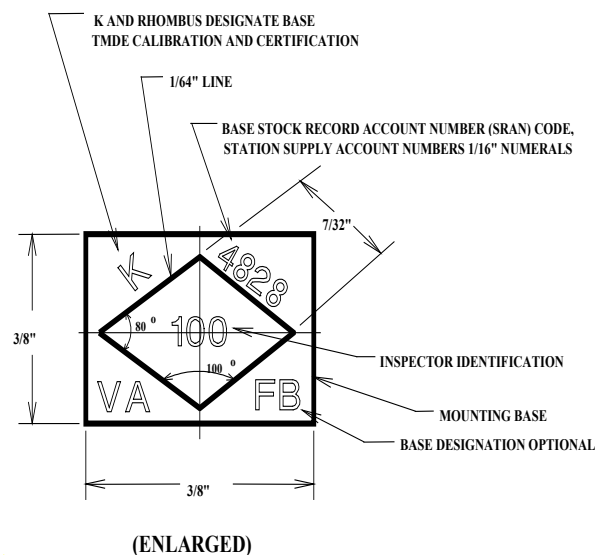
## SECTION 5

### PREPARATION AND USE OF CALIBRATION FORMS AND LABELS

#### 5.1 TMDE CERTIFICATION STAMP (K STAMP).

The K Stamp is a PMEL technician's official signature on those forms, tags, or labels described in this TO and applicable command directives. It shall be used only on these forms. Only PMEL technicians in officially designated TMDE calibration laboratories are authorized to use a TMDE Certification stamp (K-stamp as described in figure 5-1). Using an electronic version of the K stamp is authorized. The use of K stamps by PMEL technicians is mandatory on certification labels. The PMEL manager shall designate who will be issued stamps. Stamps shall be serially numbered to identify the technician or supervisor and the activity to which he or she is assigned. The K stamp shall be as illustrated in Figure 5.1. The use of a stamp with smaller or larger dimensions is optional, provided it is dimensionally proportional and does not exceed the space available on forms. The K stamps shall be obtained through local purchase.

**Figure 5-1. TMDE Certification Stamp (K Stamp).**



#### 5.2 GENERAL INSTRUCTIONS.

The forms and labels in this section are mandatory unless otherwise noted. Form illustrations in this section are samples and may not reflect the date of the current versions available. All written entries on labels and forms shall be in non-smearing permanent black or dark blue ink. Approval to use computer generated forms and labels controlled by this section not already addressed can only be obtained by AFMETCAL. Send requests to this TO's Technical Content Manager.

#### NOTE

Anyone having trouble with one of the following labels or forms, due to their manufacture, shall report it to AFMETCAL.

##### 5.2.1 Affixing Labels.

**5.2.1.1** Only authorized personnel shall affix labels.

**5.2.1.2** When affixing labels, do not cover TMDE identification data (part or serial number, etc.). Also, do not affix labels where they would prevent use of TMDE (e.g., covering controls or control markings).

**5.2.1.3** Always affix Bar Code Labels and certification labels where the user will see them during normal handling and use of the TMDE. Affix labels to a clean surface on the TMDE mainframe, if possible, not on removable parts or panels. In the following instances, it may not be possible to affix labels to the TMDE:

- a. The TMDE is too small to affix a label, or its nature will not allow affixing of a label (optical flats and gage blocks). In these cases, the label may be placed on a removable shoe tag, or if the TMDE has a container, the label may be affixed to it.
- b. The TMDE is used in an environment that might damage the labels (e.g., oil or fuels could destroy labels). Labels may be affixed away from the instruments.
- c. On complex TMDE such as test stands or checkout consoles, certification labels for components may be grouped on the end-item.

#### NOTE

When a label is not affixed to its TMDE, there shall be some definite marking or annotation (preferably part number/serial number) on either the label or TMDE that links the two together. This requirement also applies to individual TMDE items that are part of a set or kit. Individual weights, such as the MMS Weight Set, are an exception to this requirement. They shall have no part number, serial number, or color markings on them other than their size because marking would physically alter the certified parameters and damage the TMDE. For example: Identify the quantity and describe the items of the kit on the certification label (use as much detail as necessary to ensure the proper identification of these items).

**5.2.2 Removing Labels.** Only authorized personnel shall remove labels when the item is being returned to supply. Prior to returning TMDE to supply, all labels shall be removed except the Bar Code Label. (Exception: Technical Repair Centers may leave the certification labels on TMDE) The PMEL shall be notified the item has been turned in so they can remove it from their master inventory. Document the condition tags IAW TO 00-20-3.

### **5.3 TMDE BAR CODE LABELS.**

The PMEL shall ensure an original TMDE bar code label, AFTO Form 65 or 66, is initially affixed to each item of DAF TMDE (that physically exists) in the PMEL inventory unless the item fits definition of paragraph 3.2.1. Ensure after cleaning TMDE surfaces that they are dry before applying these labels because wet solvent will remove the adhesive. Ensure there is at least one/eighth inch on both ends of the label beyond the bar code marks; this allows the bar code scanner to properly read the label. Preferred location for the bar code labels is on the upper right corner of the front panel. Damaged labels shall be replaced by a new AFTO Form 65 or 66 or an equivalent computer-generated bar code using the original AFTO Form 65 or 66 bar code number. Computer generated bar codes are only authorized as a replacement. AFMETCAL distributes the AFTO Form 66 to prevent duplication of identification numbers. PMELs can obtain the forms by contacting the Technical Content Manager of this TO.

**5.3.1 AFTO FORM 65 Aluminum Stock.** The TMDE aluminum label, AFTO Form 65 (see Figure 5-2), is no longer available for distribution. Current stock on hand can be used where the AFTO Form 65 durability is an advantage and it poses no safety hazard. There may be instances, however, due to safety or TMDE function, where the polyester AFTO Form 66 shall be used instead. See the instructions for the AFTO Form 66.



**Figure 5-2. AFTO FORM 65, TMDE Bar Code Label, Aluminum Stock.**

**5.3.2 AFTO FORM 66 Polyester Stock.** The TMDE polyester label, AFTO Form 66 (see Figure 5-3), may be used wherever practical. It shall be used when using the aluminum AFTO Form 65 could be a safety hazard or prevent proper TMDE function or calibration.



**Figure 5-3. AFTO FORM 66, TMDE Bar Code Label, Polyester Stock.**

Examples include:

- a. Small RADIAC dosimeters.
- b. Small torque indicating devices.
- c. TMDE that is inserted into mainframes or incorporated into other types of equipment where the label could become a conductive foreign object damage (FOD) hazard if it came loose.
- d. TMDE used on a flightline or anywhere a dislodged label could be ingested into an aircraft engine.

## **5.4 CERTIFICATION LABELS.**

**5.4.1 General.** All equipment that requires calibration shall have a certification label attached by the calibrating work center prior to use by the Owning Work Center (OWC). The full-sized certification label (AFTO Form 108, 99) should be used whenever possible and practical. Electronic (computer-generated) versions of certification labels that are printed on blank label stock are authorized for use provided the electronic (computer-generated) form maintains the same general information and dimensions of the pre-printed form. Only one certification label shall be used for each item; all unapproved or expired labels shall be removed.

### **5.4.1.1 Exceptions.**

**5.4.1.1.1 Approved Non-DAF Source Calibrated the TMDE.** In this instance, the TMDE shall have both the non-AF certification label and an Air Force certification label with the additional required information affixed. The authority in this situation is "TO 00-20-14". Also, enter an annotation in the Air Force certification label special block, referencing the non-AF certification label or certificate, and any additional certification data associated with that calibration. When the use of a Non-Air Force Source certification label is authorized by a CMS, the Air Force certification label is not required.

**5.4.1.1.2 Non-Existent NEC End Items.** Certification labels are not required for SICL end-item part numbers designated NEC that do not physically exist (e.g., the AN/APM427) or merely designate a case, container, console, or similar housing.

**5.4.1.2 Use of Condition Tags.** When a certification label is affixed to TMDE, Air Force supply system condition tags, such as DD Forms 1574, Serviceable Label - Material, are not required unless the item is being returned to supply.

**5.4.1.3 Changes to Certification Labels.** Unless authorized, any changes to certification labels shall void the certification.

**5.4.1.3.1 User/Owner Authorized Changes.** Owning organizations may change the OWC in the IDENTIFICATION NO. block IAW paragraph 5.4.2.1.1 or 5.4.2.1.2, and CBU may be added to the DATE DUE block IAW paragraph 3.2.2b.

**5.4.1.3.2 PMEL Authorized Changes.** PMELs may re-accomplish/correct certification labels for an item certified locally or at another DAF laboratory with authorization from the Laboratory Manager, Quality Manager, or Technical Manager of the original certifying laboratory. The PMEL receiving authorization to change the calibration label will annotate PAMS or approved MIS (if not certified locally) and maintain copies of the authorization. The corrected (or new) calibration label will be stamped by the local PMEL and reference the re-accomplishment, original laboratory (if applicable) and stamp number of the original calibration technician.

**5.4.1.4 Calibration Extensions.** The PMEL manager will maintain a file copy of all approved calibration extensions until they expire and ensure a new TMDE certification label is placed on all TMDE that has been extended. Cite the email/message from the approving official in the "Special" block, along with the words "Calibration Extension". Enter the date the item was originally calibrated in the "Date Calibrated" block of the certification label. Enter the new date due calibration in the "Date Due" block. The new TMDE certification label should cite the same calibration authority as the old TMDE certification label. The PMEL manager or designated representative will enter his/her initials in the "Certified By" block of the certification label. If the "administrative" laboratory completes the certification label, notify the "certifying"

laboratory of the extension. When mission or geographic location dictates, the OWC Maintenance Superintendent or signature authority of the extension request are allowed to make these changes to the existing certification label in lieu of a new certification label provided the approval letter is maintained on file.

**5.4.2 Common Entry Instructions.** The following requirements apply to all certification labels, unless otherwise indicated. For additional requirements see specific entries under the individual labels.

**5.4.2.1 IDENTIFICATION NUMBER Block.** Enter the OWC code, and the bar code number from the AFTO Form 65 or 66 if PMEL supported, or serial number if USER supported, of the equipment being certified.

**5.4.2.1.1 Changes to OWC.** Users may change the OWC code on a certification label. When this change is made, the user shall notify the PMEL.

**5.4.2.1.2 Differing MIS Databases.** This block may be filled in by the TMDE user if an off base PMEL calibrates the TMDE, and the customer does not use the same database as the PMEL.

**5.4.2.1.3 WRM.** Also enter WRM if the TMDE will be packed and stored as war reserve material.

**5.4.2.1.4 CEE.** Also enter CEE if the TMDE will be packed and stored as contingency equipment exception.

**5.4.2.2 AUTHORITY Block.** This block applies to AFTO Forms 99 and 108 only. For those items with an AFTO Form 394 or 398, the TO listed in the PMEL master ID listing is the authority, if included; otherwise the authority is the TO listed in the applicable CMS, TO 33K-1-100-1 or TO 33K-1-100-2.

**5.4.2.2.1 Absolute Standard.** Enter ABSOLUTE STANDARD in this block when applicable.

**5.4.2.2.2 AF Calibration Procedure.** Enter the calibration procedure identified in the CMS or TO 33K-1-100-2. If an automated calibration TO or CPIN is used as the calibration authority, enter the content of the [Auto CTO] field, as formatted in the applicable CMS or 33K-1-100-2, including any sub procedure identified. If space is insufficient, the special block may be used.

**5.4.2.2.3 No AF Calibration Procedure.** If there is no calibration procedure identified, or if the AF calibration procedure has not yet been published, enter either the DOD calibration procedure, the maintenance TO number, or "COM DATA".

**5.4.2.2.4 Calibrated by Non-AF Source.** Identify it as being calibrated by a non-AF source by entering "00-20-14", as applicable.

**5.4.2.2.5 NCR.** For items designated NCR that require a Special calibration, enter "00-20-14" and list the procedure used in the special block.

**5.4.2.2.6 NEC End Item.** For See Individual Component Listing (SICL) end-items designated NEC, leave blank.

**5.4.2.2.7 NHA Components.** For items designated NHA, enter "NHA".

**5.4.2.2.8 SICW.** For items designated SICW, enter "SICW".

**5.4.2.3 SPECIAL Block.** Enter any combination of the following as necessary.

**5.4.2.3.1 Additional Calibration Data.** Indicate if there is additional calibration data.

**5.4.2.3.2 Calibration Using TO or Other Data.** Identify why the item was fully or partially calibrated using a calibration authority not listed in the CMS or TO 33K-1-100-2. If this was a result of an inadequacy in the calibration procedure, submit a recommended change (if necessary) to correct the calibration TO. If commercial data is used, also comply with paragraph 5.4.2.3.2.1.

**5.4.2.3.2.1 Com Data.** If the calibration authority is COM DATA, and the TMDE specifications are available in the CMS or TO 33K-1-100-2, annotate the special block with "specifications from" along with the actual source (i.e., TO 33K-1-

100-2 or the appropriate CMS reference). If the calibration authority is COM DATA, and the TMDE specifications are not available in the CMS or TO 33K-1-100-2, list the actual commercial data and publication/revision date that was used. If thorough research does not reveal a date of publication/revision, assign the date the research was accomplished to the data and retain on file.

**5.4.2.3.3 General Calibration TO.** If the TMDE was calibrated using one of the general calibration TOs listed in TO 33K-1-100-1 Section 3, and no accuracy is assigned in either the appropriate CMS or TO 33K-1-100-2 as displayed in the Air Force Calibration Authority Viewer (AFCAV) or the general calibration TO, enter the accuracy to which it was certified and the source data description (e.g., +/- 0.001 inch. See 'Manufacturer' catalog #29 pg. 53, model 436).

**5.4.2.3.4 Contract Calibration.** Make entries per paragraph 5.4.1 if this was a contract calibration, which resulted in calibration meeting all specifications of DAF or requirements.

**5.4.2.3.5 Calibrated IAW a CMS (i.e., F16, F15, etc.).** For TMDE calibrated IAW a CMS, note the applicable CMS when the system specifications listed in a CMS are different than the specifications listed in 33K-1-100-2 (or manufacturer listed specifications, when not listed) and/or the TMDE is subject to 33K-1-100-1, paragraph 4.1.2. For common TMDE being used to support more than one system, the TMDE must be calibrated to all applicable system requirements. This may involve multiple CMSs and/or TO 33K-1-100-2, depending on the systems being supported. If different calibration intervals or specifications are listed, use the shortest interval and tightest specification, and note which CMS was used for the interval/specification.

**5.4.2.3.6 NHA and SICL Components Calibrated as Part of the End Item.** For SICL components, use this block to denote the component is "Part of" SICL end item part number (e.g., "P/O 12345-1"). If the component is calibrated as part of the end-item (generally designated NHA) and is not attached to the end-item or is to be removed from the end-item after calibration, also include the end-item bar code number or serial number. This annotation is not required on items that are the SICL End Item and designated as SICW in the Cal Responsibility.

**5.4.2.3.7 Insufficient Space.** If there is insufficient space on the certification label for all entries, recommend affixing the certification label to a plain manila shoe tag. The certification label serves as the authority for the data written on the manila tag. The manila tag shall be attached to the TMDE item. A second alternative is to enter additional data on a calibration data sheet when the plain manila tag interferes with use of the instrument. A third alternative is to utilize an additional label to adhere to the TMDE item. **NOTE: This label shall not be a certification label but may be generated out of blank stock where the color and size may or may not match the certification label.** In all cases, reference the additional data in the special block of the certification label.

**5.4.2.3.8 SICL End Item.** For SICL end items where the components are calibrated as a part of the end item, enter "SICL end item" in this block. This annotation is not required on items that are a SICL End Item and designated as SICW in the Cal Responsibility.

**5.4.2.3.9 SICW.** For SICW items, enter either the Bar Code Number or the part number and serial number of the item it is calibrated with. The SPECIAL block of the calibrated item will list either the bar code numbers or part numbers and serial numbers of all items calibrated with the item.

**5.4.2.3.10 Calibration Factors.** When TMDE has software adjustments affecting calibration that are accessible to the user consider adding the following statement to the certification label: "Cal Void if calibration factors do not match accompanying cal chart."

**5.4.2.3.11 Calibrated by Non-AF Source.** Enter annotation referencing the non-AF certification label or certificate, and any additional certification data associated with that calibration.

**5.4.2.4 CERTIFIED BY Block.** The K stamp of the PMEL calibration technician, or the inspection/production stamp or initials of the calibration technician or other PWCs shall be entered in the block.

**5.4.2.5 DATE CALIBRATED Block.** Enter the date the TMDE was calibrated. Enter date as YYYYMMDD.

**5.4.2.5.1 Non-AF Certification.** When authorized to use manufacturer's certification, use date on manufacturer's certificate when provided, rather than date unit was received or placed into service.

**5.4.2.5.2 Absolute Standard.** For an absolute standard, enter the date the certification label is completed.

**5.4.2.5.3 SICL End Item.** For SICL end item, complete if the SICL end item has a calibration interval. If there is no SICL end item calibration, enter NEC.

**5.4.2.5.4 SICW.** For SICW items, enter SICW.

**5.4.2.6 DATE DUE Block.** Enter the date the equipment is due for calibration. Enter date as YYYYMMDD. The equipment is overdue at the start of the following day. For example, an item due calibration on 13 June becomes overdue at 0001 hours, 14 June. Calibration due dates may be extended under certain circumstances as defined in paragraph 3.4.

**5.4.2.6.1 AFPSL Extensions.** The AFPSL is authorized to extend the due date to agree with a schedule or need date for special project TMDE being assembled for a PATEC, TFCU or RASCAL.

**5.4.2.6.2 Interval with Operating Hours.** In cases where the calibration interval is based on either a specified interval or a number of operating hours, enter the date the item would be due for recalibration under the specified interval and the operation hours at which the item would require recalibration.

**5.4.2.6.3 Absolute standard or ICO.** Enter ABS or ICO instead of a date, as applicable.

**5.4.2.6.4 WRM or CEE.** Leave this block blank if the TMDE will be packed or stored as WRM or CEE. Once the item is removed from the WRM/CEE package or storage, the user shall annotate their TMDE inventory listing and the certification label with the new due date. User shall also notify the PMEL of the new date due. The calibration due date shall be computed using the assigned calibration interval starting from the date the TMDE is removed from the WRM or CEE status.

**5.4.2.6.5 SICW.** For SICW items, enter SICW. SICW items shall not generate a separate calibration due date; this date will be attached to the item it is calibrated with.

**5.4.3 AFTO FORM 108, TMDE Certification Label.** An AFTO Form 108 (see Figure 5-4) can only be used if all specifications of the calibration authority are met. The calibrating technician or other authorized person shall fill out this label as described in the general instructions or in the following manner, as applicable.

|                                                       |                    |                 |                       |                 |
|-------------------------------------------------------|--------------------|-----------------|-----------------------|-----------------|
| PREVIOUS EDITION WILL BE USED<br>AFTO FORM 108 Nov 84 | IDENTIFICATION NO. |                 | AUTHORITY (T O., ETC) |                 |
|                                                       | SPECIAL            |                 |                       |                 |
|                                                       | CERTIFIED BY       | DATE CALIBRATED | CERTIFIED BY          | DATE CALIBRATED |
|                                                       |                    | DATE DUE        |                       | DATE DUE        |
| TMDE CERTIFICATION                                    |                    |                 |                       |                 |

**Figure 5-4. AFTO FORM 108, TMDE Certification Label (White).**

**5.4.4 AFTO FORM 394, TMDE Certification Label.** AFTO Form 394 (see Figure 5-5) can only be used if all specifications of the calibration authority are met. The calibrating technician or other authorized person shall fill out this label as described in the general instructions.

|                       |                    |          |
|-----------------------|--------------------|----------|
| CERTIFIED BY          | TMDE CERTIFICATION |          |
|                       | ID.NO. _____       |          |
|                       | SPECIAL            |          |
|                       | DATE CAL           | DATE DUE |
| AFTO FORM 394, MAR 86 |                    |          |

**Figure 5-5. AFTO FORM 394, TMDE Certification Label (White).**

**5.4.5 AFTO FORM 99, Limited/Special TMDE Certification Label.** The yellow AFTO Form 99 (see Figure 5-6) is used for standards and TMDE certified with a limited or special calibration as defined in Section 3 of this TO. Also, certain 33K TOs may direct use of this label because they do not verify all, or portions of, main functions and ranges. It is the responsibility of the user to read and apply whatever is in the SPECIAL block. This label shall also be used for equipment designated as NHA, NPC or CBU. The calibrating technician or other authorized person shall fill out this label in the following manner and as described in the general instruction, as applicable.

|                                                       |                       |                 |                       |  |
|-------------------------------------------------------|-----------------------|-----------------|-----------------------|--|
| AFTO FORM 99, JUN 07<br>PREVIOUS EDITION WILL BE USED | IDENTIFICATION NUMBER |                 | AUTHORITY (T.O. ETC.) |  |
|                                                       | SPECIAL               |                 |                       |  |
|                                                       |                       |                 |                       |  |
| CERTIFIED BY                                          |                       | DATE CALIBRATED | USER APPROVAL         |  |
|                                                       |                       | DATE DUE        |                       |  |

**Figure 5-6. AFTO FORM 99, Limited/Special TMDE Certification Label (Yellow).**

**5.4.5.1 SPECIAL Block.** In addition to the information listed in the General Entry Instruction in this Section, this block shall be used for the following purposes:

- To indicate the accuracy to which the item was calibrated.
- To identify the basic function that was measured and the ranges or parameters certified (e.g., FUNCTION: 0-50 VAC). If it is more convenient, the exceptions may be entered (e.g., FUNCTION: Except 0-500V DC). In general, cite the 33K Table 1 specification that is being limited (e.g., VDC, VAC, etc.)
- To identify special accuracy, ranges, points calibrated, or other information of value to user.

**5.4.5.2 DATE CALIBRATED Block.** Enter date calibrated, NHA or leave blank as applicable.

**5.4.5.3 DATE DUE Block.** Enter date due, ICO, NHA, NPC, date due/CBU or leave blank as applicable.

**5.4.5.4 USER APPROVAL Block.** The representative designated IAW paragraph 3.6f above shall sign or initial this block to indicate that the user agrees with the limited or special calibration performed.

**5.4.6 AFTO FORM 398, Limited TMDE Certification Label.** The yellow AFTO Form 398 (see Figure 5-7) is used for standards and TMDE certified with a limited or special calibration as defined in Section 3 of this TO. Also, certain 33K TOs may direct use of this label because they do not verify all, or portions of, main functions and ranges. It is the responsibility of the user to read and apply whatever is in the SPECIAL block. This label shall also be used for equipment designated as NHA, NPC or CBU. The calibrating technician or other authorized person shall fill out this label in the following manner and as described in the general instruction, as applicable.

|              |                            |          |     |
|--------------|----------------------------|----------|-----|
| CERTIFIED BY | LIMITED TMDE CERTIFICATION |          | INI |
|              | ID.NO. _____               |          |     |
|              | SPECIAL                    |          |     |
|              | DATE CAL                   | DATE DUE |     |

**Figure 5-7. AFTO FORM 398, Limited TMDE Certification Label (Yellow).**

**5.4.6.1 SPECIAL Block.** This block shall contain specific limitations or special calibration data. It will normally be used to direct the reader to see an AFTO Form 249 or other data sheet, since it is unlikely that all the instrument limitations (or special data) can be documented on the AFTO Form 398.

**5.4.6.2 DATE CAL Block.** Enter date calibrated, NHA or leave blank as applicable.

**5.4.6.3 DATE DUE Block.** Enter date due, ICO, NHA, NPC, date due/CBU as applicable.

**5.4.6.4 INI Block.** The representative designated IAW Paragraph 3.6.f above shall initial this block to indicate that the user agrees with the limited or special calibration performed by the PMEL. Note: On versions of this label without an INI block, initial in the label border adjacent to the Date Due block.

**5.4.7 Certification Labels for SPRAMS items.** AFTO Forms 99/108/394/398 (as applicable) shall be used on individual TRUs for SPRAMS. The labels shall be completed per the general instructions except as follows:

**5.4.7.1 IDENTIFICATION NO Block.** Also enter “SPRAMS”.

**5.4.7.2 SPECIAL Block.** Once the item is removed from SPRAMS, the user shall enter the date the item was removed.

**5.4.7.3 DATE CALIBRATED Block.** Enter the original calibration date.

**5.4.7.4 DATE DUE Block.** Leave this block blank. Once the item is removed from SPRAMS, the user shall enter the new due date or NHA then notify the scheduler of the new date due. The calibration due date shall be computed using the assigned calibration interval starting from the date the TMDE is removed from SPRAMS.

## **5.5 AFTO FORM 256, NO CALIBRATION REQUIRED LABEL.**

The AFTO Form 256 (see Figure 5-8) shall be affixed to items of TMDE listed as NCR in a CMS or TO 33K-1-100-2. Electronic (computer-generated) versions of certification labels that are printed on blank label stock are authorized for use provided the electronic (computer-generated) form maintains the same general information and dimensions of the pre-printed form.



**Figure 5-8. AFTO FORM 256, No Calibration Required Label.**

**5.5.1 Affixing and Validation Responsibility.** Organizations that possess NCR TMDE are responsible for affixing and validating the AFTO Form 256 for these items.

**5.5.2 Validation.** This label shall be validated with a PMEL K stamp, or the inspection/ production stamp or initials of the user of the TMDE.

**5.5.3 PMEL NCR TMDE.** On NCR TMDE designated PMEL or AFPSL responsibility in the CMS or TO 33K-1-100-2, annotate the OWC and bar code label number (if assigned) on the AFTO Form 256.

**5.5.4 Small NCR TMDE.** On NCR TMDE too small to affix the label, a white dot can be used in place of the AFTO Form 256.

## **5.6 NIST REPORT OF MEASUREMENT.**

This report is completed by NIST on all Air Force measurement standards calibrated by NIST. A copy of the report shall remain with the standard.

## **5.7 CALIBRATION CERTIFICATES OR CALIBRATION REPORTS.**

**5.7.1 AFPSL Calibration Certificate or Report.** The AFPSL completes this certificate or report when a formal certificate/report is required or requested for base measurement standards, Air Force measurement standards and TMDE. When issued, the certificate or report shall contain the required information in paragraph 5.7.3.



**5.7.2 PMEL Calibration Certificate or Report.** The PMEL completes this certificate or report when requested by a user or as directed by the calibration TO or other directive. When issued, the certificate or report shall contain the required information in paragraph 5.7.3.

**5.7.3 Calibration Certificate and Report Format.** The format of the certificate or report shall be specifically designed for each type of calibration, but the headings shall be standardized and include all information necessary for the interpretation of the calibration results. Each calibration certificate or report shall be accurate, clear, unambiguous, and objective, in accordance with any instructions in the calibration authority. Where the certificate or report contains results of calibrations performed by other than the PMEL, these results shall be clearly identified. When issued, the certificate or report shall contain, as a minimum (NOTES shown below are to assist the PMEL's interpretation and verification of the requirement):

- a. Title, e.g., "Calibration Report" or "Calibration Certificate".
- b. Name and address of laboratory, and the location where the calibrations were carried out, if different from the address of the laboratory.
- c. Unique identification of the calibration report or certificate (e.g., the serial number), and on each page an identification in order to ensure that the page is recognized as a part of the calibration report or certificate, and a clear identification of the end of the calibration report or certificate.
- d. Name and address of customer.

**NOTE**

A requirement for the name and address of customer may be waived with customer coordination/concurrence.

- e. Identification of the method used.

**NOTE**

The calibration report or certificate may identify the method used in a manner similar to the following examples: a calibration procedure number; an ASTM, IEC, or other standard specification; or a narrative description of the method used (i.e., compared with laboratory standard X at 10 points between 10% and 100% of full scale).

- f. A description of, the condition of, and unambiguous identification of the item(s) calibrated.
- g. Date of receipt of the calibration item(s) where this is critical to the validity and application of the results, and the date(s) of performance of the calibration.
- h. Reference to the sampling plan and procedures used by the laboratory or other bodies where these are relevant to the validity or application of the results.

**NOTE**

The calibration report or certificate might not reference a sampling plan as sampling plans rarely apply to calibration; sampling plans normally are used in testing laboratories.

- i. The calibration results with, where appropriate, the units of measurement.
- j. The name(s), function(s) and signature(s) or equivalent identification of person(s) authorizing the calibration report or certificate.
- k. Where relevant, a statement to the effect that the results relate only to the items calibrated.
- l. The conditions (e.g., environmental) under which the calibrations were made that have an influence on the measurement results.
- m. The uncertainty of measurement or a statement of compliance with an identified metrological specification or clauses thereof.

**NOTE**

A calibration report or certificate that contains a statement of compliance to original manufacturer's specifications (with any exceptions or limitations noted) may satisfy this requirement.

- n. Evidence that the measurements are traceable.

**5.7.4 Calibration Certificate or Report Amendments.** Amendments to the calibration certificate or report after issue shall be made in the form of a separate document including the statement, "Supplement to Calibration Report (or calibration certificate)" and shall include the identifying number of the original report or certificate. Supplements shall meet all the requirements of original reports or certificates.

**5.7.4.1** Amendments are required for any event such as the identification of defective calibration equipment that casts doubt on the validity of results given in any calibration report, certificate, or amendment. Such amendments shall quantify the magnitude of error in the calibration results.

## **5.8 TMDE CALIBRATION DATA.**

The AFTO Form 249 has been deleted but may be used until exhausted. The PMEL should use a locally generated chart meeting the requirements of this TO (see paragraph 5.4.2.3.1, Additional Calibration Data) and the applicable calibration procedure. Specifically, identify the bar code, part number, serial number, and calibration date. Additionally, add the name, initials, or K stamp of certifying technician.

## **5.9 TMDE CALIBRATION CORRECTION CHART.**

The AFTO Form 250 has been deleted but may be used until exhausted. The PMEL should use a locally generated chart meeting the requirements of this TO (see paragraph 5.4.2.3.1, Additional Calibration Data) and the applicable calibration procedure. Specifically, identify the bar code, part number, serial number, and calibration date. Additionally, add the name, initials, or K stamp of certifying technician.

## **5.10 MEASUREMENT STANDARD DECALS.**

Equipment that is certified as an Air Force Measurement Standard or a Base Measurement Standard shall be identified with the appropriate decal established for this purpose. TO 33K-1-100-2 identifies Base Measurement Standards.

**5.10.1 Gold Decal.** The gold Air Force Measurement Standard Decal and a certification label shall be affixed to standards possessed and used by the AFPSL as a basic measurement standard for the DAF.

**5.10.2 Blue Decal.** The blue Base Measurement Standard Decal and a certification label shall be affixed to standards that are possessed and used by a PMEL as a Base Measurement Standard and designated as such in TO 33K-1-100-2. PMEL personnel who certify or possess the standard shall affix the decal. The base measurement standard decal shall not be affixed to working standards calibrated by the possessing PMEL. The blue decals may be obtained from the AFPSL.

## **5.11 AFTO FORM 255, CERTIFICATION VOID IF BROKEN.**

The AFTO Form 255 (see Figure 5-9) label, or an alternate method listed below, shall be applied to operator accessible controls or adjustments on standards and other TMDE, which if moved, will invalidate the calibration. If a panel must be removed to access controls or adjustments, they are not considered operator accessible. Apply this label sparingly and reserve for critical locations or TMDE susceptible to tampering. If label is broken or shows signs of tampering, the certification is void. Equipment shall be scheduled for certification unless situations addressed in exceptions paragraph apply. This label shall be validated with a K stamp of PMEL technician. Any non-PMEL organization applying the label can use inspection or production stamp or the initials of the responsible certifying technician. Do not cover the face of this label with any kind of tape that would defeat the purpose of the label. Alternate methods, such as sealant or manufacturer's calibration/warranty void seals, may be used at the option of the responsible calibrating work center to seal TMDE if the AFTO Form 255 cannot be applied satisfactorily.



**Figure 5-9. AFTO FORM 255, Certification Void If Broken Label**  
(Previous version can be used until stock exhausted.)

**5.11.1 Exceptions.** If the seal is broken accidentally, or to perform organizational maintenance such as cleaning or replacing fuses, pilot lights, batteries, or other minor hardware items, an identification tag such as an AFTO Form 350 shall be attached noting the reason, date, and OWC supervisor's signature. Recertification shall not be accomplished if calibration accuracy is not in question.

## **5.12 AFTO FORM 45, REQUEST FOR CALIBRATION RESPONSIBILITY DETERMINATION.**

### **NOTE**

Ensure the *Request for Calibration Responsibility Determination*, AFTO Form 45 (see Figure 5-10), is **not** submitted for equipment owned and used by departments outside the DAF (see par 4.9 for TMDE considered leased/borrowed). Such a customer is responsible to give the PMEL the capability to support new equipment, if the PMEL does not already have it.

The PMEL, with the assistance of the user of new TMDE, shall initiate the *Request for Calibration Responsibility Determination*, AFTO Form 45 (see Figure 5-10), to ensure that the AFMETCAL Program can support the particular item of TMDE. This information is required to enable the PMELs, AFMETCAL and the inventory managers to obtain support equipment and write any calibration procedures for calibration and maintenance of the item. If inadequate data are submitted with the form and determination of responsibility cannot be made, the request may be returned. Many TMDE items also have accessories that require calibration; therefore, when submitting requests for calibration determinations, include data on any accessories that may require calibration so they can be included in the calibration determination. Electronic input of the AFTO Form 45 and supporting tech data through METWEB is preferred. Alternatively, the initiating PMEL shall complete the top portion of the form and forward the form to AFMETCAL or email to [AFTO22.45@afmetcalaf.mil](mailto:AFTO22.45@afmetcalaf.mil) (See METWEB "TO SUPPORT HOME" page). The PMEL shall retain a copy. When received, AFMETCAL shall complete the form (normally within 60 days of receipt of all required information) and make a copy available electronically on METWEB. Blocks marked with an asterisk (\*) are mandatory and shall be completed.

**5.12.1 PMEL PORTION Blocks.** The blocks in the request portion of the form shall be completed by the PMEL, with the assistance and coordination of the user (see AFI 23-101 and AFMAN 23-122), in the following manner:

- a. **MODEL/PART NUMBER/TYPE.** (Mandatory) Enter the manufacturer's model designation, part number or similar data. Provide all data available.
- b. **MANUFACTURER.** (Mandatory) Enter the name of the manufacturer of the item.
- c. **SERIAL NUMBER.** (Mandatory) Enter serial number of the item.
- d. **NOMENCLATURE.** (Mandatory) Enter nomenclature of item.
- e. **APPLICABLE MAINTENANCE MANUAL.** Identify the maintenance TO that applies to the unit.
- f. **NATIONAL STOCK NUMBER.** (Mandatory) Enter the national stock number of the item. If not available, enter date stock number was requested.
- g. **QUANTITY.** Enter the quantity to be supported by the PMEL, if known.
- h. **WEAPON SYSTEM(s) OR END ITEM SUPPORTED.** (Mandatory) Identify the system or end item being supported. (e.g., F-16C or AN/ALQ-131V3)
- i. **ALLOWANCE SOURCE CODE.** (Mandatory) Enter the allowance source code (ASC) corresponding to system or item supported that was used to obtain this item. If none, explain in remarks the specific authority used to obtain the item (Maintenance TO, commercial data, etc.).
- j. **LOCAL PURCHASE.** (Mandatory) Check the block for items locally purchased. The user is responsible for providing the PMEL any items required for maintenance and calibration of any item locally purchased. Before

buying any items locally, the potential owner shall contact the PMEL to determine if PMEL support will require any additional equipment, supplies or data. See user responsibilities in Section 3.

- k. WORK STOPPAGE. Check if this refers to work stoppage due to a lack of this item, not the inability to calibrate this item.
- l. ITEM ON HAND. Check if the item is on hand in the PMEL or is available on call. Use the PROJECTED CAL DUE DATE to inform AFMETCAL of the month and year the calibration procedure will be needed.
- m. PROJECTED CAL DUE DATE (MO/YR). Enter projected date the item is expected to require calibration again.
- n. SUPPORTING PMEL / PHONE / ADDRESS / BASE / COMMAND. (Mandatory) Enter the name, phone, address, and command of the PMEL submitting the request
- o. USER NAME / PHONE / EMAIL / OWC/AGENCY / CMD/ORG / BASE/CITY. (Mandatory) Enter the user's point of contact name, phone, and email. Also enter the owning work center code, agency, command, organization, base, and city.
- p. NAME / PHONE / EMAIL OF REQUESTING INDIVIDUAL. Enter name, Defense Switched Network (DSN) or commercial telephone number, and email address of PMEL point of contact in case further data is necessary.
- q. SUPPORTING DATA SUBMITTED. (Mandatory) Check the appropriate block. Identify specific items of commercial data submitted along with this form in the REMARKS block. If you reference a website, provide the site address in the REMARKS block. AFMETCAL needs all available data to make the responsibility determination. It is also needed to write an adequate calibration procedure if the request results in a determination that the PMEL is responsible for calibration and repair of the item. If enough data is not submitted, a calibration responsibility determination cannot be made, and the request shall be returned as "INSUFFICIENT DATA" without a final determination. AFMETCAL shall ensure that all data are returned to the PMEL/user as soon as possible. SUPPORTING DATA RECEIVED and DATE DATA RECEIVED will be filled in by AFMETCAL once the form and data arrive.
- r. REMARKS. One type of comment to include here is a calibration responsibility determination that the PMEL or user would recommend and include your reasons for the recommendation. Your PMEL knows who can best support the item. Include remarks regarding the actual measurement to be made, the owning organization, its capability to support the item, and what system the equipment is to be used on. Include any other comments the PMEL thinks may help the evaluator make the calibration determination. Include applicable work unit code(s) (WUC) whenever they exist for that part number.
- s. REQUEST DATE. Enter the date of the request.

**5.12.2 AFMETCAL PORTION Blocks.** AFMETCAL personnel shall complete the following blocks:

- a. AFMETCAL CONTROL #. Enter the applicable control number.
- b. OPR ORG. Enter the organizational symbol for AFMETCAL office that shall serve as OPR for this request.
- c. OPR NAME / PHONE / EMAIL. Enter the name, DSN or commercial telephone numbers, and email address of the person that shall be OPR for this request.
- d. DATE RECEIVED. Enter date received by the OPR ORG.
- e. RESPONSIBILITY DETERMINATION. Check the block or fill in the responsibility determination.
- f. NEW PROCEDURE REQUIRED. Check the appropriate block.
- g. WUC IS. Enter the assigned WUC.
- h. CALIBRATION INTERVAL IS. Enter the initial calibration interval for this part number or model.
- i. CALIBRATION AUTHORITY IS. Enter the calibration authority if calibrated.
- j. MAINTENANCE MANUAL IS. Enter one, if one exists. It may provide useful data.
- k. REMARKS. This block is used to explain rationale behind the entry in one of the previous two blocks. Give specific reasons why a determination cannot be made or explain why a specific determination was made.
- l. ITEM. Check appropriate blocks.
- m. NON-TMDE. Check if item is not TMDE according to definition in TO 00-20-14 and is not PMEL responsibility. If this block is checked, the item under consideration is the responsibility of the USER for support.
- n. SUPERVISOR. Written name of supervisor of person called out above.
- o. RESPONSE DATE. Enter the date the calibration responsibility determination is made.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                    |                                                     |  |                                                                                    |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--|------------------------------------------------------------------------------------|--|
| <b>REQUEST FOR CALIBRATION RESPONSIBILITY DETERMINATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |                                                     |  | OMB NUMBER<br>0704-0188                                                            |  |
| Public reporting burden for this collection of information is estimated to average 6 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information including suggestions for reducing this burden to Department of Defense, Washington Headquarters Services, DIOR, 1215 Jefferson Davis Highway, Suite 1204, Arlington VA 22202-4302 and to OMB, Paperwork Reduction Project (0704-0188), Washington DC, 20503. Please DO NOT RETURN your form to either of these addresses. Send your completed form to 562 CMSG/GBSA, 813 Irving-Wick Dr W, Bldg 2, Heath Ohio 43056-1199. |                                                                                                                                                                    |                                                     |  |                                                                                    |  |
| <b>PMEL PORTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                    |                                                     |  | <b>(*) MANDATORY</b>                                                               |  |
| MODEL/PART NUMBER/TYPE*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                    | MANUFACTURER*                                       |  | SERIAL NUMBER*                                                                     |  |
| NOMENCLATURE*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                    |                                                     |  | APPLICABLE MAINTENANCE MANUAL*                                                     |  |
| NATIONAL STOCK NUMBER*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | QUANTITY                                                                                                                                                           | WEAPON SYS(s) OR END ITEM SUPPORTED*                |  | ALLOWANCE SOURCE CODE*                                                             |  |
| LOCAL PURCHASE*<br><input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | WORK STOPPAGE<br><input type="checkbox"/>                                                                                                                          | ITEM ON HAND<br><input type="checkbox"/>            |  | PROJECTED CAL DUE DATE (MO/YR)                                                     |  |
| SUPPORTING PMEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                    |                                                     |  | USER                                                                               |  |
| PHONE*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                    | NAME*                                               |  |                                                                                    |  |
| ADDRESS*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                    | PHONE*                                              |  |                                                                                    |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                    | EMAIL*                                              |  |                                                                                    |  |
| BASE*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                    | OWC/AGENCY*                                         |  |                                                                                    |  |
| COMMAND*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                    | CMD/ORG*                                            |  |                                                                                    |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                    | USER BASE/CITY*                                     |  |                                                                                    |  |
| REQUESTING INDIVIDUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                    | SUPPORTING DATA SUBMITTED* <input type="checkbox"/> |  |                                                                                    |  |
| NAME*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                    | SUPPORTING DATA RECEIVED* <input type="checkbox"/>  |  |                                                                                    |  |
| PHONE*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                    | DATE DATA RECEIVED*                                 |  |                                                                                    |  |
| EMAIL*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                    |                                                     |  |                                                                                    |  |
| REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                    |                                                     |  |                                                                                    |  |
| MAIL TO:<br>562 CMSG/GBSA, 813 IRVING-WICK DR W, BLDG 2, HEATH OHIO 43056-1199                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                    |                                                     |  | REQUEST DATE*                                                                      |  |
| <b>562 CMSG PORTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                    |                                                     |  |                                                                                    |  |
| OPR NAME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                    | PHONE                                               |  |                                                                                    |  |
| OPR ORG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                    | EMAIL                                               |  |                                                                                    |  |
| CONTROL NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                    | DATE RECEIVED                                       |  |                                                                                    |  |
| RESPONSIBILITY DETERMINATION IS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <input type="checkbox"/> CANNOT BE MADE DUE TO INSUFFICIENT TECHNICAL INFORMATION. A DETERMINATION WILL BE MADE IF YOU SUBMIT THE NEEDED INFORMATION. SEE REMARKS. |                                                     |  | NEW PROCEDURE REQUIRED<br><input type="checkbox"/> YES <input type="checkbox"/> NO |  |
| WUC IS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CALIBRATION INTERVAL IS                                                                                                                                            | CALIBRATION AUTHORITY IS                            |  | MAINTENANCE MANUAL IS                                                              |  |
| REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                    |                                                     |  |                                                                                    |  |
| ITEM <input type="checkbox"/> WILL BE <input type="checkbox"/> WILL NOT BE <input type="checkbox"/> ALREADY IS <input type="checkbox"/> LISTED IN T.O. 33K-1-100 <input type="checkbox"/> LISTED IN CMS _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                    |                                                     |  |                                                                                    |  |
| NON-TMDE <input type="checkbox"/> ITEM IS NOT TMDE ACCORDING TO DEFINITION IN TO 00-20-14, DOES NOT REQUIRE CALIBRATION AND IS "USER" RESPONSIBILITY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                    |                                                     |  |                                                                                    |  |
| SUPERVISOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                    |                                                     |  | RESPONSE DATE                                                                      |  |

AFTO FORM 45, 2007XXXX

PREVIOUS EDITION IS OBSOLETE

Figure 5-10. AFTO FORM 45, Request For Calibration Responsibility Determination.

**5.13 AFTO FORM 57, PMEL TECHNICAL ORDER INFORMATION.**

The AFTO Form 57 (see Figure 5-11) is for the PMEL to record data relative to a calibration procedure that pertains to their peculiar situation. This form is not required with each TO. It is to be completed to the degree (and when) PMEL personnel feel is necessary or as needed. Also, it shall not be used in place of a recommended change action. The following guidance is provided for entries on this form:

**NOTE**

The technician shall verify information entered on the AFTO Form 57 prior to using such information.

- a. Block 1, MAINTENANCE TO INFORMATION. Enter the part number and maintenance TO for the different models or part numbers covered by the procedure in the appropriate blocks.
- b. Block 2, ADDITIONAL ITEMS REQUIRED FROM THE OWC. Enter any additional ancillary items or equipment that must be supplied by the OWC to facilitate repair or calibration of the TMDE entered in Block 1.
- c. Block 3, LOCATION OF TEST FIXTURES. Enter the PMEL storage location of test fixtures required for repair or calibration of the TMDE entered in Block 1.
- d. Block 4, SUITABLE SUBSTITUTES FOR STANDARDS REQUIRED IN CALIBRATION PROCEDURES. Enter the part number of all TMDE listed in the procedure that is not locally available and the part number of the substitute standard in the appropriate blocks.

**NOTE**

Ensure required uncertainty ratios are maintained when substituting standards during calibration.

- e. Block 5, AFTO FORM 22 INFORMATION. Enter all recommended change information as necessary to identify deficiencies in the procedure. Enter date submitted, paragraph affected, and name of technician submitting the recommended change as a minimum.

**NOTE**

Approval of a recommended change is not justification to deviate from published technical data IAW TO 00-5-1.

- f. Block 6, ALTERNATE PROCEDURE INFORMATION. Enter any information that shall allow the technician to locate or perform approved alternate procedures.
- g. Block 7, PECULIARITIES OF TMDE OR CALIBRATION PROCEDURE. Enter any information that will assist the technician during calibration of the TMDE.
- h. Block 8, HIGH FAILURE COMPONENTS. Enter any components that are considered high failure items or components that are difficult to identify (Part Number and NSN) in the appropriate blocks.

**5.13.1 Continuation Sheets.** Continuation sheets can be attached as necessary.

| PMEL TECHNICAL ORDER INFORMATION                  |                                |                                                                          |                                |
|---------------------------------------------------|--------------------------------|--------------------------------------------------------------------------|--------------------------------|
| 1. MAINTENANCE TECHNICAL ORDER INFORMATION        |                                |                                                                          |                                |
| A. PART NUMBER                                    | B. MAINTENANCE TECHNICAL ORDER | A. PART NUMBER                                                           | B. MAINTENANCE TECHNICAL ORDER |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
| 2. ADDITIONAL ITEMS REQUIRED FROM THE OWNING WORK |                                |                                                                          |                                |
| 3. LOCATION OF TEST FIXTURES                      |                                | 4. SUITABLE SUBSTITUTES FOR STANDARDS REQUIRED IN CALIBRATION PROCEDURES |                                |
|                                                   |                                | A. TMDE CALLED FOR                                                       | B. SUBSTITUTE ITEM             |
|                                                   |                                |                                                                          |                                |
| 5. AFTO FORM 22 INFORMATION                       |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
| 6. ALTERNATE PROCEDURE INFORMATION                |                                | 7. PECULIARITIES OF TMDE OR CALIBRATION PROCEDURE                        |                                |
|                                                   |                                |                                                                          |                                |
| 8. HIGH FAILURE COMPONENTS                        |                                |                                                                          |                                |
| A. PART NUMBER                                    | B. NOMENCLATURE                | C. NATIONAL STOCK NUMBER                                                 |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |
|                                                   |                                |                                                                          |                                |

AFTO FORM 57, 19960301 (EF-V2)

Figure 5-11. AFTO FORM 57, PMEL Technical Order Information.





**SECTION 6**  
**RESERVED**



## SECTION 7

### PMEL ASSESSMENT PROGRAM

#### 7.1 GENERAL.

As directed by AFMAN 21-113, AFMETCAL maintains a system to assess calibration laboratories for effectiveness and the capability to perform measurements that are safe, accurate, reliable, and traceable through the AFPSL to the International System of Units through NIST or other AFMETCAL approved sources. The AFMETCAL Assessment Program shall not be, or perceived to be, a substitute for government oversight of contracted PMELs. Contract compliance may require analysis of assessment documents; however, determinations of effectiveness are exclusive to the AFMETCAL Assessment Program.

#### 7.2 ASSESSMENT INTERVALS.

The Evaluation Team will perform an on-site assessment in accordance with the AF Inspection System, AFMAN 21-113, and the PMEL Assessment Program defined in this TO. This may align with the MAJCOM/IG Unit Effectiveness Inspection Capstone, Continual Evaluation, or other Site Visits.

#### 7.3 ASSESSMENT NOTIFICATION.

Notifications will be delivered in accordance with AFI 90-201 and applicable MAJCOM requirements. Assessment notifications will be issued approximately 15-45 calendar days prior to an assessment.

#### 7.4 ASSESSMENT PROCESS.

There are two stages in the PMEL assessment process: Virtual Assessment and On-site Assessment.

**7.4.1 Virtual Assessment.** Virtual Assessments are conducted to add the critical no-notice element to the assessment process. Laboratories and MCFs will not be notified of the virtual assessment. This assessment may be accomplished at any point between on-site assessments or just prior to an on-site assessment. Findings will be forwarded to the appropriate MFM and will be used to feed the risk-based, on-site sampling strategy. The findings of any virtual assessment completed prior to an on-site assessment will not be forwarded until the completion of the on-site assessment.

**7.4.1.1** Virtual assessment data shall be uploaded and updated to the designated AFMETCAL SharePoint site as outlined in the Management Internal Control Toolset (MICT) self-assessment communicator titled AFMAN 21-113 Air Force Metrology and Calibration Program. Any additional virtual assessment data shall be requested through the respective PMEL and MFM as needed. The requested virtual assessment data may include, but is not limited to:

- a. Current QM. If a significant portion of the QM information is sub-located in another policy document or continuity book, include those documents with the virtual assessment data.
- b. Master inventory data. This data shall be provided with the following headings in spreadsheet format (not required if using PAMS):
  - (1) ID Number
  - (2) Part Number
  - (3) Work Unit Code
  - (4) Calibration Measurement Summary identifier
  - (5) Noun/nomenclature
  - (6) Date Calibrated
  - (7) Date Due Calibration
  - (8) Calibration Interval
  - (9) Certifying Technician K-Stamp Number
  - (10) Owning Work center
  - (11) Performing Work center
  - (12) K-Area
  - (13) Calibration Technical Order number or Technical Data identifier
  - (14) Status (AWM, INW, AWP, Awaiting pickup, etc.)
  - (15) Locally Calibrated Equipment Owned by the PMEL
- c. Copies of root cause analysis for all PNCs and critical QNCs, with any resulting corrective actions and follow-up documentation.
- d. A listing by K-Stamp number of the technicians qualified in each measurement area (if not using PAMS).

- e. Management Review reports with resulting corrective actions and follow-up documentation since previous assessment.
- f. Previous 12 months of PAS Data (if not using PAMS).

**7.4.2 On-site Assessment.** All PMEL MS, QP and facility/ECS documentation since the last assessment shall be available on-site.

**7.4.2.1** The on-site assessment shall assess and/or validate the following criteria:

- a. Compliance and effectiveness of management system requirements.
- b. Compliance and effectiveness of quality program requirements.
- c. Compliance with facility and environmental requirements.
- d. If applicable, corrective actions resulting from a previous assessment.

**7.4.2.2** Measurement Capability Assessment (MCA) is the process that will be used to evaluate the effectiveness of both the quality program and management system. This shall include, but not be limited to, processes for calibrating TMDE; application of metrology principles/procedures; and condition/use of TMDE. The assessment team shall sample processes and items based upon supported inventory, scope of calibrations performed and defined measurement area disciplines (MADs). Each TFCU, JETSC, RASCAL, and on-site operation shall also be subject to assessment.

**7.4.2.3** The assessment team will outline the assessment plan and the certification recommendations to PMEL personnel. The assessment team will provide an executive summary out-brief to the owning Group Commander, equivalent or designee. On-site assessment result briefs are tentative until the final assessment report is published by the AFMETCAL Director.

**7.4.3 Assessment Criteria.** The following criteria shall be used to determine certification status.

- a. MS. The MS shall meet all requirements of this TO. The PMEL management demonstrates sound decision-making ability within the MS, during the MCA, and through participation in the PT/MAP that supports their specified local mission and the PMEL Repair Network.
- b. QP. The QP shall meet all requirements of this TO. The PMEL QP demonstrates technical capability and proficiency to evaluate measurement area disciplines within QP, during the MCA and participation in the PT/MAP that supports their specified local mission and the PMEL Repair Network.
- c. Facility and ECS. The facilities and ECS shall comply with operational requirements of this TO and applicable DAF directives.

**7.4.4 Assessment Results.** Findings shall be compiled from Virtual and On-site Assessments to determine the capability and effectiveness of a PMEL to perform safe, accurate, reliable, and traceable measurements. The compiled results are reported to the AFMETCAL Director and/or a designated appointee for certification determination. The AFMETCAL Director and/or designated appointee maintains responsibility of the final determination.

## **7.5 CERTIFICATION STATUS.**

At the completion of an on-site assessment each PMEL will receive two distinct grades: the certification of compliance grade and an effectiveness grade.

**7.5.1** The certification of compliance grade shall be specified in the final assessment report as one of the following categories:

**7.5.1.1 Certified.** A PMEL whose MS, QP, facilities and ECS demonstrate the PMEL has the capability to support their specified local mission and the PMEL Repair Network shall be issued a Certificate of Compliance at the conclusion of the onsite assessment. The final assessment report and the certificate are proof of compliance with this TO and other applicable DAF directives. The Certificate of Compliance shall remain in effect unless revoked by AFMETCAL. All previous certificates are void when final certification status is certified, withheld or recommended for closure. Whenever a significant change occurs within the PMEL, the 'Certified' status shall be reviewed by the PMEL's MFM and AFMETCAL to ensure certification requirements can be maintained. Such changes are: if the contractor changes, the type of manning changes (i.e., civil service, military, etc.) or the PMEL relocates to a different facility (facility may include RASCAL units and on-site operations used as an extension of the main PMEL). AFMETCAL certification is proof of AFMETCAL Program compliance and should not be confused with third-party accreditation under commercial guidelines.

**7.5.1.2 Certification Pending.** The PMEL failed to meet some assessment criteria, but with continued action or minor improvement has the capability to support their specified local mission and the PMEL Repair Network. A previously issued Certificate of Compliance remains in effect during this period. Certification will remain pending until improvements have been made and validated by the MFM, the Assessment Team performs a reassessment, and the AFMETCAL Director approves certification. Additional actions may be required as a result of this certification status. The final assessment report will detail the required actions (i.e., CAP, reassessment type, timeline). A final status of pending will change to withhold if the required data or actions are not accomplished.

**7.5.1.3 Certification Withheld.** The PMEL failed to demonstrate the capability to support and posed a significant risk to their specified local mission and the PMEL Repair Network. A previously issued Certificate of Compliance is no longer valid. However, calibrations may continue only with increased local and MAJCOM surveillance of the problems or measurement areas provided the level of risk is evaluated and accepted. Additionally, on-site visits from AFMETCAL may be required to re-assess the PMEL.

**7.5.1.4 Not certified, recommended for closure.** Problems in MS, QP, facilities and ECS prevent the PMEL from making safe, accurate, reliable, and traceable measurements; resolution is not expected.

**7.5.2** The effectiveness grade shall be specified in the final assessment report as one of the following categories:

**7.5.2.1 Highly Effective.** A rating of Highly Effective can only result from a Certified status. This rating indicates the PMEL meets/exceeds the criteria for an effective rating and most or all of the following are consistently true:

- a. The PMEL has increased capability to perform safe, accurate, reliable, and traceable measurements in support of their specified local mission and of the PMEL Repair Network.
- b. Widespread evidence exists of high proficiency.
- c. Performance or operations exceed expectations and mission requirements.
- d. Resources and programs are efficiently managed and mostly deficiency-free.
- e. The facility and ECS are compliant and local efforts have been made to increase efficiency and effectiveness.
- f. Programs and processes are nearly deficiency-free and produce highly reliable results.
- g. The MCA presented low risk in the measurement area disciplines and efficient mission or PMEL repair network support.
- h. The QP is institutionalized, used to measure and report improvements within the program, and provides actionable feedback to management.
- i. Root-Cause Analysis is often used to find and fix problems.
- j. An effective Management System is in place with clear indications of management support, planning, use of risk management, and continuous improvement efforts at all levels.
- k. Risk-based criteria are often applied when making decisions within the MS.
- l. The PMEL has embraced a culture of critical self-assessment. Problems are identified, management is aware of the issues and solid corrective action plans are in place.
- m. Processes in place are mature and well established. Deficiencies are rare.

**7.5.2.2 Effective.** A rating of Effective can only result from a Certified status. This rating indicates most of the following are generally true:

- a. The PMEL has the capability to perform safe, accurate, reliable, and traceable measurements in support of their specified local mission and of the PMEL Repair Network.
- b. Personnel are proficient.
- c. Performance or operations meet expectations and mission requirements.
- d. Resources and programs are managed with few deficiencies.
- e. The facility and ECS are compliant or mitigating procedures are in place.
- f. Critical processes are documented, measured and repeatable.
- g. The MCA presented few significant deficiencies in the measurement area disciplines and did not present an unacceptable mission or repair network degradation.
- h. The QP provides an accurate, adequate, and relevant picture of performance.
- i. Root-Cause Analysis is usually used to find and fix problems.
- j. The Management System is present and continuous process improvement efforts are evident.
- k. Risk-based criteria are often considered when making decisions within the MS.
- l. The PMEL has embraced a culture of self-assessment. Problems are identified, management is aware of the issues and corrective action is attempted.

- m. Processes in place are sufficient to meet mission requirements. Deficiencies may exist, but do not impede or limit mission accomplishment.

**7.5.2.3 Marginally Effective.** A rating of Marginally Effective can result from a Certified or Certification Pending status and may require a CAP as directed by the final report. This rating indicates the PMEL does not meet the criteria for an effective rating, and/or most of the following are consistently true:

- a. The PMEL has limited capability to perform safe, accurate, reliable, and traceable measurements in support of their specified local mission and of the PMEL Repair Network.
- b. Personnel have limited proficiency.
- c. Performance or operations are limited in meeting expectations and mission requirements.
- d. Resources and programs are insufficiently managed with some deficiencies.
- e. The facility and ECS are not compliant, and the impact presents moderate risk to mission.
- f. Processes are not well-established causing inconsistent results.
- g. The MCA presented deficiencies in the measurement area disciplines and presented a moderate risk to the mission or PMEL repair network.
- h. The QP provides an accurate (though limited) picture of performance.
- i. Little effort is given to using Root-Cause Analysis to find and fix problems.
- j. The Management System has some elements but is not working in a cohesive process. Little to no evidence exists of continuous process improvement efforts.
- k. Risk-based criteria is sometimes considered when making decisions within the MS.
- l. The PMEL has not embraced a culture of critical self-assessment. Problems are not routinely identified; management is not always aware of the issues and/or corrective action is not sufficient.
- m. Processes in place limit the ability to meet mission requirements. Deficiencies exist which may impede or limit mission accomplishment.

**7.5.2.3 Ineffective.** A rating of Ineffective can result from a Certification Pending or Certification Withheld status and requires a CAP as directed by the final report. This rating indicates the PMEL is not effective and does not meet criteria for a marginally effective rating and/or most of the following are consistently true:

- a. The PMEL does not demonstrate the capability to perform safe, accurate, reliable, and traceable measurements in support of their specified local mission and of the PMEL Repair Network.
- b. Personnel have inadequate proficiency.
- c. Performance or operations do not meet expectations and mission requirements.
- d. Resources and programs are not well managed and significant deficiencies exist.
- e. Programs and processes are not well-established causing risk to the mission.
- f. The facility and ECS are not compliant, mitigating procedures are insufficient or non-existent, or the impact presents high risk to mission.
- g. The MCA presented significant deficiencies in the measurement area disciplines and a high risk to the mission or repair network.
- h. The QP does not provide an accurate, adequate, or relevant picture of performance.
- i. No effort is given to using Root-Cause Analysis to find and fix problems.
- j. The Management System is not evident or is unproductive. No evidence exists of continuous process improvement efforts.
- k. Risk-based criteria is rarely considered when making decisions within the MS.
- l. The PMEL has not embraced a culture of critical self-assessment. Problems are not identified; management is not aware of the issues and/or corrective action is not sufficient.
- m. Processes are not carried out in a competent manner or produce unreliable results.
- n. The number and severity of deficiencies preclude or seriously limit mission accomplishment and increase risk to operations. Evidence exists of systemic non-compliance, widespread disregard for prescribed procedures or inadequate proficiency of personnel.

**7.5.3 Corrective Action Plan (CAP).** The final assessment report will dictate if a CAP is required, regardless of Certification Status. If directed, the CAP shall include detailed milestones. The CAP shall be delivered to the MFM within 45 days from the date of the final assessment report. The CAP may be coordinated with the AFMETCAL Evaluation Team, but the MFM shall approve the finalized CAP and send a copy to the AFMETCAL Evaluation Team. The MFM may request additional data as needed to monitor CAP milestones until the PMEL is 'Certified'. The MICT SAC shall be updated to reflect the current certification status and updated accordingly to reflect CAP progress.

**7.5.3.1 CAP Timeline/Milestones.** If a CAP is required, all CAP milestones shall be completed within 6 months from the final certification assessment report. Extension requests beyond this timeline will be coordinated through the MFM and AFMETCAL Evaluation Team for consideration. CAP timeline/milestone extension requests beyond 12 months will only be approved under extenuating circumstances and shall be coordinated with and approved by the MFM.

**7.5.4 Certification Reassessment.** The final assessment report following a Certification Pending status will dictate the type of reassessment (virtual/on-site) forecasted. When deficiencies have been remedied and validated by the MFM, the MFM will notify the Assessment Team the PMEL is ready for reassessment. The Evaluation Team schedules/performs a virtual or on-site reassessment for validation and provides status recommendation to the AFMETCAL Director.

## **7.6 ASSESSMENT REPORTS.**

The assessment team shall prepare and distribute a report for each assessment.

**7.6.1 Report Content.** The report shall concisely and clearly describe the compliance and effectiveness of the MS, QP, and Facility/ECS.

**7.6.2 Report Distribution.** AFMETCAL shall distribute a copy of the final report to the responsible unit commander, PMEL, MFM, and IG, normally within 45 days after the end of the on-site assessment. An electronic copy of all final reports shall be maintained at AFMETCAL.

## **7.7 APPEAL PROCESS.**

The owning MFM may appeal the final AFMETCAL Certification Status to AFMETCAL within 30 days of the final report. The AFMETCAL Director may convene a Certification Panel for advice. The subsequent decision of the AFMETCAL Director is final.





## SECTION 8

### PMEL FACILITY OPERATIONAL REQUIREMENTS

#### NOTE

This section does not contain PMEL design criteria. They are contained FC 4-218-01F, *Air Force Criteria for Precision Measurement Equipment Laboratory Design and Construction*, DAFMAN 32-1084, *Standard Facility Requirements*, and AFI 32-1024, *Standard Facility Requirements*. PMEL managers and their senior leadership should encourage local civil engineering to coordinate with/seek assistance from Air Force Civil Engineer Center (AFCEC) Field Support (CEM) Civil Engineer Maintenance, Inspection and Repair Team (CEMIRT) in the early design stages of any PMEL facility or environmental control system construction, modification, or upgrade project. Support can be requested via the CEMIRT Gateway at <https://portal.afcec.hedc.af.mil/CO/cemirtgateway/SitePages/Home.aspx>.

#### 8.1 GENERAL REQUIREMENTS.

This section contains PMEL facility operating parameters and/or notes concerning environment, power, lighting, air filtration, air locks, controlled areas, cleaning area, flooring, ceilings, and other subjects. One purpose of this section is to ensure that PMELs placed in buildings not built in accordance with FC 4-218-01F and DAFMAN 32-1084 meet minimum requirements. This section also contains some of the general information needed to determine space requirements within some of the laboratory areas. The data in this section provides guidance for the day-to-day operation of the PMEL as it relates to the facility. PMEL management shall make every effort to establish and maintain the PMEL facility in a condition facilitating proper performance of calibrations/verifications.

**8.1.1 Design Criteria.** FC 4-218-01F and DAFMAN 32-1084 contain the design criteria for Type II() and Type III PMELs on DAF installations and shall be complied with when modifying or repairing an existing PMEL or designing a new PMEL facility. PMEL designs and layouts shall be evaluated on an individual basis and shall depend upon mission requirements. These criteria shall not be used as sole justification to improve facilities if conditions of existing facilities do not adversely affect the environmental requirements or mission performance of the PMEL. Assistance for determining facility requirements may be obtained from the AFMETCAL Plans & Analysis Section. Each PMEL shall maintain a file reflecting any deviation from FC 4-218-01F or DAFMAN 32-1084, which was approved by AFMETCAL and/or HQ AFCEA.

#### 8.1.2 Coordination of Plans for Facility Projects.

**8.1.2.1** When any PMEL or major command first decides to build a new PMEL or modify an existing PMEL, AFMETCAL Plans & Analysis Section shall be provided all justification available so AFMETCAL can update long range planning and ensure interservice coordination of applicable projects. AFMETCAL shall forward appropriate comments and recommendations to the command headquarters.

**8.1.2.2** When a project is approved for construction of a new facility or modification/improvement of an existing facility, the PMEL manager shall ensure that the command's PMEL functional manager is placed on distribution and provided with a copy of the design criteria, specifications, and drawings.

**8.1.2.3 PMEL Power Requirements.** PMEL power requirements are listed in FC 4-218-01F. Equipment, which is critically sensitive to line voltage variation, shall require additional voltage regulation through use of bench type regulators and/or filters.

**8.1.2.4 Grounding Requirements.** The ground in the PMEL shall be checked to ensure it has retained its validity per AFMAN 32-1065. Measurement techniques for grounds are shown in TO 31-10-24. PMEL management shall maintain a copy of inspection forms provided by Civil Engineering IAW AFMAN 32-1065.

**8.1.2.5 Geophysical Surveys.** Most PMELs require certain geophysical surveys to be performed. AFI 14-205 gives information about obtaining these surveys. These surveys are only required when any of the following sub-paragraph measurement capabilities are required. Surveys are as follows:

- a. World Geodetic System of 1984 (only if Loran C is still in use): Geodetic positions, Latitudes and Longitude in degrees, minutes, and seconds. Required in order to perform accurate time measurements. Loran C requires a minimum 10 arc seconds accuracy for latitude and longitude.
- b. International Gravity Standardization Net of 1971 (IGSN 71): Gravity information in milligals (mGals). The required accuracy for use with dead weight piston pressure standards is 5 mGals. As long as the PMEL does not move, there is no need for an update. If your PMEL has moved to a new location, then your force and pressure standards shall be recalibrated using the new gravity value before they can be used to obtain maximum accuracy.
- c. Astronomic Data. Astronomic azimuth in degrees, minutes, and seconds. Needed if the PMEL is supporting north-seeking gyrocompass units. Accuracy needed is 10 arc seconds or less.

## **8.2 PMEL ENVIRONMENT.**

TO 33K-1-100-2 identifies the measurement restrictions of measurement standards. To ensure measurement accuracy, it is necessary to maintain continuous control of the environment in which the measurement standards are maintained. Temperature, humidity, dust, vibration, and electromagnetic interference can each have a detrimental effect on the ability of a standard to ensure measurement accuracy of TMDE. Each of the environmental factors impact calibration and repair operations to a different degree. It is a fact that most standards change value with a change in temperature. The temperature limits placed on the PMEL ensure that the measurement confidence level meets the needs of some of the more precise equipment. Humidity control is becoming a very significant environmental need in the PMELs, especially at the lower humidity levels because of the potential for electrostatic discharge (ESD) damage to integrated micro circuitry, as well as its interference with sensitive measurements. Therefore, a PMEL whose environment goes outside the limits for temperature and humidity cannot ensure calibration accuracy and/or presents an operating hazard to some element of the measurement setup. The calibration procedures may provide other restrictions.

**8.2.1 PMEL Environmental Control System (ECS) Effectiveness.** The ECS shall maintain the laboratory environment so that each room used for calibration meets requirements at least 90% of the time, over the previous 12-month period. When the PMEL ECS cannot maintain environmental specifications 90% of the time or more, aggressive action on the part of PMEL/upper-level management is required. Excessive out-of-tolerance periods can contribute to equipment degradation, production slow-downs or stoppages. PMELs may seek assistance from the Civil Engineering Maintenance Inspection Repair Team (CEMIRT) through their local Civil Engineering for assistance with long-term out-of-tolerance conditions.

**8.2.2 ECS Outages.** Laboratory management shall ensure nothing listed as environmentally sensitive is to be calibrated during environmental outages, unless special provisions are made, documented, and the documentation kept on file. An Environmental Outage Log\* shall summarize environmental data for each environmental monitor to include:

- a. Total hours recorded.
- b. Total hours temperature above applied limits.
- c. Total hours temperature below applied limits.
- d. Total hours relative humidity above applied limits.
- e. Total hours relative humidity below applied limits.
- f. Total hours overlap (both temperature and relative humidity outside limits).
- g. Total hours of missing data (e.g., dry pens, clock stopped, chart overrun).

\*Not required if recorded in PAMS Environmental Log or IAW TO 33K5-4-9002-10.

**8.2.3 PMEL Operational Environment.** The following operational environmental requirements apply only to a PMEL area where calibration/repair is performed. Particular care shall be taken when calibration/repair is performed at a site other than the permanent PMEL premises. Keep in mind these are operational requirements, not design requirements.

**8.2.3.1** Present measurement needs dictate the operational temperature in the PMEL facility shall be controlled as follows, unless the calibration procedure specifies other restrictions:

**8.2.3.1.1** Calibration and Repair area. A temperature of 73°F ±6°F (22.8°C ±3.3°C) in the calibration and repair areas of Type II( ) or III PMELs, except for the 68°F dimensional calibration area.

**8.2.3.1.2** Dimensional 68°F Calibration and Repair area. A temperature of 68°F ±1.0°F (20°C ±0.56°C) shall be maintained in the 68°F dimensional area of PMELs where 68°F dimensional areas are authorized (see paragraph 12.4). The 68°F dimensional calibration area criteria is based on the need for a closely controlled environmental area for calibration and use of higher accuracy dimensional TMDE. By international agreement the true size and shape of an object is that which exists at a uniform temperature of 68°F (20°C).

**8.2.3.2** An optimum operational humidity level is 35% RH. The relative humidity shall not be allowed to go lower than 20% RH or exceed 50% RH in any Type II( ) or Type III PMEL.

**8.2.3.3** Temperature and humidity shall be monitored and recorded continuously. Span calibrations are encouraged for recorders used to monitor the environment in calibration areas. The span shall cover the operational requirement of the PMEL. The following rules apply to temperature and humidity recorders:

- a. Recording devices monitoring 68°F calibration and repair areas shall be calibrated a minimum of  $\pm 0.5^\circ\text{F}$ ,  $\pm 5\%$  RH.
- b. Recording devices monitoring 73°F calibration and repair areas shall be calibrated a minimum of  $\pm 2.0^\circ\text{F}$ ,  $\pm 5\%$  RH.
- c. If a computerized system is used to record environmental data, the time between data samples shall not exceed 15 minutes.
- d. Daily or weekly charts shall be used when not using computerized or roll type recording devices.

**8.2.3.3.1** A recording device is required for each 2000 ft<sup>2</sup> of calibration and repair area or per room if smaller than 2000 ft<sup>2</sup>.

**8.2.3.3.2** Recording device(s) shall be located in close proximity to areas where the most stringent calibrations are performed. Locate the recording device so that exterior wall temperature, high heat producing equipment and supply air do not adversely affect the recorder indication.

**8.2.3.3.3** Psychrometer readings (accomplished IAW established calibration procedures for hygrothermographs) shall be accomplished weekly (usually at the time of chart change) to check the accuracy of the recorder. The start and stop dates, standard thermometer reading and psychrometer readings (wet bulb and calculated humidity) shall be annotated on each chart.

**8.2.3.3.4** For those recorders not requiring weekly chart changes (e.g., continuous roll type graph or other electronic recorder), a psychrometer check is only required after any of the following instances:

- a. Quarterly
- b. Graph paper is changed (if the recorder uses graph paper)
- c. If any malfunction is suspected

If logging psychrometer readings on recorders of this type is impractical, a log may be substituted for recording the readings. Should the psychrometer readings indicate a recorder is outside its rated accuracy for either temperature or relative humidity, the recorder shall then be recalibrated. It is also recommended that the environmental charts for each area monitored be filed in separate folders to facilitate periodic review. Span calibrations are encouraged for recorders used to monitor the environment in calibration areas. The span shall cover the operational requirement of the PMEL.

#### NOTE

If the temperature and humidity chart recorder meet required accuracy and is calibrated through the entire span of the PMEL applicable temperature and humidity range, psychrometer readings are not required.

**8.2.3.3.5** Some recorders are slow responding. Differences between the psychrometer and recorder do not necessarily mean adjustments are required. Ensure the psychrometer is used properly and take a number of readings. Adjust the recorder only if repeated comparisons indicate the need.

**8.2.3.4** Because the measurement accuracy of many standards is affected by environmental changes, the environment shall be maintained within the foregoing environmental criteria (limits). This includes periods when the PMEL is not in operation, but items are being "soaked" prior to calibration. When this is not possible, the following applies:

- a. When the environmental system regains control, the PMEL environment shall be allowed to reach stability. Humidity out-of-tolerance conditions do not require a soak time, but metallic surfaces may require cleaning during and after periods of high humidity. The stable condition starts when the temperature excursions remain within the  $73^\circ\text{F} \pm 6^\circ\text{F}$  range ( $68^\circ\text{F} \pm 1^\circ\text{F}$  for the dimensional area) and repeatable measurements can be made. This time is sufficient for all measurements except as follows:
  - (1) Optical Flats (7 inches in diameter and larger): Do not calibrate or use without a "soak" time of 3 hours for every hour the environment is out of tolerance, up to a maximum of 24 hours "soak" time.

- (2) Linear dimensions (1 inch and shorter): For accuracies better than  $\pm 0.0001$  inch, provide a "soak" time of one and one-half hours for each hour the environment is out of specification to a maximum of 12 hours prior to calibration or use of instruments.
- (3) Linear dimensions (longer than 1 inch): For accuracies of  $\pm 0.0001$  inch/inch through  $\pm 0.001$  inch/inch, provide a "soak" time of one and one-half hour for each hour the environment is out of tolerance up to a maximum of 12 hours prior to calibration or use. For accuracies better than  $\pm 0.0001$  inch/inch, provide a "soak time" of three hours for each hour the environment is out of tolerance up to a maximum of 24 hours.
- (4) In each temperature case identified above, the total "soak" time shall be equal to that specified, plus any waiting time that may be required to achieve repeatability of readings.
- b. PMEL operations affected by humidity shall cease as long as the humidity is out of specifications. No trouble shooting/repair shall be performed on ESD vulnerable (CMOS-MOSFETs-MOS) solid-state equipment when the humidity falls below 20% unless the following occurs: The use of ion generators may provide an acceptable environment for a localized area such as a workbench with a low ESD potential. This would permit repair and calibration on the bench even though the rest of the calibration and repair area is below the 20% humidity level. The PMEL shall periodically monitor the operation of ion generators to ensure the ion generator is providing safe ESD environmental conditions. ESD control procedures are described in MIL-HDBK-263, MIL-STD-1686, and TO 00-25-234.

### **8.3 TMDE ENVIRONMENTAL RESTRICTIONS.**

PMELs shall conform to the maintenance TO, calibration TO, or manufacturer's data regarding any environmental limitations or temperature coefficients of any item used in a measurement setup. The calibration TO may state specific environmental conditions which shall be met for the calibration.

### **8.4 DUST LEVELS.**

Dust is an enemy that makes it difficult to make accurate measurements in some measurement areas. The easiest way to keep dust out of the PMELs calibration and repair area is to have a positive airflow from the environmental system and/or locate the PMEL within an area of a building, which is relatively dust-free. Ensuring the filters in the ECS are clean is another major factor in keeping dust levels down. Vacuum cleaning, use of lint free dust cloths, and reduction of horizontal plane surfaces minimizes the effects of dust. Devices producing dust or residue particles from grinding or abrasion shall not be installed in, or used in, the calibration and repair area of the PMEL.

### **8.5 NOISE.**

It is recommended that the noise level in the PMEL be no greater than 70 dB. AFI 48-127, *Occupational Noise and Hearing Conservation Program*, Chapter 3, identifies hazardous noise exposure limits.

### **8.6 VIBRATION.**

The PMEL areas where calibration is performed shall be maintained as vibration free as possible. The cause of vibration, which affects the repeatability of TMDE, shall be identified and reduced to acceptable levels. The PMEL shall notify the AFMETCAL Technology Applications and Electronic Engineering sections of any vibration problems that cannot be solved. Refer to FC 4-218-01F.

### **8.7 ELECTROMAGNETIC INTERFERENCE (EMI)/RADIO FREQUENCY INTERFERENCE (RFI).**

EMI that causes degradation of equipment performance and disruption of calibration shall be suppressed or eliminated to the maximum extent possible.

**8.7.1 New PMEL Facilities.** The effects of EMI to PMEL equipment shall be minimized through proper site selection. EMI from overhead high voltage power lines, radio/TV transmitters and aircraft operations may introduce calibration errors. The EMI limits for a proposed PMEL site from 10 kHz to 26.5 GHz should be no greater than 1 V/m. Proper PMEL layout and construction should reduce internal EMI below 0.1 V/m. When this is not possible, additional shielding precautions/enclosures may be needed within the PMEL should EMI become an issue. An electromagnetic compatibility (EMC) survey shall be performed for each proposed PMEL facility site. The results of the EMC survey shall be used to determine site suitability. The surveys shall be performed by the 85 EIS/SCYM in accordance with AFI 17-221, *Spectrum Interference Resolution Program*. Upon completion of the EMC survey, a formal engineering report shall be supplied to both the PMEL customer and AFMETCAL Plans & Analysis Section.

**8.7.2 Existing PMEL Facilities.** EMI that causes degradation of equipment performance or disruption of calibration at existing sites shall be eliminated or suppressed to the maximum extent possible. The PMEL shall report apparent EMI

problems to the 85 EIS/SCYM in accordance with AFI 17-221. The 85 EIS shall provide on-site assistance in determining the source(s) of the EMI and propose solutions for the elimination or suppression of the EMI. Upon completion of the EMI survey a formal engineering report shall be provided to both the PMEL customer and AFMETCAL Plans & Analysis Section.

**8.7.3 Obtaining Related Assistance.** An EMI or EMC survey shall be obtained through a request to the 85 EIS/SCYM at the following location:

85 EIS/SCYM  
670 Maltby Hall Dr, Ste 234  
Keesler AFB MS 39534-2633

Voice: DSN 597-3926 Comm 228-377-3926  
FAX: DSN 597-3956 Comm 228-377-3956

## **8.8 LIGHTING.**

**8.8.1 General Lighting.** The general lighting for all occupied areas shall be balanced to minimize shadows and produce uniform illumination.

**8.8.2 Minimum lighting.** Minimum lighting levels are specified to reduce the need for additional lighting fixtures in the calibration and repair area because they become dust collectors and can interfere with setting up a measurement capability. Proper illumination is required to perform precision measurements.

**8.8.3 Uniform Illumination.** Uniform illumination is defined as a distribution of light at the measured height where the maximum and minimum general illumination in the immediate work area (e.g., bench top, test set up) is not more than one-sixth above or below the average illumination in the area.

**8.8.4 Measurement of Lighting.** The average room illumination for general lighting shall be measured after 100 hours of use in the calibration and repair area at the level of the horizontal working surface of the PMEL bench and in the administrative areas at desktop level. The surface shall be free of items and adjacent personnel during the measurement to minimize shadows on the light meter sensor. The illumination level can drop drastically during the initial burn-in time.

|                                   | <u>Absolute Minimum</u> |
|-----------------------------------|-------------------------|
| a. Calibration/Repair/Dimensional | 50 Foot Candle (FC)     |
| b. Cleaning                       | 50 FC                   |
| c. Office and Administrative      | 50 FC                   |
| d. Technical Library              | 50 FC                   |
| e. Training Room                  | 50 FC                   |
| f. Receiving, Issue               | 50 FC                   |
| g. Bench Stock                    | 20 FC                   |
| h. AWM/AWP Holding                | 10 FC                   |
| i. Stairways, Corridors, Airlock  | 20 FC                   |
| j. Elevators, Shipping Dock       | 20 FC or *              |
| k. Restroom, Utility, Storage     | 10 FC or *              |

\* Not less than 1/5 of the brightest adjacent traffic area.

**8.8.5 Supplementary Lighting.** Additional general lighting shall be used for tasks that are difficult to perform within the ambient light level.

**8.8.6 Lighting Fixtures.** Fluorescent lamps shall be used wherever possible to conserve energy. Relamping of fluorescent luminaires shall be done using new fluorescent lamps of a white or daylight variety. They shall have an initial rated lumen output after 100 hours use of not less than 98% of the Illuminating Engineering Society rating for a cool white T-12 medium bipin lamp in the luminaire size lamp. Areas having special spectral emission requirements shall be lamped accordingly.

**8.8.7 Energy Conservation.** Lighting may be turned off during non-working hours providing:

- Due consideration is given to the effects on air conditioning reheat systems to ensure that energy is used efficiently by turning off the lights.
- Environmental limits can be maintained with the lighting turned off.

## **8.9 DOORS & WINDOWS.**

All windows and unused doors or other openings shall be sealed to prevent non-conditioned air infiltration and dust contamination from outside. Doors opening into the PMEL shall be designed and arranged to maintain effective dust and temperature controls. The doors for personnel passage shall be held to a minimum consistent with fire and safety regulations. A building with no windows in the calibration and repair area of the building is preferred because of the potential effect of the direct sunlight on measurement. There may be windows in the calibration and repair area as long as they are not in outside walls of the building because direct sunlight in the PMELs could cause higher temperature in small areas.

## **8.10 AIR LOCKS.**

Air locks shall be constructed at the entrance to the calibration and repair area of the PMEL. An exception can be made if the PMEL is contained inside another environmentally controlled building and the PMEL calibration and repair area has positive air pressure in relation to the outside environmentally controlled area. Shoe cleaners with a self-contained vacuum system (or external vacuum system) shall be located where they will be most effective in reducing contamination that might be carried into the calibration and repair area of the PMEL. Carpeting shall not be installed in air locks due to possibility of the carpeting interfering with the operation of air lock doors.

## **8.11 CONTROLLED AREAS.**

**8.11.1 Dedicated Laser Room.** Any laser system that would interfere with or cause a safety problem for adjoining operations shall be placed in a dedicated room or area. The base bioenvironmental engineer can assist in determining the need for a separate room. Also refer to AFI 48-139, *Laser and Optical Radiation Protection Program*. Size and configuration of the area depends upon the laser system. The HP 5528A is a typical laser not requiring a dedicated room.

**8.11.2 Other Measurement Area.** The PMEL shall control access to areas where unnecessary presence (electrostatic) or excessive body heat (68 degree room) may affect quality.

## **8.12 FLOORING.**

Selection of a floor covering for the calibration area is based on the characteristics of durability, resilience, ease of maintenance, and electrical insulation properties. Light colors are recommended due to their reflection. Carpeting is not permitted due to dust collection and ESD concerns. A high grade of commercial vinyl plastic floor covering material of continuous length is preferred. Installed asphalt, rubber, vinyl asbestos or vinyl tile that is in serviceable condition may continue to be used. Deteriorating vinyl asbestos tile may become a hazard. Contact your local bio-environmentalist if the vinyl asbestos tile begins to deteriorate. Tile seams shall be flush and tight. Coved corners and edges are desirable to simplify cleaning. The continuous length floor covering provides a higher resistance to ground than the individual tiles and it simplifies the cleanup of hazardous materials such as mercury spills. Floors shall not be waxed due to dust and safety hazards. An electrostatic dissipative sealant may be used.

## **8.13 FURNITURE AND FIXTURES.**

All furniture and fixtures shall be of a design that prevents accumulation of dust and facilitates cleaning. Working surfaces shall be resistant to heat, chemicals, chipping, or other deterioration. Cabinets or lockers will be flush mounted, if possible.

## **8.14 FMS TYPE IV AND TYPE II() PMEL FACILITY REQUIREMENTS.**

The following Type IV PMEL and supporting Type II() PMEL facility requirements are in support of FMS customers only; the USAF does not utilize Type IV PMELs. Air locks and shoe cleaners are not required for Type IV PMELs.

### **8.14.1 F-15 Type IV PMEL Requirements.**

#### **8.14.1.1 Electrical Power.**

- a. 200 wye/115 +3-7 VAC, 3 Phase, 4 wire, 400 ±20 Hz, 10 KVA.
- b. 120 VAC ±10% 1 Phase, 3 wire, 60 ±6 or 50 ±3 Hz, 20 KVA.
- c. 240 VAC ±10% 1 Phase, 3 wire, 60 ±6 or 50 ±3 Hz, 10 KVA.
- d. Grounding shall be in accordance with FC 4-218-01F, measurement techniques per TO 31-10-24.

#### **8.14.1.2 Lighting.** (See paragraph 8.8.)

**8.14.1.3 Temperature and Humidity Controls.** A temperature of 73 °F ±9 °F and humidity control of 15 to 70% RH shall be maintained in the calibration and repair areas. Temperature and humidity shall be continuously monitored and recorded.

**8.14.1.4 Floor Space.** Minimum space to support one ESS is 1,634 ft<sup>2</sup>. Two ESSs require 2234 ft<sup>2</sup>. This may vary if the unit has been mobilized. If the Type IV is consolidated into the Type II(), this is the additional space that could be required in the Type II().

**8.14.1.5 Additional Requirements.**

**8.14.1.5.1** Capping of unmated connectors is not required.

**8.14.1.5.2** A portable or central vacuum cleaning system is required to clean TMDE and for general housecleaning. Cleaning TMDE within the calibration/repair area is acceptable, but is restricted to use of vacuum, small brushes, and small hand-held nontoxic spray-and-wipe cleaning. Where toxic fumes, excessive dust, or other safety hazards exist, cleaning shall be accomplished in an approved area outside the calibration and repair area. This includes devices containing mercury, such as manometers and barometers.

**NOTE**

Eating or drinking in the Type IV is permitted only in a designated area set aside for that purpose. A designated eating area is not authorized in the Type IV if an existing break area is convenient. No calibration or repair is permitted in that area.

**8.14.2 F-16 Type IV PMEL Requirements.**

**8.14.2.1 Electrical Power.**

- a. 200 wye/115 +3-7 VAC, 3 Phase, 4 wire, 400 ±20 Hz, 10 KVA
- b. 120 VAC ±10% 1 Phase, 3 wire, 60 ±6 or 50 ±3 Hz, 20 KVA.
- c. 240 VAC ±10%, 1 Phase, 3 wire, 60 ±6 or 50 ±3 Hz, 10 KVA.
- d. Grounding shall be in accordance with FC 4-218-01F, measurement techniques per TO 31-10-24.

**8.14.2.2 Temperature and Humidity Controls.** Temperature shall be maintained between 73 ±9 °F and humidity shall be maintained between 15 and 70% RH. These temperature and humidity limits differ from those of the Type II() PMEL in order to permit use of the least restrictive facility requirements. These limits allow full use of the equipment. Temperature and humidity shall be continuously monitored and recorded.

**8.14.2.3 Floor Space.** Minimum area for one set is 1040 ft<sup>2</sup>, two sets is 1,600 ft<sup>2</sup>. If the Type IV is consolidated into the Type II(), this is the additional space that could be required in the Type II().

**8.14.2.4 Lighting.** (See paragraph 8.8.)

**8.14.2.5 Additional Requirements.**

**8.14.2.5.1** Capping of unmated connectors is not required.

**8.14.2.5.2** A portable or central vacuum cleaning system is required to clean TMDE and for general housecleaning. Cleaning TMDE within the calibration/repair area is acceptable, but is restricted to use of vacuum, small brushes, and small hand-held nontoxic spray-and-wipe cleaning. Where toxic fumes, excessive dust, or other safety hazards exist, cleaning shall be accomplished in an approved area outside the calibration repair area. This includes devices containing mercury, such as manometers and barometers.

**NOTE**

Eating or drinking in the Type IV is permitted only in a designated area set aside for that purpose. A designated eating area is not authorized in the Type IV if an existing break area is convenient. No calibration or repair is permitted in that area.

**8.14.3 F-15 Supporting Type II() PMEL Requirements.** A 16' X 15' floor area is required within the calibration and repair area to accommodate a 6' X 3' granite surface plate and a line-of-site area for optical calibration of a HUD Mount Alignment

Adapter. Floor loading beneath the surface plate shall be capable of supporting the surface plate (3,000 pounds), a granite angle block (1,500 pounds) plus other ancillary equipment. Total loading is for 5,000 pounds. The surface plate is mounted on either a tripod or a quadraped stand with four-inch rectangular feet. Casters are normally provided on the stand.

**8.14.3.1** An increase in workload of approximately 300 items of F-15 related support equipment could be expected in the Type II( ).

**8.14.4 F-16 Supporting Type II( ) PMEL Requirements.** Type II( ) PMELs supporting the F-16 may require additional floor space to accommodate a 3' X 6' surface plate and stand. A surface plate of this size shall require a working area approximately 15' X 16'. Floor loading beneath surface plate shall be capable of supporting approx. 5000 pounds (3500-pound surface plate plus ancillary blocks).

**8.14.4.1** Additional workspace may be required in a Type II( ) PMEL supporting an F-16 ESS. An increase in workload of approximately 200 items of F-16 related support equipment can be expected in the Type II( ).

### **8.15 ENVIRONMENTAL LIMITS FOR RASCAL.**

RASCAL environmental operating limits are as follows:

- a. Temperature =  $73 \pm 6^{\circ}\text{F}$ .
- b. Humidity = 20% to 50% RH.



## SECTION 9

### QUALITY PROGRAM (QP)

#### 9.1 QUALITY PROGRAM (QP).

The purpose of the QP in the AFMETCAL Program is to provide an overall picture of laboratory quality, capability, and risk in order to ensure weapon system safety, accuracy, reliability, and traceability (SART). The QP shall identify laboratory and personnel processes that do not conform to established practices, as well as any end-of-line product quality deficiencies. Data obtained through the QP shall be utilized to assist the MS in the development of corrective actions to mitigate problems/risk. All QP activity shall be thoroughly documented and maintained on file for two years, or since previous assessment, whichever is greater.

#### 9.2 QP REVIEWS.

QP reviews are conducted to sample PMEL product quality and procedural compliance. PMEL Quality Assurance personnel collect data which PMEL managers/supervisors use to proactively monitor and control the quality of PMEL products and processes. By using QP data to identify the most significant problems and negative trends, managers can implement corrective actions to eliminate or minimize these problems and trends. PQA evaluators shall not review their own products or processes. Data from the AFMETCAL Assessment Program may be utilized for analysis but shall not count towards minimum requirements of the Quality Program.

##### 9.2.1 Types of Review.

**9.2.1.1 Quality Review (QR).** A QR is a complete review of end-of-line quality. Actions resulting from NCs during QRs are generally reactive in nature and point management toward areas that require further investigation using root cause analysis. The QR may identify both QNCs and PNCs.

**9.2.1.1.1** All MDC Action Taken codes are subject to QR except for: C, D, M, S, X, W and Y. PAMS or alternate MIS must be set to sample 3 percent of production subject to QR. At any given time or period of time the QR rate may drop below or be above the 3 percent threshold. This is normal and is due to the nature of algorithms used to make QR selections. No action needs to be taken when this occurs. If PAMS is not used, the alternate MIS must sample at 3 percent, over the previous 12-month period. All calibrations including PMEL-owned calibration standards, ATE and TMDE calibrated on-site are subject to the same selection criteria and method of review. The intent is to randomly select and complete QRs at an average rate of 3 percent of production. Alternate methods of selecting additional QRs, at management's discretion, shall be documented.

**9.2.1.1.2** The QR includes the following:

- a. Safety compliance.
- b. Physical condition.
- c. Documentation (e.g., MDC, forms, labels, task qualification, etc.).
- d. Traceability (e.g., standards, reports, technical data, environment, facility, etc.).

**9.2.1.1.3** QRs may include a parameter verification at management's discretion. This decision should be based on the findings of the mandated requirements of the QR as stated above and data gathered through other quality indicators, such as PRs, MAD reviews, etc.

**9.2.1.1.4** A QR may be bypassed only when extraordinary reasons prevent completion. PMEL Management shall document (1) the local policy for bypassing QRs and (2) a method of pre-selecting TMDE for QR that are inherently difficult to sample after certification (e.g., off-base, priority, jet engine test stand, etc.).

**9.2.1.2 Process Review (PR).** A PR provides a step-by-step view into a specified area(s) of a process. The selected process should be observed as it normally occurs, including evaluation of other objective data (MIS entry, interviews, etc.). This highlights errors of a technician and/or laboratory process. The knowledge gained from a PR is preventative in nature and may identify areas of improvement. The actions resulting from process nonconformities are proactive in nature and are an integral part of the continuous process improvement cycle. The PR is the most valuable tool to collect information to determine the health of the laboratory. The PR may identify nonconformities (PNCs) as well as end-of-line quality

nonconformities (QNCs). It is recommended to use this tool to the greatest extent possible and it may be used to evaluate functions independent of calibration processes. A PR may assess, among other things, the effectiveness of a corrective action, the laboratory's capability within a measurement area, training program effectiveness, technician proficiency, logistic functions, etc.

**9.2.1.2.1** Management must define and document their laboratory's MADs to ensure a complete review of laboratory capability is accomplished. The typical PMEL MADs include Capacitance, Resistance, Voltage, Frequency, Power, Phase Noise, Pressure, Vacuum, Force, Dimensional, Mass, Temperature, Torque, and Flow. The PMEL shall include at a minimum complex and/or high-risk Weapon Systems TMDE, unique capabilities identified by Note Codes in TO 33K-1-100-2, and any other locally identified critical areas. Management shall ensure PR selections maximize the evaluation of MADs and covers each MAD every 12 months.

**9.2.1.2.1.1** Management must develop a risk-based sampling plan and define/document their criteria for a MAD to be considered completely assessed in the Quality Manual. Examples of maximizing MAD selections include (but are not limited to) sampling varying levels of complexity, selecting multiple technicians, and selecting equipment that is critical to the laboratory/customer's mission. For each MAD, a determination of risk must be made and documented. Increased risk requires risk management and shall be documented.

**9.2.1.2.1.2.** Management may choose to bypass sampling a MAD based on low demand and/or low risk; however, the justification (to include any kind of risk analysis) shall be analyzed in the Management Review under the Quality Program Effectiveness section.

**9.2.1.2.2** Classification of PRs.

**9.2.1.2.2.1** Annual. All K-stamp holders shall participate in at least two PRs every 12 months. Management shall ensure these PRs evaluate training and metrology skills appropriate to the technician's position (e.g., a full calibration of an item they typically calibrate). Reviews on technicians who are assigned a K-stamp only to create certification labels for items calibrated by non-AF agencies (e.g., N49/59/64) shall include observation of the items listed in paragraph 9.2.1.1.3b. The PR may be selected from any status (AWM, ACP, AQR, etc.) provided all items listed in paragraph 9.2.1.1.2 can be observed. An annual PR may be used to evaluate a measurement area discipline.

**9.2.1.2.2.2** Targeted. At management's discretion, target any portion of any process. Examination of a process may involve one or more PRs as necessary to collect sufficient data. A targeted PR may be used to evaluate a measurement area discipline, a K-stamp holder, an incoming process, etc.

**9.2.1.2.3** Evaluator Proficiency Evaluation (EPE). An EPE provides management with an opportunity to assess the technical ability and proficiency of QA personnel. Management must ensure that QA personnel have the ability to identify/detect NCs and provide an accurate picture of the laboratory health so management can determine associated risks. An EPE consists of a Manager (or designated individual) evaluating QA personnel's training and metrology skills appropriate to the QAE's position (i.e., a full review of an item they typically observe or complete). An EPE will be conducted on QA personnel (including augmentees or alternates) and must pass prior to performing unsupervised evaluations and inspections. The frequency following the initial EPE is at the discretion of management. EPEs should be used as a tool by management to assess the effectiveness of the Quality Program.

### **9.3 NONCONFORMITY (NC).**

NC codes are used to identify distinct product quality deficiencies and process errors through QP activity. An item or process may have more than one NC and each NC shall be assigned a NC code. NCs discovered during PRs or critical NCs discovered during QRs may be areas of improvement to increase laboratory performance.

#### **9.3.1 Quality NC (QNC).**

A QNC is a specific occurrence of a condition which does not conform to specifications or other standards and adversely affects end-of-line safety, accuracy, reliability, or traceability. A QNC can be identified during QRs and PRs. Quality NCs that do not affect safety, accuracy, reliability, or traceability are considered non-critical and the AFMETCAL program does not require PMELs to report them in their PMEL activity summary. This classification is provided for the convenience of PMEL management.

**9.3.2 Process NC (PNC).**

A PNC is an observed process deficiency that affects, or could affect, safety (equipment or personnel), accuracy, reliability, or traceability. A PNC may be identified during QRs or PRs. Identifying PNCs aids in preventing QNCs. Analysis of PNCs is used by internal PMEL management to improve PMEL processes and by AFMETCAL to verify the laboratory's Quality Program effectiveness. PNC codes not beginning with the letter "L" may not affect safety, accuracy, reliability, or traceability, and may be used at management's discretion to evaluate functions independent of calibration processes.

**9.3.3 Review Documentation.**

The results of each review, with or without NCs, shall be recorded in the PAMS QP log or alternate MIS (if PAMS is not available). The following information must be entered: ID Number, Part Number, Serial Number, Work Order/Job Control Number (JCN), Employee Name or Number, Closeout Date, NC Code and RC Code.

**9.4 ROOT CAUSE (RC) ANALYSIS.**

Accurate RC code classification of identified NCs is essential to effective corrective action/process improvement. Make every effort to determine the true RC of the NC and not just a symptom. Avoid simply restating the NC. Usually one "action or inaction" appears to be responsible, but a good RC analysis will address what caused the "action or inaction." By eliminating or reducing the most significant cause of problems, management improves overall PMEL quality performance. RC codes shall be assigned to all critical NCs. Typically, supervisors, PQA evaluators, and technicians perform the analysis. The assigned NC/RC codes shall be analyzed by management during the MR as a component of QP effectiveness.

**9.4.1** As a step in RC analysis for all critical NCs identified during a QR, the PQA evaluator shall observe the certifying technician perform the suspect process without divulging the nonconforming condition. Since the PQA only observes the suspect process with the intent to discover a RC, this should not be documented as a PR. Should QA discover NC(s) unrelated to the original NC, an additional PR may be generated to record the new-found NC(s).

**9.4.2** Corrective actions, recalls and follow-ups must be performed and documented, as applicable. Further investigation/follow-up may be performed as a result of identifying repeated, recurring NC/RC combinations. The minimum steps required for RC analysis are below.

- a. State the nonconforming condition. Be clear and concise and provide authoritative references whenever possible. Do not include information that pertains to the RC.
- b. Document observable facts, not opinions or assumptions, made while observing the suspect process.
- c. List all possible causes for the NC.
- d. Document most probable RC selection and concisely explain how and why the RC code was selected. In addition, concisely explain rationale for possible RC codes that were not selected.
- e. Document actions taken to correct the nonconformity and any necessary follow-ups to include additional findings.
- f. For critical NCs state whether recall was initiated or reasoning for not performing a recall. If recall is initiated, document any follow-up actions and any additional findings.
- g. Document actions taken to prevent the nonconformity from reoccurring and any necessary follow-ups to include additional findings (consider technician, section, and laboratory, etc.), if applicable.
- h. For critical NCs, PMEL TMDE Flight Chief (or equivalent) shall approve completed RC analysis (sign, initial, or PMEL/MIS entry).
- i. Forward the RC documentation per local procedures.

**9.4.3** Management must define and document a maximum acceptable timeline for the completion of RC analysis. Quick, thorough, and accurate documentation of the NC, RC and its corrective action must be captured in a timely manner to facilitate an effective QP.

**9.5 PROCESS REVIEW PLAN (PRP).**

The PRP shall be completed monthly to ensure effective MAD coverage in order to assess laboratory proficiency and capability. The PRP is used to project the QP plan for PRs (consider ARs, MADs, EPEs and other targeted PRs) to be completed for the coming month. A PMEL may perform fewer or more PRs as the data dictates, however they must still comply with minimum PR requirements outlined in this TO.

## 9.6 NONCONFORMITY AND ROOT CAUSE CODES.

Codes shall be assigned to each identified critical nonconformity and its respective RC. The purpose of assigning codes is to group similar nonconformities and root causes for analysis. Standard nonconformity and RC codes are listed in table 9-1. These specific codes shall not be changed. Select only the appropriate verbiage from the code description to clearly identify the NC/RC code used.

**Table 9-1. Summary of QP Codes.**

| <b>Critical Quality Nonconformity (QNC) Codes:</b> |                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A01</b>                                         | The item is unsafe or hazardous to use (e.g., foreign object inside, over-fused, does not comply with TO 33-1-32).                                                                                                                                                                                                                                   |
| <b>A02</b>                                         | The item has an inoperative/intermittent function, or the item is completely inoperative; the item has a physical defect that prevents use of a certified function.                                                                                                                                                                                  |
| <b>A04</b>                                         | The item does not meet calibration tolerances/uncertainties for all parameters certified.                                                                                                                                                                                                                                                            |
| <b>A07</b>                                         | The documentation on/provided with the item has an error (missing a TO directed limitation/annotation; certification label date due extended beyond prescribed interval; item certified better than calibration authority specifications; CMS incorrectly applied).                                                                                  |
| <b>A08</b>                                         | The items documentation in the PAMS/MIS has an error (missing one or more critical standards from the PAMS/MIS traceability listing/report of measurement; the PAMS/MIS date due is extended beyond prescribed interval)                                                                                                                             |
| <b>A09</b>                                         | Test Uncertainty Ratio not maintained or required standard not used/certified at time of calibration.                                                                                                                                                                                                                                                |
| <b>Non-Critical QNC Codes:</b>                     |                                                                                                                                                                                                                                                                                                                                                      |
| <b>B05</b>                                         | The item has a physical defect that does not affect the certified functions, or the item is not clean.                                                                                                                                                                                                                                               |
| <b>B07</b>                                         | The item has a documentation error (missing a TO directed annotation; missing one or more items of TMDE from the PAMS/MIS traceability listing/report of measurement).                                                                                                                                                                               |
| <b>Process Nonconformity (PNC) Codes:</b>          |                                                                                                                                                                                                                                                                                                                                                      |
| <b>L01</b>                                         | Technical data not used, wrong technical data used or technical data not current. Technical data used contained an error requiring a recommended change. Technical data used was incorrect, including calibration procedure, maintenance TO, commercial data, CPIN, etc. Automated program/calibration software not validated/verified or incorrect. |
| <b>L02</b>                                         | Cyber Security procedures not followed or requirements not met.                                                                                                                                                                                                                                                                                      |
| <b>L03</b>                                         | Incoming or outgoing inspections were not performed or incorrectly performed (FOD, TO 33-1-32, TO 33K-2-11, etc.).                                                                                                                                                                                                                                   |
| <b>L04</b>                                         | Safety of personnel not considered or maintained.                                                                                                                                                                                                                                                                                                    |
| <b>L05</b>                                         | Complete calibration could not be performed. (Reserved for AFMETCAL Evaluation use only)                                                                                                                                                                                                                                                             |
| <b>L06</b>                                         | Standard or calibration accessory not used or improperly used. Overdue or uncalibrated standard used for quantitative measurements (CBU, NCR, ICO w/o source data, etc.).                                                                                                                                                                            |
| <b>L07</b>                                         | Environment ignored or not monitored; environmental data not properly documented (environmental charts, certification labels, etc.); item used or calibrated outside the allowable operating environmental conditions.                                                                                                                               |
| <b>L08</b>                                         | Standard incorrectly substituted or test uncertainty ratio not maintained.                                                                                                                                                                                                                                                                           |
| <b>L09</b>                                         | Technician does not demonstrate task qualification/proficiency.                                                                                                                                                                                                                                                                                      |
| <b>L10</b>                                         | Critical step, note or caution improperly performed, not accomplished, ignored, or misinterpreted.                                                                                                                                                                                                                                                   |
| <b>L11</b>                                         | Technician failed to identify out-of-tolerance condition or incorrectly identified in-tolerance indication as out of tolerance.                                                                                                                                                                                                                      |
| <b>L14</b>                                         | Calculations not performed or improperly performed.                                                                                                                                                                                                                                                                                                  |
| <b>L15</b>                                         | Calibration factors or correction charts not used, improperly used or not accomplished.                                                                                                                                                                                                                                                              |
| <b>L16</b>                                         | Documentation improperly performed/omitted.                                                                                                                                                                                                                                                                                                          |
| <b>Q01</b>                                         | PQA failed to identify a critical Quality NC                                                                                                                                                                                                                                                                                                         |
| <b>Q02</b>                                         | PQA failed to identify a Process NC                                                                                                                                                                                                                                                                                                                  |
| <b>Review Status Codes:</b>                        |                                                                                                                                                                                                                                                                                                                                                      |
| <b>N00</b>                                         | No Defect.                                                                                                                                                                                                                                                                                                                                           |
| <b>N01</b>                                         | Review in process - not completed.                                                                                                                                                                                                                                                                                                                   |

| <b>Table 9-1. Summary of QP Codes – Continued.</b> |                                                                                                                                                     |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Root Cause (RC) Codes:</b>                      |                                                                                                                                                     |
| <b>C00</b>                                         | No root cause necessary (use when NC Code is N00)                                                                                                   |
| <b>C01</b>                                         | Wrong or invalid calibration technical data used.                                                                                                   |
| <b>C02</b>                                         | Calibration technical data in error (qualified technician would not detect).                                                                        |
| <b>C03</b>                                         | Calibration technical data ambiguous.                                                                                                               |
| <b>E03</b>                                         | Facility issues/concerns.                                                                                                                           |
| <b>F01</b>                                         | Component/equipment failure caused degradation or hard failure (normal process would not find).                                                     |
| <b>F02</b>                                         | Failure related to previous maintenance or repair (normal inspection would not find).                                                               |
| <b>F03</b>                                         | Failure related to thermal heating (not evident during normal warm-up).                                                                             |
| <b>H01</b>                                         | Human error: When all equipment, technical data, training, and other factors are adequate. The technician made a mistake, oversight, omission, etc. |
| <b>H02</b>                                         | Improper Action: When all equipment, technical data, training, and other factors are adequate. The technician disregarded policy.                   |
| <b>I01</b>                                         | Established laboratory process inadequate or faulty.                                                                                                |
| <b>S01</b>                                         | Standard out of tolerance (normal use would not detect).                                                                                            |
| <b>S04</b>                                         | Standard intermittently malfunctioning (common metrology practices would not detect).                                                               |
| <b>S05</b>                                         | Accessories or calibration fixtures faulty (common metrology practices would not detect).                                                           |
| <b>T01</b>                                         | OJT insufficient (trainer demonstrates task proficiency).                                                                                           |
| <b>T02</b>                                         | OJT insufficient (trainer does not demonstrate task proficiency).                                                                                   |
| <b>T07</b>                                         | Technician does not demonstrate task proficiency.                                                                                                   |
| <b>U01</b>                                         | User responsibility not accomplished.                                                                                                               |



## **SECTION 10**

### **MANAGEMENT SYSTEM (MS)**

#### **10.1 MANAGEMENT SYSTEM (MS).**

The PMEL Management System (MS) is defined as the organizational structure, responsibilities, procedures, processes, and resources for the calibration of TMDE. The elements of a PMEL's MS shall be documented in a quality manual. All MS data shall be thoroughly documented and maintained on file for two years, or from the previous assessment, whichever is greater.

#### **10.2 QUALITY MANUAL (QM).**

The QM is a document that defines the PMEL's policies and objectives for, and commitment to, metrology principles and quality calibration services. The QM shall be issued under authority of the PMEL Manager and made available for use by all personnel. The purpose of the QM is twofold. First, it provides an organized way of communicating how the PMEL's MS is structured and implemented, both to personnel working in the PMEL and to external parties such as base and corporate management, customers, and assessors. Second, it provides a basis for orienting new personnel on their roles and responsibilities under the PMEL MS and how it is implemented in the PMEL. The QM shall be reviewed and documented as deemed necessary by PMEL management. At a minimum, the QM shall include or reference the location of the following information:

- a. Training. The PMEL training requirements, methods used to obtain the required training, who shall review the records, and the interval between qualification/training records review.
- b. Quality Programs. Local quality assurance objectives and practices. This includes specific management policies for meeting requirements of AFMAN 21-113 and this TO.
- c. Recall and Notification. The decision process used to determine:
  - (1) If TMDE should be subject to recall.
  - (2) If notification of the customer/user is necessary when a PMEL standard or customer-owned TMDE is found to be out of tolerance or when a faulty process (that may result in an out-of-tolerance condition) is identified.
- d. Exceptions and Limitations. PMEL management's policies and processes for calibration interval exceptions, limitations, and special calibrations.

#### **10.3 MANAGEMENT INTERNAL CONTROL TOOLSET (MICT).**

The TMDE (AFMAN 21-113) Self-Assessment Communicator (SAC) in MICT is to be used as a living document. At a minimum, the SAC shall be completed once every 12 months or within 12 months from contract start date. However, when a change occurs within the laboratory that affects a SAC item (to include upload requirements), the SAC item shall be updated in a timely manner and maintained to reflect the current state of the PMEL. The MICT SAC is not a comprehensive checklist of the compliance items directed by AFMETCAL, contract requirements, or local policy. A locally developed checklist (using MICT) may be created and maintained at the manager's discretion to supplement the PMEL's self-assessment program.

#### **10.4 INTERNAL AUDIT (IA).**

An IA is required for PMELs that do not have access to the AFMAN 21-113 TMDE SAC in MICT. An IA is a complete review of operations to verify continued compliance with the MS. PMELs shall perform and document IAs at least once every 12 months, or within 12 months from contract start date for new contractor operated PMELs. Personnel possessing the knowledge and skills necessary to understand the process being audited as well as the process of auditing shall conduct the audit. If possible, auditors should be selected from personnel independent of the area being audited. For scheduling purposes, management may elect to complete portions of the audit throughout the year, providing the entire MS is covered within a 12-month period. The IA may be accomplished in conjunction with the management review at local management discretion. As a minimum, IA documentation shall include:

- a. MS compliance (defined by local Quality Manual policies and objectives).
- b. TO 00-20-14 compliance.
- c. Assigned action items, POCs, ECDs and documented follow-ups for identified deficiencies.

#### **10.5 MANAGEMENT REVIEW (MR).**

An MR is an executive overview of the MS and shall be accomplished at least once every 12 months, with a static closeout date of 31 July of each year and uploaded to MICT no later than 31 Aug of each year. If contract turnover occurs (new

contractor), the MR will be accomplished within 6 months from contract start date with subsequent MRs aligned to the static closeout date. The MR is a two-fold document that should review the previous year's key indicators and processes from a macro-level to assess the overall risk to the PMEL MS. Each component of the MR must be evaluated and assigned a risk level defined in paragraph 10.8. The MR should also be used as a planning document for the coming year to improve the overall PMEL operation. Problems identified during the MR should be assigned action items, POCs and ECDs if necessary. If actions are already underway to address problems, the MR should reference the source tracking methods to avoid redundant documentation of action items. Follow-ups should be performed and documented as necessary in between MRs. MR documentation shall include review and evaluation of (as a minimum):

- a. The outcome of internal/external audits/assessments, etc., (since the last MR). A review of compliance and effectiveness shall occur. Address any problems that may have generated follow up actions.
- b. Quality program effectiveness. Analyze all QP components and support QP effectiveness determination.
- c. Customer relations program summary. Utilize customer surveys/feedbacks, priority requests, items deferred/on hold and any additional criteria impacting customer relations/support.
- d. Changes in the volume and type of the work. At a minimum, address the impact that such changes have on the PMEL.
- e. Equipment/capability needs/shortfalls (if required, include plan to restore self-sufficiency). At a minimum, evaluate self-sufficiency, laboratory capability and ability to provide primary mission support.
- f. Facility needs/shortfalls. Identify concerns and impact to PMEL performance or ability to provide support. Document actions/measures if not already documented elsewhere.
- g. Training/proficiency needs/shortfalls (for the coming year at a minimum). Analyze TPEs, task coverage, work force changes and impacts. Utilize QP data and other training related information as necessary.
- h. Effectiveness and status of other process improvement efforts. Review other process improvement activities/initiatives and evaluate effectiveness.

## **10.6 PMEL ACTIVITY SUMMARY (PAS).**

A monthly PMEL activity summary shall include the following:

**10.6.1 PAMS Management Indicator Report.** For those organizations who do not utilize PAMS the following shall replace the PAMS Management Indicator:

- a. Key laboratory metrics.
  - (1) Equipment availability rate (current rate and standard).
  - (2) Total overdue TMDE assets.
  - (3) Total input.
  - (4) Total output.
- b. QR.
  - (1) QR pass rate (percentage).
  - (2) Total QRs completed.
  - (3) Total QRs w/QNCs (multiple QNCs on a single JCN count as a single fail).
  - (4) List of QR items with critical NCs (bar code number, part number, nomenclature, NC, and RC).
- c. PR.
  - (1) List of PR items with critical NCs (bar code number, part number, nomenclature, NC, and RC).
  - (2) List Targeted PRs performed (area targeted and result).
- d. List of all bypassed reviews (bar code number, part number, nomenclature, and justification).

**10.6.2** Any additional requirements may be set based on local operational needs.

## **10.7 TRAINER PROFICIENCY EVALUATION (TPE).**

The purpose of the TPE is to provide an overall picture of laboratory training (not required for contract PMELs). TPEs are intended to focus on evaluating training effectiveness and provide objective data to improve overall training. TPEs collect data which PMEL managers use to proactively monitor and control the quality of PMEL training. By using TPE data to identify the most significant problems and negative trends with training, managers can implement process improvement actions to eliminate or minimize these problems and trends.

**10.7.1 Evaluation.** A TPE is intended as a tool for the Manager to evaluate trainer proficiency as deemed necessary. It provides management with an opportunity to assess the educational ability and effectiveness of personnel performing training. A TPE consists of a Manager (or designated evaluator) evaluating trainers performing training.



**10.7.2 Documentation.** PMEL managers must establish and define requirements to evaluate and document TPEs.

## **10.8 RISK MANAGEMENT.**

As detailed in AFI 90-802, *Risk Management*, risk management is a decision-making process to systematically evaluate possible courses of action, identify risks and benefits, and determine the best courses of action for any given situation. When appropriately applied to the AFMETCAL Program's Quality Program and Management System, risk management enables PMEL managers to maximize capabilities while limiting risks through application of a simple, systematic process. Appropriate use of risk management increases the PMEL's ability to accomplish their mission and activity while preserving limited resources safely and effectively. AFI 90-802 and DAFPAM 90-803 provide the framework for operational risk management while the following guidelines tailor that framework to the AFMETCAL Program's Quality Program and Management System. **Note:** These guidelines do not address risk management guidelines tied to and should not be used for Acquisition and Sustainment Life Cycle Management.

### **10.8.1 Definitions.**

**10.8.1.1 Risk Management.** The systematic process of identifying hazards, assessing risk, making control decisions, implementing control decisions, and supervising and reviewing the activity for effectiveness.

**10.8.1.2 Risk Assessment.** The process of detecting conditions/events and their causes, and systematically assessing the associated risks.

**10.8.1.3 Hazard (referred to as a condition or event throughout this document).** Any condition or event with the potential to cause mission degradation or failure due to effects on the PMEL's ability to perform safe, accurate, reliable, or traceable measurements; the PMEL's capability; or the Calibration Network.

**10.8.1.4 Probability.** Any condition or event with the potential to cause mission degradation or failure due to effects on the PMEL's ability to perform safe, accurate, reliable, or traceable measurements; the PMEL's capability; or the Calibration Network.

**10.8.1.5 Severity.** The expected consequences of an event in terms of mission impact.

**10.8.1.6 Risk.** The probability and severity of loss or adverse impact from exposure to various conditions and/or events.

**10.8.2** The 5-Step Risk Management Process as described in AFI 90-802 and DAFPAM 90-803 should be used as the primary guidelines in accomplishing risk management within the PMEL.

**10.8.2.1** For Step 2, Assess the Risk, the following Risk Assessment Matrix is provided to aid in determining how risk to the QP and MS should be described. Additionally, example interpretations of the table are listed below. These examples are not all inclusive and the local management should use the table as appropriate to describe their own assessed risk.

**Table 10-1. Risk Assessment Matrix.**

| Risk Assessment Matrix |                                      |                                           | PROBABILITY              |            |          |
|------------------------|--------------------------------------|-------------------------------------------|--------------------------|------------|----------|
|                        |                                      |                                           | Likelihood of Occurrence |            |          |
|                        |                                      |                                           | Probable                 | Occasional | Remote   |
| SEVERITY               | Impact of Event/Condition on Mission | Critical (Unacceptable Impact to Mission) | High                     | High       | Moderate |
|                        |                                      | Moderate (Undesirable Impact to Mission)  | High                     | Moderate   | Low      |
|                        |                                      | Negligible (Acceptable Impact to Mission) | Moderate                 | Low        | Low      |
|                        |                                      |                                           | Risk Assessment Levels   |            |          |
|                        |                                      |                                           | High                     | Moderate   | Low      |

**10.8.2.1.1 Low Risk.** Likelihood of non-compliance occurring is remote based on the PMEL's proven proficiency/capability, Management System and Quality Program's proven ability to detect, and present or future task coverage forecast. If left uncorrected, impact on customers' mission is negligible.

**10.8.2.1.2 Moderate Risk.** Likelihood of non-compliance is occasional based on task proficiency, capability workarounds, unforecasted present or future task coverage, and Management System and Quality Program's suspect ability to detect. If left uncorrected, impact on customers' mission is moderate.

**10.8.2.1.3 High Risk.** Likelihood of detected or undetected non-compliance is probable based on the PMEL's unproven proficiency, lack of present or future task coverage, and/or Management System and Quality Program's inability to currently detect. If left uncorrected, impact to customers' mission is critical.

## SECTION 11

### PROFICIENCY TESTING/MEASUREMENT ASSURANCE PROGRAM (PT/MAP) and ACCEPTANCE TESTING

#### 11.1 PROFICIENCY TESTING/MEASUREMENT ASSURANCE PROGRAM (PT/MAP).

PMELs shall be required to complete proficiency testing as directed by AFMETCAL. Proficiency testing results may be based on PMEL compliance with completion and return of Exchange Standard Intercomparison Data Sheets found in TO 33K-2-11, data collected upon PMEL receipt and processing of exchange or transfer standards, or testing performed on a unique measurement artifact. The PMEL Assessment Program shall consider the results of proficiency testing as objective evidence, along with virtual and on-site assessment findings, of the PMEL's ability to perform competent calibrations.

**11.1.1** Proficiency Testing on unique artifacts is the process used to assess laboratory performance by means of comparing and evaluating calibrations or tests on a measurement artifact. If a PMEL is selected for Proficiency Testing on a unique artifact, paragraphs 11.1.1.1 through 11.1.1.4 will apply. The objective is to assess both the technical competence of laboratory personnel and the PMEL's measurement capability as well as reliability and condition of measurement standards following shipping.

**11.1.1.1** Measurement artifacts for selected measurement areas shall be distributed to PMELs by the AFPSL based on a schedule determined by AFMETCAL. The number of proficiency tests required of each PMEL annually will vary depending on the availability of measurement artifacts but shall not exceed six (6) artifacts per PMEL annually.

**11.1.1.2** Artifact testing shall be scheduled by the AFPSL on behalf of AFMETCAL. PMELs shall be notified in advance as to the approximate arrival date of the measurement artifact. Instructions for performing the test, reporting results, notifying AFMETCAL and the AFPSL and shipping will be included with the artifact.

**11.1.1.3** The PMEL shall be required to perform measurements on an artifact using the same calibration methods used to calibrate customer equipment or methods directed by AFMETCAL. The PMEL shall report results as indicated in the PT/MAP instructions and data sheets provided by AFMETCAL. The PMEL is expected to complete proficiency testing of a measurement artifact within the time allotted in the PT/MAP instructions. The measurement artifact shall not leave the PMEL unless otherwise directed by the PT/MAP instructions.

**11.1.1.4** Unless otherwise stated in the PT/MAP instructions, the performance of the proficiency test is judged by calculating the error, normalized with respect to the uncertainty, of the measurement as follows:

$$E_{\text{normal}} = \left| \frac{Value_{\text{PMEL}} - Value_{\text{ref}}}{\sqrt{Uncertainty_{\text{PMEL}}^2 + Uncertainty_{\text{ref}}^2}} \right|$$

Where:

$E_{\text{normal}}$  = normalized error of the PMEL

$Value_{\text{PMEL}}$  = the value as measured by the PMEL

$Value_{\text{ref}}$  = the value as measured by the AFPSL (reference laboratory)

$Uncertainty_{\text{ref}}$  = the overall uncertainty of the AFPSL (reference laboratory)

$Uncertainty_{\text{PMEL}}$  = the overall uncertainty of the PMEL (e.g., Measurement uncertainty that would be provided to a customer)

#### NOTE

All Values and Uncertainties shall be expressed in the same units. To pass the proficiency test the PMEL shall have a value for  $E_{\text{normal}}$  less than 1 (e.g.,  $E_{\text{normal}} < 1$ ).

**11.1.2** AFMETCAL shall notify the PMEL of their proficiency testing results.

#### 11.2 ACCEPTANCE TESTING.

AFMETCAL equipment procurements often require Acceptance Testing at the PMELs. It is imperative that Acceptance Testing be completed within the contractual timeframes to preclude payment for equipment that does not meet AF requirements. Required acceptance testing procedures and timelines will be included in the applicable Equipment Support

Plan (ESP). PMELs shall perform and report Acceptance Testing in accordance with the appropriate ESP. Acceptance Testing status will be tracked via the PMEL Dashboard on MetWeb.

## SECTION 12

### AIR FORCE METROLOGY AND CALIBRATION LABORATORIES

#### 12.1 MAJOR COMMAND (MAJCOM) AND AGENCY CODES.

Table 12-1 contains MAJCOM and agency codes\* derived from the AFCA Table Management Distribution System (TMDS) website: <https://trdmws.maf.ustranscom.mil/> (search for Air Force Major Command Code).

**Table 12-1. MAJCOM and Agency Codes.**

| MAJCOM/AGENCY                                                    | CODE* |
|------------------------------------------------------------------|-------|
| Air Combat Command (ACC)                                         | 1C    |
| Air Education and Training Command (AETC)                        | 0J    |
| AF Global Strike Command (AFGSC)                                 | GS    |
| AF Materiel Command (AFMC)                                       | 1M    |
| AF Reserve Command (AFRC)                                        | 0M    |
| AF Special Operations Command (AFSOC)                            | 0V    |
| AF Intelligence, Surveillance, and Reconnaissance Agency (AFISR) | 0U    |
| Air Mobility Command (AMC)                                       | 1L    |
| Air National Guard (ANG)                                         | 4Z    |
| Pacific Air Forces (PACAF)                                       | 0R    |
| US Air Force Academy (USAF A)                                    | 0B    |
| US Air Forces in Europe (USAFE)                                  | 0D    |
| US Air Forces Central (AFCENT)                                   | 3C    |
| US Space Force (USSF)                                            | 1S    |

#### 12.2 AIR FORCE PRIMARY STANDARDS LABORATORY (AFPSL).

Table 12-2 contains AFPSL basic information. Address is AFPSL, 813 Irving-Wick Dr. W, Ste 4M, Heath OH 43056-6118.

**Table 12-2. AFPSL Site Information.**

| SITE NUMBER | LAB LOCATION       | SITE ID | BASE GEOLOC Code* | PMEL Type | MAJ COM | CMD Code | Remarks  |
|-------------|--------------------|---------|-------------------|-----------|---------|----------|----------|
| A001        | AFMETCAL Heath, OH | SL      | RRTC              | AFPSL     | AFMC    | 1M       | Contract |

\* Derived from AFCA TMDS website: <https://trdmws.maf.ustranscom.mil/>

#### 12.3 METROLOGY & CALIBRATION FLIGHT (MCF) SITES.

The Metrology & Calibration Flight (MCF) is a specialized calibration flight located at the ALCs, Oklahoma City, OK OC-ALC), SITE ID: OC; Warner Robins, GA (WR-ALC), SITE ID: WM; and Ogden, UT (OO-ALC), SITE ID: OM, formerly known as the DIMF. MCF and their operational requirements are detailed in Section 13.

**12.4 PRECISION MEASUREMENT EQUIPMENT LABORATORIES (PMELs).**

Table 12-3 contains the assigned PMEL NUMBER and basic site information for all DAF PMELs.

**Table 12-3. PMEL Site Information.**

| PMEL LOCATION               | SITE NUMBER | SITE ID | BASE GEOLOC Code | PMEL Type | Lead MAJ COM | CMD Code | Remarks  |
|-----------------------------|-------------|---------|------------------|-----------|--------------|----------|----------|
| Altus AFB OK                | P001        | AT      | AGGN             | IIB       | AETC         | 0J       | CS       |
| Al-Udeid AB, QA             | P078        | AU      | ALDA             | III       | AFCENT       | 3C       | Mil      |
| Andersen AFB, GU            | P002        | AN      | AJYJ             | IIB       | PACAF        | 0R       | Contract |
| Andrews AFB MD              | P003        | DC      | AJXF             | IIB       | AMC          | 1L       | Contract |
| Arnold AFB TN               | P004        | AR      | ANZW             | IIC       | AFMC         | 1M       | Contract |
| Aviano AB IT                | P005        | AV      | ASHE             | IIB       | USAFE        | 0D       | Mil      |
| Barksdale AFB LA            | P006        | LA      | AWUB             | IIB       | AFGSC        | GS       | Contract |
| Cannon AFB NM               | P009        | CC      | CZQZ             | IIB       | AFSOC        | 0V       | Contract |
| Charleston AFB SC           | P012        | CH      | DKFX             | IIB       | AMC          | 1L       | Mil      |
| Columbus AFB MS             | P014        | CO      | EEPZ             | IIB       | AETC         | 0J       | Contract |
| Davis Monthan AFB AZ        | P015        | DM      | FBNV             | IIB       | ACC          | 1C       | Mil      |
| Duluth IAP MN               | P017        | DU      | FMKM             | IIB       | ANG          | 4Z       | CS       |
| Dyess AFB TX                | P018        | DY      | FNWZ             | IIB       | AFGSC        | GS       | Contract |
| Edwards AFB CA              | P019        | ED      | FSPM             | IIC       | AFMC         | 1M       | Mil      |
| Eglin AFB FL                | P020        | EG      | FTFA             | IIC       | AFMC         | 1M       | Mil      |
| Ellsworth AFB SD            | P022        | EL      | FXBM             | IIB       | AFGSC        | GS       | Contract |
| Elmendorf AFB AK            | P023        | AK      | FXSB             | IIA       | PACAF        | 0R       | Contract |
| F.E. Warren AFB WY          | P024        | FE      | GHLN             | IIB       | AFGSC        | GS       | Contract |
| Fairchild AFB WA            | P025        | FC      | GJKZ             | IIB       | AMC          | 1L       | Contract |
| Feltwell RAF GB             | P026        | FW      | MSET             | IIA       | USAFE        | 0D       | Contract |
| Forbes Field KS             | P027        | FO      | GUQG             | IIB       | ANG          | 4Z       | CS       |
| Hickam AFB HI               | P029        | HK      | KNMD             | IIB       | PACAF        | 0R       | CS       |
| Hill AFB UT                 | P030        | HU      | SGQM             | IIA       | AFMC         | 1M       | Contract |
| Holloman AFB NM             | P031        | HO      | KWRD             | IIC       | AETC         | 1C       | Mil      |
| Kadena AB JP                | P032        | KJ      | LXEZ             | IIA       | PACAF        | 0R       | Mil      |
| Keesler AFB MS              | P033        | KS      | MAHG             | IIB       | AETC         | 0J       | CS       |
| Kirtland AFB NM             | P034        | NM      | MHNV             | IIB       | AFGSC        | GS       | CS       |
| Kunsan AB KR                | P035        | KU      | MLWR             | IIB       | PACAF        | 0R       | Mil      |
| Langley AFB VA              | P037        | FF      | MUHJ             | IIB       | ACC          | 1C       | Mil      |
| Little Rock AFB AR          | P039        | LR      | NKAK             | IIB       | AMC          | 1L       | Mil      |
| Luke AFB AZ                 | P040        | LF      | NUEX             | IIB       | AETC         | 0J       | Mil      |
| MacDill AFB FL              | P041        | MD      | NVZR             | IIB       | AMC          | 1L       | Contract |
| Malmstrom AFB MT            | P042        | MM      | NZAS             | IIB       | AFGSC        | GS       | Contract |
| Maxwell AFB Gunter Annex AL | P043        | MG      | JUBJ             | IIB       | AETC         | 0J       | Contract |
| McChord AFB WA              | P044        | MC      | PQWY             | IIB       | AMC          | 1L       | Mil      |
| McConnell AFB KS            | P045        | ML      | PRQE             | IIB       | AMC          | 1L       | Contract |
| McGuire AFB NJ              | P046        | NJ      | PTFL             | IIB       | AMC          | 1L       | Mil      |
| Minot AFB ND                | P047        | MT      | QJVF             | IIB       | AFGSC        | GS       | Contract |
| Misawa AB JP                | P048        | MJ      | QKKA             | IIB       | PACAF        | 0R       | Mil      |
| Moody AFB GA                | P049        | FT      | QSEU             | IIB       | ACC          | 1C       | Contract |

**Table 12-3. PMEL Site Information-Continued.**

| <b>PMEL<br/>LOCATION</b>                   | <b>SITE<br/>NUMBER</b> | <b>SITE<br/>ID</b> | <b>BASE<br/>GEOLOC<br/>Code</b> | <b>PMEL<br/>Type</b> | <b>Lead<br/>MAJ<br/>COM</b> | <b>CMD<br/>Code</b> | <b>Remarks</b> |
|--------------------------------------------|------------------------|--------------------|---------------------------------|----------------------|-----------------------------|---------------------|----------------|
| Mt Home AFB ID (with RASCAL #1)            | P050                   | MO                 | QYZH                            | IIB                  | ACC                         | 1C                  | Mil            |
| Nellis AFB NV                              | P051                   | NL                 | RKMF                            | IIB                  | ACC                         | 1C                  | Mil            |
| Offutt AFB NE                              | P053                   | OF                 | SGBP                            | IIB                  | ACC                         | 1C                  | Contract       |
| Osan AB KR                                 | P054                   | OS                 | SMYU                            | IIB                  | PACAF                       | 0R                  | Mil            |
| Otis ANGB MA                               | P055                   | MA                 | SPBQ                            | IIC                  | ANG                         | 4Z                  | CS             |
| Patrick AFB/<br>Cape Canaveral AFS FL      | P056                   | PA                 | SXHT/<br>DBEH                   | IIC                  | AFMC                        | 1M                  | Contract       |
| Peterson AFB CO                            | P057                   | PE                 | TDKA                            | IIB                  | AFMC                        | 1M                  | Contract       |
| Ramstein AB DE                             | P059                   | RS                 | TYFR                            | IIB                  | USAFE                       | 0D                  | Mil            |
| Randolph AFB TX                            | P060                   | RA                 | TYMX                            | IIB                  | AETC                        | 0J                  | CS             |
| Robins AFB GA                              | P061                   | WG                 | UHJC                            | IIA                  | AFMC                        | 1M                  | Contract       |
| Scott AFB IL                               | P062                   | SC                 | VDYD                            | IIB                  | AMC                         | 1L                  | Contract       |
| Selfridge ANGB MI                          | P063                   | MI                 | VGLZ                            | IIB                  | ANG                         | 4Z                  | CS             |
| Seymour Johnson AFB NC (with<br>RASCAL #2) | P064                   | SJ                 | VKAG                            | IIB                  | ACC                         | 1C                  | Mil            |
| Shaw AFB SC                                | P065                   | SW                 | VLSB                            | IIB                  | ACC                         | 1C                  | Mil            |
| Sheppard AFB TX                            | P066                   | SH                 | VNVP                            | IIB                  | AETC                        | 0J                  | Contract       |
| Thule AB GL                                | P067                   | TH                 | WWCX                            | IIB                  | AFMC                        | 1M                  | Contract       |
| Tinker AFB OK                              | P068                   | TO                 | SHDF                            | IIA                  | AFMC                        | 1M                  | Contract       |
| Travis AFB CA                              | P069                   | TR                 | XDAT                            | IIB                  | AMC                         | 1L                  | Mil            |
| Tyndall AFB FL                             | P070                   | TY                 | XLWU                            | IIB                  | ACC                         | 1C                  | Contract       |
| USAF Academy CO                            | P071                   | AF                 | XQPZ                            | IIB                  | USAFA                       | 0B                  | CS             |
| Vance AFB OK                               | P073                   | VN                 | XTLF                            | IIB                  | AETC                        | 0J                  | Contract       |
| Vandenberg AFB CA                          | P074                   | HV                 | XUMU                            | IIC                  | AFMC                        | 1M                  | Contract       |
| Whiteman AFB MO                            | P075                   | KC                 | YWHG                            | IIB                  | AFGSC                       | GS                  | Contract       |
| Wright-Patterson AFB OH                    | P076                   | WP                 | ZHTP                            | IIC                  | AFMC                        | 1M                  | Contract       |
| Yokota AB JP                               | P077                   | YJ                 | ZNRE                            | IIB                  | PACAF                       | 0R                  | Mil            |

**Remarks:** CS - Civil Servant Operated (primarily)  
MIL - Military Operated (primarily)  
Contract - Contractor Operated





## SECTION 13

### METROLOGY & CALIBRATION FLIGHT (MCF)

#### 13.1 PURPOSE.

The Metrology & Calibration Flight (MCF) is a specialized calibration activity physically located at the ALCs. The MCF will obtain full AFMETCAL Program traceability through support from the co-located Type IIA PMEL. The MCF performs depot- and intermediate-level maintenance and calibration as defined in the statutory framework paragraphs of AFMCI 21-102, *Depot Maintenance Management*. The MCF is aligned under the ALC Maintenance Support Group (MXSG).

#### 13.2 OPERATION.

For the purposes of this TO and AFMAN 21-113, the MCF Flight Chief is synonymous with PMEL Manager, and/or TMDE Flight Chief. The ALCs and MCF shall comply with AFMAN 21-113 and this TO unless exempted below in paragraph 13.3.

**13.2.1 Calibration Locations.** MCF technicians perform on-site calibrations and are embedded with work centers spread across the depot complex. The MCF does not operate from a physical laboratory, and as such, technicians shall meet all on-site calibration and environmental requirements defined in Section 3, Paragraph 3.9.5 of this TO.

**13.2.2 Workload.** MCF workload is defined below in paragraph 13.4. MCF calibration technicians are authorized to use TMDE Certification Stamps.

**13.2.3 Training.** MCF personnel shall be qualified/trained and shall maintain training records IAW AF Sustainment Center (AFSC) or local policy. The MCF Flight Chief may request PMEL supplemental course quotas through the AFMC MFM. The MCF shall develop a training program, to include OJT and classroom training. This training will address PMEL measurement disciplines and advanced/unique training requirements at the ALC.

**13.2.4 Management System (MS) and Quality Program (QP).** The MCF is not part of the Type IIA PMEL and shall have its own separate MS and QP.

**13.2.5 TMDE Coordinator Training.** The MCF shall establish a TMDE Coordinator training program separate from the Type IIA and maintain a database or log to track training events.

**13.2.6 PATEC, JETSC and Base Measurement Standards.** The MCF may own and/or operate a PATEC or JETSC if approved by the AFMC MFM and AFMETCAL. In rare cases, the MCF may also be authorized to own and use a Base Measurement Standard. Base Measurement Standards must be approved for on-site use via email by AFMETCAL. This approval, which may be granted permanently, requires documented evidence the Base Measurement Standard will be stored and used in a controlled environment suitable for the standard. Email approval from AFMETCAL will be stored with the Base Measurement Standard or with the MCF Quality Program files.

#### 13.3 EXCEPTIONS.

**13.3.1 TO 00-20-14 Exceptions.** TO 00-20-14 requirements for PMELs apply to the MCF except as noted below.

##### 13.3.1.1 SECTION 4 CALIBRATION ASSISTANCE AND SUPPORT AGREEMENTS.

Section 4 paragraphs and sub-paragraphs apply to the MCF. Calibration assistance from the ALC IIA PMEL or AFMETCAL may be requested when a MCF lacks the capability to calibrate assigned TMDE. The MCF requiring support shall notify their respective supervisor of the part number(s) requiring calibration and the reason they cannot support the item(s). The MCF may request assistance as directed in paragraph 4.1 with the following exceptions:

- a. Section 4 Paragraph 4.1.b does not apply to MCF
- b. Section 4 Paragraph 4.1.c does not apply to MCF
- c. Section 4 Paragraph 4.1.2 does not apply to MCF
- d. Section 4 Paragraph 4.3 does not apply to MCF

**13.3.1.2 SECTION 7 PMEL ASSESSMENT PROGRAM.**

Section 7 paragraphs and sub-paragraphs apply to the MCF. AFMETCAL shall assess and certify the MCF separately from the Type IIA PMEL. The MCF shall maintain certification IAW Section 7 of this TO except the following:

- a. Section 7 Paragraph 7.4.2.1.c does not apply to MCF.
- b. Section 7 Paragraph 7.4.3.c does not apply to MCF.

**13.3.1.3 SECTION 8 PMEL FACILITY OPERATIONAL REQUIREMENTS.**

Section 8 paragraphs and sub-paragraphs do not apply to the MCF. The MCF must comply with the environmental requirements and additional care noted in paragraph 3.9.5.

**13.3.1.4 SECTION 9 QUALITY PROGRAM.**

Section 9 paragraphs and sub-paragraphs apply to the MCF. Measurement Area Disciplines (MADs) shall be identified in the Quality Manual. Along with the applicable standard MADs defined in paragraph 9.2.1.2.1, the MCF shall identify unique, complex, or other high-risk calibrations to be included in the MAD listing. Due to the unique nature of the ALC mission, the MCF must ensure those measurements and calibrations that present the most risk to the MCF and ALC operation are defined. TMDE calibrations involving the use of a Locally Developed Procedure may meet this criteria.

**13.3.1.5 SECTION 11 PROFICIENCY TESTING/MEASUREMENT ASSURANCE PROGRAM.**

Section 11 paragraphs and sub-paragraphs do not normally apply to the MCF. However, if the MCF is performing calibrations of items identified with a calibration responsibility of PMEL IAW paragraph 3.1.5.3, and those items have been designated part of the proficiency testing program, including usage of exchange standards and/or artifact standards, the MCF is required to complete the proficiency testing and follow all directives that apply to the PMELs in Section 11.

**13.4 MCF WORKLOAD DEFINITION.**

TMDE designated as “MCF” calibration responsibility in TO 33K-1-100-1, TO 33K-1-100-2, and/or CMS TOs shall be calibrated by the MCF. The MCF may request calibration assistance from the Type IIA PMEL if required. TMDE designated “PMEL” responsibility is supported by the Type IIA PMEL. Items assigned (or which may be assigned) a Calibration Responsibility of MCF include:

**13.4.1 On-site, non-portable TMDE located at the ALC.** This includes non-portable ALC Automatic Test Stations (ATS) and Automatic Test Equipment (ATE). This does not include any on-site calibrations located outside the ALC unless approved by AFMETCAL.

**13.4.2 Equipment embedded in ALC test stands and/or stations.** This includes pressure gauges, panel meters, temperature monitors, or other TMDE that are mounted in test stands or stations and the removal of such could potentially disrupt depot operations or damage the test stand. In this case, the entire test stand is considered MCF responsibility and all associated TMDE will be calibrated on-site.

**13.4.3 Peculiar TMDE in the following categories:**

**13.4.3.1 Proprietary TMDE solely used at the ALCs.** Documentation demonstrating the proprietary limitations imposed by the manufacturer must be maintained by the MCF.

**13.4.3.2 One-of-a-kind TMDE that differs significantly from common TMDE in length of calibration or repair, difficulty of calibration or repair, or requiring special training not normally found at a base-level PMEL.** An example of this includes Infrared Target Simulator (IRTS) repairs performed at OO-ALC.

**13.4.3.3 TMDE critical to ALC production line.** MCF Calibration Responsibility will only be assigned for this equipment if the Type IIA contract turnaround/priority response time is insufficient to meet ALC production demands. Normally in these circumstances, the MCF may submit a request to deviate from the Calibration Responsibility Determination. AFMETCAL may permanently approve a MCF calibration responsibility for TMDE, which when removed for calibration, causes recurring ALC production issues.

**13.5 ADDITIONAL REQUIREMENTS.**

**13.5.1 Calibration Responsibility.** TMDE designated as “PMEL” calibration responsibility must be sent to the Type IIA PMEL or the Type IIA PMEL must perform the on-site calibration. If TMDE is designated as “PMEL” calibration responsibility and the TMDE meets one of the MCF workload definitions above, then the MCF may contact AFMETCAL to request a change to the calibration responsibility. If TMDE is easily removable from a console, test station, or test stand and that TMDE is not considered peculiar or does not meet the definition in paragraphs 13.4.1 and/or 13.4.2, then it shall be calibrated by the Type IIA PMEL.

**13.5.2 Responsibility Determination Exceptions.** The MCF may request to deviate from the “PMEL” calibration responsibility determination, but they must comply with paragraph 3.1.5.3 of this TO. Calibration of a USER designated item by a MCF is not considered a deviation of the calibration responsibility determination.

**13.5.3 MDC and Inventory Reporting Requirements.** MDC Data must be provided by the 10th of each month for the previous month MDC (e.g., June data due by 10 July). The complete MCF Inventory data shall be provided Semiannually no later than 31 January and 31 July. Data shall be provided in accordance with the instructions found on the AFMETCAL SharePoint.



## SECTION 14

### ACQUISITION MANAGEMENT AND INTEGRATION CENTER (AMIC) AIR FORCE ENTERPRISE CONTRACTED (AFEC) PMEL SERVICES

#### 14.1 PURPOSE.

The Air Force Enterprise Contracted (AFEC) PMEL Services contract is administered by Air Combat Command's (ACC) Acquisition Management and Integration Center (AMIC). AFEC PMEL is a multi-command, single contract providing an enterprise approach to contracted PMEL services. AMIC maintains the roles of Functional Services Manager (FSM), Program Manager (PM), Contracting Officer (CO), and contract oversight via Contracting Officer's Representatives (COR) IAW AFI 63-138, *Acquisition of Services*. MAJCOM Functional Managers and PMEL Host Wings maintain responsibility for maintenance of facilities, HVAC, and the PMEL's equipment inventory.

#### 14.2 OPERATION.

The Contractors shall comply with Performance Work Statement requirements, AFMAN 21-113, and this TO unless exempted below in paragraph 14.3.

**14.2.1 Government Roles.** The following positions comprise the Government's Contract Management Team.

**14.2.1.1 Contracting Officer (CO).** CO duties are detailed in the Federal Acquisition Regulation (FAR), subpart 1.602-2, *COR*, and AFFARS MP5346.103, *Contracting Officer Responsibilities*. The CO is the only USG agent authorized to award or modify contracts, and is the only person authorized to contractually obligate the USG.

**14.2.1.2 Program Manager (PM).** The PM assists customers with requirements development and definition, PWS modifications, deficiency resolution, and portfolio management.

**14.2.1.3 Quality Assurance Program Coordinator (QAPC).** The QAPC (also known as Quality Manager) integrates the contract quality requirements into the quality assurance program. The QAPC trains and supervises the CORs.

**14.2.1.4 Contracting Officer's Representatives (COR).** The COR is responsible for providing continuous technical oversight of the Contractor's performance. The purpose is to ensure that the Contractor meets the performance standards contained in the contract.

**14.2.2 Contractor Program Management Office (PMO).** The Contractor's Program Management Office is comprised of, at minimum, the Contractor's Program Manager and Quality Manager. They are responsible for developing, documenting, implementing, maintaining, and continuously improving a single comprehensive Quality Management System (QMS) which includes all locations operated by the Contractor under the AFEC PMEL program. For the purposes of this TO and AFMAN 21-113, QMS is a combination of the Management System and the Quality Program.

**14.2.3 Enterprise Approach.** The Contractor shall utilize a single QMS. Quality Manuals, QMS documents, processes and procedures shall mirror each other with minimal adjustments for location specific entries. The QMS should be implemented in a way that all locations operated by the Contractor can be viewed as elements of a single PMEL.

**14.2.4 COR audits/AFMETCAL certification.** Contractors will perform all calibration process and product audits in the presence of the Government COR, or when requested, AFMETCAL evaluator. These calibration process and product audits shall be performed by the Contractor following the Contractor's normal process as outlined in the Quality Manual.

#### 14.3 EXCEPTIONS.

**14.3.1 TO 00-20-14 Exceptions.** TO 00-20-14 requirements apply to AFEC PMELs except as noted below:

##### 14.3.1.1 SECTION 3 OPERATION.

Paragraph 3.10.2 Overdue equipment notifications are managed IAW the PWS.

**14.3.1.2 SECTION 4 CALIBRATION ASSISTANCE AND SUPPORT AGREEMENTS.**

AFEC PMEL contracts are awarded to companies using an enterprise approach to calibration support. Contractors are permitted to temporarily shift calibration support to sites within their network to maintain mission support and meet customer needs. Any changes in support not approved by the Government (AMIC PMs) in writing, will be accomplished at the Contractor's expense. The responsibility for monitoring and enforcing changes in support falls solely to AMIC.

Paragraph 4.4 PMEL Support Agreements. AMIC has the sole responsibility for establishing, maintaining, and monitoring support agreements between contract PMELs and their customers.

**14.3.1.3 SECTION 7 PMEL ASSESSMENT PROGRAM.**

Not all host wings align contracted PMELs under an organization on base. The contracted PMEL may not be assigned to a squadron, group, or directorate with an appropriate individual to provide an executive summary out-brief IAW paragraph 7.4.2.3 of this TO.

**14.3.1.4 SECTION 9 QUALITY PROGRAM (QP).**

Annual Process Reviews and EPEs. Due to the enterprise QMS requirement, contract employees can perform any laboratory function at any location under the Contractor's QMS without requiring additional PRs and EPEs at each location, providing they've met the requirements at their primary duty location.

Paragraph 9.2.1.2.2.1, Annual Process Reviews. Contract employees who have had two ARs can perform calibration functions at any site operated by the Contractor, without requiring two new ARs at each location.

Paragraph 9.2.1.2.3, Evaluator Proficiency Evaluation (EPE). Contract Quality Personnel who have met the initial EPEs requirement can perform quality functions at any site operated by the Contractor, without requiring additional EPEs at each location.

Contractor Corporate Program Managers and Quality Managers who perform EPEs on PMEL quality personnel do not require EPEs themselves. They are the Contractor equivalent to Flight Chief/PMEL Managers.

**14.3.1.5 SECTION 10 MANAGEMENT SYSTEM (MS).**

Paragraph 10.3, Management Internal Control Toolset (MICT). Not all host wings allow Contractor access to MICT. For those PMELs without access to MICT, an internal audit shall be accomplished IAW paragraph 10.4 of this TO.

Paragraph 10.6, PMEL Activity Summary (PAS). Contractors do not participate in the MSEP, or report performance and quality data to the host wing.

Paragraph 10.7, Trainer Proficiency Evaluation (TPE). Contractors are not required to perform TPEs. Contractors are not required to have a formal training programs/processes.